```
tau1 = 1.0e-5*ddth1*(2751*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th5)**2 - 6811.4*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th5) - 12771.375*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th5) +
    6442.0*sin(th1)**2*sin(th2)**2*sin(th3)**2*cos(th5)**2 + 22616.25*sin(th1)**2*sin(th2)**2*sin(th3)**2*cos(th5) + 65802.4*sin(th1)**2*sin(th2)**2*sin(th3)**2 -
    4752*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th5) + 30807.0*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*cos(th3) + 6811.4*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th3)*cos(th4) -
    7382.0*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4)*cos(th5) - 22616.25*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4) -
    3746.27*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3) - 10217.1*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5) - 6811.4*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th4)*cos(th5)**2 -
    12771.375*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th4)*cos(th5) + 1158.0*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th4) + 6634.1*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th5) +
    37484.832*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3) + 18093.0*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th5) + 102231.36*sin(th1)**2*sin(th2)**2*sin(th3) +
    8611*sin(th1)**2*sin(th2)**2*sin(th4)**2*cos(th3)**2 + 4752*sin(th1)**2*sin(th2)**2*sin(th4)*sin(th5)*cos(th3)**2*cos(th4) + 9036.72*sin(th1)**2*sin(th2)**2*sin(th4)*cos(th3)**2 +
    24645.6*sin(th1)**2*sin(th2)**2*sin(th4)*cos(th3) + 6442.0*sin(th1)**2*sin(th2)**2*sin(th5)**2*cos(th3)**2*cos(th4)**2 + 6811.4*sin(th1)**2*sin(th2)**2*sin(th5)*cos(th3)**2*cos(th4)**2*cos(th5) -
    6634.1*sin(th1)**2*sin(th2)**2*sin(th5)*cos(th3)**2*cos(th4) - 18093.0*sin(th1)**2*sin(th2)**2*sin(th5)*cos(th3)*cos(th4) + 2751*sin(th1)**2*sin(th2)**2*cos(th3)**2*cos(th4)**2*cos(th5)**2 +
    295*sin(th1)**2*sin(th2)**2*cos(th3)**2*cos(th4)**2 - 3746.27*sin(th1)**2*sin(th2)**2*cos(th3)**2*cos(th4)*cos(th5) + 339.68*sin(th1)**2*sin(th2)**2*cos(th3)**2*cos(th4) +
    10870.35*sin(th1)**2*sin(th2)**2*cos(th3)**2 - 10217.1*sin(th1)**2*sin(th2)**2*cos(th3)*cos(th4)*cos(th5) + 926.4*sin(th1)**2*sin(th2)**2*cos(th3)*cos(th4) + 47668.8*sin(th1)**2*sin(th2)**2*cos(th3) +
    78713.88*sin(th1)**2*sin(th2)**2 - 4752*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th5) + 30807.0*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*cos(th2) +
    6811.4*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th2)*cos(th4) - 7382.0*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4)*cos(th5) -
    22616.25*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4) - 3746.27*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2) - 6811.4*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th4)*cos(th5)**2
    - 12771.375*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th4)*cos(th5) + 1158.0*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th4) + 6634.1*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th5) +
    37484.832*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2) + 17222*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)**2*cos(th2)*cos(th3) + 9504*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3)*cos(th4)
    + 18073.44*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th2)*cos(th3) + 24645.6*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th2) +
    12884.0*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)**2*cos(th2)*cos(th3)*cos(th4)**2 - 5502*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)**2*cos(th2)*cos(th3) +
    13622.8*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4)**2*cos(th5) - 13268.2*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4) +
    13622.8*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th5) + 25542.75*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3) -
    18093.0*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th4) + 4001.6975*sin(th1)**2*sin(th2)*sin(th3)*sin(th5) + 5502*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)**2*cos(th5)**2 +
    590*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)**2 - 7492.54*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)*cos(th5) +
    679.36*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4) - 12884.0*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th5)**2 - 45232.5*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th5) -
     109864.1*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3) - 10217.1*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th4)*cos(th5) + 926.4*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th4) +
    47668.8*sin(th1)**2*sin(th2)*sin(th3)*cos(th2) - 7086.425*sin(th1)**2*sin(th2)*sin(th3)*cos(th5) - 40040.616*sin(th1)**2*sin(th2)*sin(th3) +
    4752*sin(th1)**2*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th5) - 30807.0*sin(th1)**2*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2 - 9652.86*sin(th1)**2*sin(th2)*sin(th4)*cos(th3) -
    6811.4*sin(th1)**2*sin(th2)*sin(th5)**2*cos(th2)*cos(th3)**2*cos(th4) + 7382.0*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) +
    22616.25*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4) + 3746.27*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2 + 10217.1*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3) +
    7086.425*sin(th1)**2*sin(th2)*sin(th5)*cos(th3)*cos(th4) + 6811.4*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5)**2 + 12771.375*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) -
     1158.0*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th4) - 6634.1*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th5) - 37484.832*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2 -
    18093.0*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)*cos(th5) - 102231.36*sin(th1)**2*sin(th2)*cos(th2)*cos(th3) + 16575.6*sin(th1)**2*sin(th2)*cos(th2) +
    4001.6975*sin(th1)**2*sin(th2)*cos(th3)*cos(th4)*cos(th5) - 362.84*sin(th1)**2*sin(th2)*cos(th3)*cos(th4) - 18670.28*sin(th1)**2*sin(th2)*cos(th3) - 61363.2*sin(th1)**2*sin(th2) +
    8611*sin(th1)**2*sin(th3)**2*sin(th4)**2*cos(th2)**2 + 4752*sin(th1)**2*sin(th3)**2*sin(th4)*sin(th5)*cos(th2)**2*cos(th4) + 9036.72*sin(th1)**2*sin(th3)**2*sin(th4)*cos(th2)**2 +
    6442.0*sin(th1)**2*sin(th3)**2*sin(th5)**2*cos(th2)**2*cos(th4)**2 + 6811.4*sin(th1)**2*sin(th3)**2*sin(th5)*cos(th2)**2*cos(th4)**2*cos(th5) -
    6634.1*sin(th1)**2*sin(th3)**2*sin(th5)*cos(th2)**2*cos(th4) + 2751*sin(th1)**2*sin(th3)**2*cos(th2)**2*cos(th4)**2*cos(th5)**2 + 295*sin(th1)**2*sin(th3)**2*cos(th2)**2*cos(th4)**2 -
    3746.27*sin(th1)**2*sin(th3)**2*cos(th2)**2*cos(th4)*cos(th5) + 339.68*sin(th1)**2*sin(th3)**2*cos(th2)**2*cos(th4) + 10870.35*sin(th1)**2*sin(th3)**2*cos(th2)**2 +
    4752*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th5) - 30807.0*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)**2*cos(th3) - 9652.86*sin(th1)**2*sin(th3)*sin(th4)*cos(th2) -
    6811.4*sin(th1)**2*sin(th3)*sin(th5)**2*cos(th2)**2*cos(th3)*cos(th4) + 7382.0*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) +
    22616.25*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4) + 3746.27*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th3) + 7086.425*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)*cos(th4) +
    6811.4*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)**2*cos(th5)**2 + 12771.375*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) - 1158.0*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th4)
    - 6634.1*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th5) - 37484.832*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3) + 4001.6975*sin(th1)**2*sin(th3)*cos(th2)*cos(th4)*cos(th5) -
    362.84*sin(th1)**2*sin(th3)*cos(th2)*cos(th4) - 18670.28*sin(th1)**2*sin(th3)*cos(th2) + 6442.0*sin(th1)**2*sin(th4)**2*sin(th5)**2 + 6811.4*sin(th1)**2*sin(th4)**2*sin(th5)*cos(th5) +
    2751*sin(th1)**2*sin(th4)**2*cos(th5)**2 + 295*sin(th1)**2*sin(th4)**2 - 4752*sin(th1)**2*sin(th4)*sin(th5)*cos(th4) + 2751*sin(th1)**2*sin(th5)**2*cos(th2)**2*cos(th3)**2 -
    6811.4*sin(th1)**2*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th5) - 12771.375*sin(th1)**2*sin(th5)*cos(th2)**2*cos(th3)**2 - 4001.6975*sin(th1)**2*sin(th5)*cos(th2)*cos(th3) +
    6442.0*sin(th1)**2*cos(th2)**2*cos(th3)**2*cos(th5)**2 + 22616.25*sin(th1)**2*cos(th2)**2*cos(th3)**2*cos(th5) + 65802.4*sin(th1)**2*cos(th2)**2*cos(th3)**2 + 5632*sin(th1)**2*cos(th2)**2 +
    7086.425*sin(th1)**2*cos(th2)*cos(th3)*cos(th5) + 40040.616*sin(th1)**2*cos(th2)*cos(th3) - 6492.11*sin(th1)**2*cos(th2) + 8611*sin(th1)**2*cos(th4)**2 + 22710.053*sin(th1)**2 -
    9.094947017729280e-13*sin(th1)*cos(th1)*cos(th4) + 2751*sin(th2)**2*sin(th3)**2*sin(th5)**2*cos(th1)**2 - 6811.4*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th5) -
    12771.375*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th1)**2 + 6442.0*sin(th2)**2*sin(th3)**2*cos(th1)**2*cos(th5)**2 + 22616.25*sin(th2)**2*sin(th3)**2*cos(th1)**2*cos(th5) +
    65802.4*sin(th2)**2*sin(th3)**2*cos(th1)**2 - 4752*sin(th2)**2*sin(th3)*sin(th4)*cos(th1)**2*cos(th3)*cos(th5) + 30807.0*sin(th2)**2*sin(th3)*sin(th4)*cos(th1)**2*cos(th3) +
    6811.4*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th3)*cos(th4) - 7382.0*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5) -
    22616.25*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3)*cos(th4) - 3746.27*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3) - 10217.1*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2 -
    6811.4*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)**2*cos(th4) + 3746.27*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3) + 10217.1*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2 +
    6811.4*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)**2*cos(th5)**2 + 22616.25*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)**2*cos(th5) + 65802.4*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)**2 +
    102231.36*sin(th2)**2*sin(th3)*cos(th1)**2 + 8611*sin(th2)**2*sin(th4)**2*cos(th1)**2*cos(th3)**2 + 4752*sin(th2)**2*sin(th4)*sin(th5)*cos(th1)**2*cos(th3)**2*cos(th4) +
    9036.72*sin(th2)**2*sin(th4)*cos(th1)**2*cos(th3)**2 + 24645.6*sin(th2)**2*sin(th4)*cos(th1)**2*cos(th3) + 6442.0*sin(th2)**2*sin(th5)**2*cos(th1)**2*cos(th3)**2*cos(th4)**2 +
    6811.4*sin(th2)**2*sin(th5)*cos(th1)**2*cos(th3)**2*cos(th4)**2*cos(th5) - 6634.1*sin(th2)**2*sin(th5)*cos(th1)**2*cos(th3)**2*cos(th4) - 18093.0*sin(th2)**2*sin(th5)*cos(th1)**2*cos(th3)*cos(th4) +
    2751*sin(th2)**2*cos(th1)**2*cos(th3)**2*cos(th4)**2*cos(th5)**2 + 295*sin(th2)**2*cos(th1)**2*cos(th3)**2*cos(th4)**2 - 3746.27*sin(th2)**2*cos(th1)**2*cos(th3)**2*cos(th4)*cos(th5) +
    339.68*sin(th2)**2*cos(th1)**2*cos(th3)**2*cos(th4) + 10870.35*sin(th2)**2*cos(th1)**2*cos(th3)**2 - 10217.1*sin(th2)**2*cos(th1)**2*cos(th3)*cos(th4)*cos(th5) +
    926.4*sin(th2)**2*cos(th1)**2*cos(th3)*cos(th4) + 47668.8*sin(th2)**2*cos(th1)**2*cos(th3) + 78713.88*sin(th2)**2*cos(th1)**2 - 4752*sin(th2)*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2)*cos(th5) +
    30807.0*sin(th2)*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2) + 6811.4*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th4) -
    7382.0*sin(th2)*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th4)*cos(th5) - 22616.25*sin(th2)*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th4) -
    3746.27*sin(th2)*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2) - 6811.4*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th4)*cos(th5)**2 - 12771.375*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th4)*cos(th5)
     + 1158.0*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th4) + 6634.1*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th5) + 37484.832*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2) +
    17222*sin(th2)*sin(th3)*sin(th4)**2*cos(th1)**2*cos(th2)*cos(th3) + 9504*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) +
    18073.44*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3) + 24645.6*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th2) +
    12884.0*sin(th2)*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)**2 - 5502*sin(th2)*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3) +
    13622.8*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)**2*cos(th5) - 13268.2*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) +
```

```
13622.8*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) + 25542.75*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3) -
18093.0*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th4) + 4001.6975*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2 + 5502*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)**2*cos(th5)**2 +
590*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)**2 - 7492.54*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5) +
679.36*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) - 12884.0*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5)**2 - 45232.5*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) -
 109864.1*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3) - 10217.1*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th4)*cos(th5) + 926.4*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th4) +
47668.8*sin(th2)*sin(th3)*cos(th1)**2*cos(th2) - 7086.425*sin(th2)*sin(th3)*cos(th1)**2*cos(th5) - 40040.616*sin(th2)*sin(th3)*cos(th1)**2 +
4752*sin(th2)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th5) - 30807.0*sin(th2)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)**2 - 9652.86*sin(th2)*sin(th4)*cos(th1)**2*cos(th3) -
6811.4*sin(th2)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4) + 7382.0*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) +
22616.25*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4) + 3746.27*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2 + 10217.1*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3) +
7086.425*sin(th2)*sin(th5)*cos(th1)**2*cos(th3)*cos(th4) + 6811.4*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)*cos(th5)**2 + 12771.375*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) -
 1158.0*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4) - 6634.1*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th5) - 37484.832*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2 -
18093.0*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) - 102231.36*sin(th2)*cos(th1)**2*cos(th2)*cos(th3) + 16575.6*sin(th2)*cos(th1)**2*cos(th2) +
4001.6975*sin(th2)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5) - 362.84*sin(th2)*cos(th1)**2*cos(th3)*cos(th4) - 18670.28*sin(th2)*cos(th1)**2*cos(th3) - 61363.2*sin(th2)*cos(th1)**2 +
8611*sin(th3)**2*sin(th4)**2*cos(th1)**2*cos(th2)**2 + 4752*sin(th3)**2*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th4) + 9036.72*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2)**2 +
6442.0*sin(th3)**2*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th4)**2 + 6811.4*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th4)**2*cos(th5) -
6634.1*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th4) + 2751*sin(th3)**2*cos(th1)**2*cos(th2)**2*cos(th4)**2*cos(th5)**2 + 295*sin(th3)**2*cos(th1)**2*cos(th2)**2*cos(th4)**2 -
3746.27*sin(th3)**2*cos(th1)**2*cos(th2)**2*cos(th4)*cos(th5) + 339.68*sin(th3)**2*cos(th1)**2*cos(th2)**2*cos(th4) + 10870.35*sin(th3)**2*cos(th1)**2*cos(th2)**2 +
4752*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th5) - 30807.0*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3) - 9652.86*sin(th3)*sin(th4)*cos(th1)**2*cos(th2) -
6811.4*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4) + 7382.0*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) +
22616.25*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4) + 3746.27*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3) + 7086.425*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th4) +
6811.4*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5)**2 + 12771.375*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) - 1158.0*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)
- 6634.1*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th5) - 37484.832*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3) + 4001.6975*sin(th3)*cos(th1)**2*cos(th2)*cos(th4)*cos(th5) -
362.84*sin(th3)*cos(th1)**2*cos(th2)*cos(th4) - 18670.28*sin(th3)*cos(th1)**2*cos(th2) + 6442.0*sin(th4)**2*sin(th5)**2*cos(th1)**2 + 6811.4*sin(th4)**2*sin(th5)*cos(th1)**2*cos(th5) +
2751*sin(th4)**2*cos(th1)**2*cos(th5)**2 + 295*sin(th4)**2*cos(th1)**2 - 4752*sin(th4)*sin(th5)*cos(th1)**2*cos(th4) + 2751*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th3)**2 -
6811.4*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th5) - 12771.375*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)**2 - 4001.6975*sin(th5)*cos(th1)**2*cos(th2)*cos(th3) +
6442.0*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th5)**2 + 22616.25*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th5) + 65802.4*cos(th1)**2*cos(th2)**2*cos(th3)**2 + 5632*cos(th1)**2*cos(th2)**2 +
7086.425*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) + 40040.616*cos(th1)**2*cos(th2)*cos(th3) - 6492.11*cos(th1)**2*cos(th2) + 8611*cos(th1)**2*cos(th4)**2 + 22710.053*cos(th1)**2) + 4.0e-11*ddth1 +
1.0e-5*ddth2*(2376*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)**2*sin(th5) + 6442.0*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5)**2*cos(th4) +
6811.4*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th4)*cos(th5) - 3317.05*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5) + 2751*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th4)*cos(th5)**2 -
8316*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th4) - 1873.08*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th5) + 169.84*sin(th1)**2*sin(th2)*sin(th3)*sin(th4) -
2376*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th4)**2 - 4518.36*sin(th1)**2*sin(th2)*sin(th3)*cos(th4) + 222.2*sin(th1)**2*sin(th2)*sin(th3) - 3405.8*sin(th1)**2*sin(th2)*sin(th4)*sin(th5)**2*cos(th3)
 + 3691.0*sin(th1)**2*sin(th2)*sin(th4)*sin(th5)*cos(th3)*cos(th5) + 11308.125*sin(th1)**2*sin(th2)*sin(th4)*sin(th5)*cos(th3) + 3405.6*sin(th1)**2*sin(th2)*sin(th4)*cos(th3)*cos(th5)**2 +
6385.5*sin(th1)**2*sin(th2)*sin(th4)*cos(th3)*cos(th5) - 579.0*sin(th1)**2*sin(th2)*sin(th4)*cos(th3) - 2376*sin(th1)**2*sin(th2)*cos(th3)*cos(th4)*cos(th5) +
15403.5*sin(th1)**2*sin(th2)*cos(th3)*cos(th4) - 3405.8*sin(th1)**2*sin(th3)*sin(th4)*sin(th5)**2*cos(th2) + 3691.0*sin(th1)**2*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th5) +
11308.125*sin(th1)**2*sin(th3)*sin(th4)*sin(th5)*cos(th2) + 3405.6*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)*cos(th5)**2 + 6385.5*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)*cos(th5) -
579.0*sin(th1)**2*sin(th3)*sin(th4)*cos(th2) - 2376*sin(th1)**2*sin(th3)*cos(th2)*cos(th4)*cos(th5) + 15403.5*sin(th1)**2*sin(th3)*cos(th2)*cos(th4) -
2376*sin(th1)**2*sin(th4)**2*sin(th5)*cos(th2)*cos(th3) - 6442.0*sin(th1)**2*sin(th4)*sin(th5)**2*cos(th2)*cos(th3)*cos(th4) - 6811.4*sin(th1)**2*sin(th4)*sin(th5)*cos(th2)*cos(th3)*cos(th4)*cos(th5) +
3317.05*sin(th1)**2*sin(th4)*sin(th5)*cos(th2)*cos(th3) + 9046.5*sin(th1)**2*sin(th4)*sin(th5)*cos(th2) - 2751*sin(th1)**2*sin(th4)*cos(th2)*cos(th3)*cos(th4)*cos(th5)**2 +
8316*sin(th1)**2*sin(th4)*cos(th2)*cos(th3)*cos(th4) + 1873.08*sin(th1)**2*sin(th4)*cos(th2)*cos(th3)*cos(th5) - 169.84*sin(th1)**2*sin(th4)*cos(th2)*cos(th3) +
5108.4*sin(th1)**2*sin(th4)*cos(th2)*cos(th5) - 463.2*sin(th1)**2*sin(th4)*cos(th2) + 2376*sin(th1)**2*sin(th5)*cos(th2)*cos(th3)*cos(th4)**2 + 4518.36*sin(th1)**2*cos(th2)*cos(th3)*cos(th4) -
222.2*sin(th1)**2*cos(th2)*cos(th3) + 12322.8*sin(th1)**2*cos(th2)*cos(th4) - 1172.16*sin(th1)**2*cos(th2) + 2376*sin(th2)*sin(th3)*sin(th4)**2*sin(th5)*cos(th1)**2 +
6442.0*sin(th2)*sin(th3)*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th4) + 6811.4*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th4)*cos(th5) - 3317.05*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2 +
2751*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th4)*cos(th5)**2 - 8316*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th4) - 1873.08*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th5) +
169.84*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2 - 2376*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th4)**2 - 4518.36*sin(th2)*sin(th3)*cos(th1)**2*cos(th4) + 222.2*sin(th2)*sin(th3)*cos(th1)**2 -
3405.8*sin(th2)*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th3) + 3691.0*sin(th2)*sin(th4)*sin(th5)*cos(th1)**2*cos(th3)*cos(th5) + 11308.125*sin(th2)*sin(th4)*sin(th5)*cos(th1)**2*cos(th3) +
3405.6*sin(th2)*sin(th4)*cos(th1)**2*cos(th3)*cos(th5)**2 + 6385.5*sin(th2)*sin(th4)*cos(th1)**2*cos(th3)*cos(th5) - 579.0*sin(th2)*sin(th4)*cos(th1)**2*cos(th3) -
2376*sin(th2)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5) + 15403.5*sin(th2)*cos(th1)**2*cos(th3)*cos(th4) - 3405.8*sin(th3)*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th2) +
3691.0*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th5) + 11308.125*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2) + 3405.6*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)*cos(th5)**2 +
6385.5*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)*cos(th5) - 579.0*sin(th3)*sin(th4)*cos(th1)**2*cos(th2) - 2376*sin(th3)*cos(th1)**2*cos(th2)*cos(th4)*cos(th5) +
15403.5*sin(th3)*cos(th1)**2*cos(th2)*cos(th4) - 2376*sin(th4)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th3) - 6442.0*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) -
6811.4*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5) + 3317.05*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3) + 9046.5*sin(th4)*sin(th5)*cos(th1)**2*cos(th2) -
2751*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5)**2 + 8316*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) + 1873.08*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) -
169.84*sin(th4)*cos(th1)**2*cos(th2)*cos(th3) + 5108.4*sin(th4)*cos(th1)**2*cos(th2)*cos(th5) - 463.2*sin(th4)*cos(th1)**2*cos(th2) + 2376*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)**2 +
4518.36*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) - 222.2*cos(th1)**2*cos(th2)*cos(th3) + 12322.8*cos(th1)**2*cos(th2)*cos(th4) - 1172.16*cos(th1)**2*cos(th2)) +
1.0e-5*ddth3*(2376*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)**2*sin(th5) + 6442.0*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5)**2*cos(th4) +
6811.4*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th4)*cos(th5) - 3317.05*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5) + 2751*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th4)*cos(th5)**2 -
8316*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th4) - 1873.08*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th5) + 169.84*sin(th1)**2*sin(th2)*sin(th3)*sin(th4) -
2376*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th4)**2 - 4518.36*sin(th1)**2*sin(th2)*sin(th3)*cos(th4) + 222.2*sin(th1)**2*sin(th2)*sin(th3) - 3405.8*sin(th1)**2*sin(th2)*sin(th4)*sin(th5)**2*cos(th3)
 + 3691.0*sin(th1)**2*sin(th2)*sin(th4)*sin(th5)*cos(th3)*cos(th5) + 11308.125*sin(th1)**2*sin(th2)*sin(th4)*sin(th5)*cos(th3) + 3405.6*sin(th1)**2*sin(th2)*sin(th4)*cos(th3)*cos(th5)**2 +
6385.5*sin(th1)**2*sin(th2)*sin(th4)*cos(th3)*cos(th5) - 579.0*sin(th1)**2*sin(th2)*sin(th4)*cos(th3) - 2376*sin(th1)**2*sin(th2)*cos(th3)*cos(th4)*cos(th5) +
15403.5*sin(th1)**2*sin(th2)*cos(th3)*cos(th4) - 3405.8*sin(th1)**2*sin(th3)*sin(th4)*sin(th5)**2*cos(th2) + 3691.0*sin(th1)**2*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th5) +
11308.125*sin(th1)**2*sin(th3)*sin(th4)*sin(th5)*cos(th2) + 3405.6*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)*cos(th5)**2 + 6385.5*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)*cos(th5) -
579.0*sin(th1)**2*sin(th3)*sin(th4)*cos(th2) - 2376*sin(th1)**2*sin(th3)*cos(th2)*cos(th4)*cos(th5) + 15403.5*sin(th1)**2*sin(th3)*cos(th2)*cos(th4) -
2376*sin(th1)**2*sin(th4)**2*sin(th5)*cos(th2)*cos(th3) - 6442.0*sin(th1)**2*sin(th4)*sin(th5)**2*cos(th2)*cos(th3)*cos(th4) - 6811.4*sin(th1)**2*sin(th4)*sin(th5)*cos(th2)*cos(th3)*cos(th4)*cos(th5) +
3317.05*sin(th1)**2*sin(th4)*sin(th5)*cos(th2)*cos(th3) - 2751*sin(th1)**2*sin(th4)*cos(th2)*cos(th3)*cos(th4)*cos(th5)**2 + 8316*sin(th1)**2*sin(th4)*cos(th2)*cos(th3)*cos(th4) +
1873.08*sin(th1)**2*sin(th4)*cos(th2)*cos(th3)*cos(th5) - 169.84*sin(th1)**2*sin(th4)*cos(th2)*cos(th3) + 2376*sin(th1)**2*sin(th5)*cos(th2)*cos(th3)*cos(th4)**2 +
4518.36*sin(th1)**2*cos(th2)*cos(th3)*cos(th4) - 222.2*sin(th1)**2*cos(th2)*cos(th3) + 2376*sin(th2)*sin(th3)*sin(th4)**2*sin(th5)*cos(th1)**2 +
6442.0*sin(th2)*sin(th3)*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th4) + 6811.4*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th4)*cos(th5) - 3317.05*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2 +
```

```
2751*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th4)*cos(th5)**2 - 8316*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th4) - 1873.08*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th5) +
169.84*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2 - 2376*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th4)**2 - 4518.36*sin(th2)*sin(th3)*cos(th1)**2*cos(th4) + 222.2*sin(th2)*sin(th3)*cos(th1)**2 -
3405.8*sin(th2)*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th3) + 3691.0*sin(th2)*sin(th4)*sin(th5)*cos(th1)**2*cos(th3)*cos(th5) + 11308.125*sin(th2)*sin(th4)*sin(th5)*cos(th1)**2*cos(th3) +
3405.6*sin(th2)*sin(th4)*cos(th1)**2*cos(th3)*cos(th5)**2 + 6385.5*sin(th2)*sin(th4)*cos(th1)**2*cos(th3)*cos(th5) - 579.0*sin(th2)*sin(th4)*cos(th1)**2*cos(th3) -
2376*sin(th2)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5) + 15403.5*sin(th2)*cos(th1)**2*cos(th3)*cos(th4) - 3405.8*sin(th3)*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th2) +
3691.0*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th5) + 11308.125*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2) + 3405.6*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)*cos(th5)**2 +
6385.5*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)*cos(th5) - 579.0*sin(th3)*sin(th4)*cos(th1)**2*cos(th2) - 2376*sin(th3)*cos(th1)**2*cos(th2)*cos(th4)*cos(th5) +
15403.5*sin(th3)*cos(th1)**2*cos(th2)*cos(th4) - 2376*sin(th4)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th3) - 6442.0*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) -
6811.4*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5) + 3317.05*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3) - 2751*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5)**2 +
8316*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) + 1873.08*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) - 169.84*sin(th4)*cos(th1)**2*cos(th2)*cos(th3) +
2376*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)**2 + 4518.36*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) - 222.2*cos(th1)**2*cos(th2)*cos(th3)) +
1.0e-5*ddth4*(-2376*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th5) + 15403.5*sin(th1)**2*sin(th2)*sin(th3)*sin(th4) + 3405.6*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)**2*cos(th4) -
3691.0*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th4)*cos(th5) - 11308.125*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th4) - 3405.8*sin(th1)**2*sin(th2)*sin(th3)*cos(th4)*cos(th5)**2 -
6385.875*sin(th1)**2*sin(th2)*sin(th3)*cos(th4)*cos(th5) + 579.0*sin(th1)**2*sin(th2)*sin(th3)*cos(th4) + 6442.0*sin(th1)**2*sin(th2)*sin(th4)**2*sin(th5)**2*cos(th3) +
6811.4*sin(th1)**2*sin(th2)*sin(th4)**2*sin(th5)*cos(th3)*cos(th5) + 2751*sin(th1)**2*sin(th2)*sin(th4)**2*cos(th3)*cos(th5)**2 + 8906*sin(th1)**2*sin(th2)*sin(th4)**2*cos(th3) +
4518.36*sin(th1)**2*sin(th2)*sin(th4)*cos(th3) + 12322.8*sin(th1)**2*sin(th2)*sin(th4) + 6442.0*sin(th1)**2*sin(th2)*sin(th5)**2*cos(th3)*cos(th4)**2 +
6811.4*sin(th1)**2*sin(th2)*sin(th5)*cos(th3)*cos(th4)**2*cos(th5) - 3317.05*sin(th1)**2*sin(th2)*sin(th5)*cos(th3)*cos(th4) - 9046.5*sin(th1)**2*sin(th2)*sin(th5)*cos(th4) +
2751*sin(th1)**2*sin(th2)*cos(th3)*cos(th4)**2*cos(th5)**2 + 8906*sin(th1)**2*sin(th2)*cos(th3)*cos(th4)*cos(th5) + 1873.19*sin(th1)**2*sin(th2)*cos(th3)*cos(th4)*cos(th5) +
169.84*sin(th1)**2*sin(th2)*cos(th3)*cos(th4) - 5108.7*sin(th1)**2*sin(th2)*cos(th4)*cos(th5) + 463.2*sin(th1)**2*sin(th2)*cos(th4) + 6442.0*sin(th1)**2*sin(th3)*sin(th4)**2*sin(th5)**2*cos(th2) +
6811.4*sin(th1)**2*sin(th3)*sin(th4)**2*sin(th5)*cos(th2)*cos(th5) + 2751*sin(th1)**2*sin(th3)*sin(th4)**2*cos(th2)*cos(th5)**2 + 8906*sin(th1)**2*sin(th3)*sin(th4)**2*cos(th2) +
4518.36*sin(th1)**2*sin(th3)*sin(th4)*cos(th2) + 6442.0*sin(th1)**2*sin(th3)*sin(th5)**2*cos(th2)*cos(th4)**2 + 6811.4*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)*cos(th4)**2*cos(th5) -
3317.05*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)*cos(th4) + 2751*sin(th1)**2*sin(th3)*cos(th2)*cos(th4)**2*cos(th5)**2 + 8906*sin(th1)**2*sin(th3)*cos(th2)*cos(th4)**2 -
1873.19*sin(th1)**2*sin(th3)*cos(th2)*cos(th4)*cos(th5) + 169.84*sin(th1)**2*sin(th3)*cos(th2)*cos(th4) + 2376*sin(th1)**2*sin(th4)*cos(th2)*cos(th3)*cos(th5) -
15403.5*sin(th1)**2*sin(th4)*cos(th2)*cos(th3) - 4826.43*sin(th1)**2*sin(th4) - 3405.6*sin(th1)**2*sin(th5)**2*cos(th2)*cos(th3)*cos(th4) +
3691.0*sin(th1)**2*sin(th5)*cos(th2)*cos(th3)*cos(th4)*cos(th5) + 11308.125*sin(th1)**2*sin(th5)*cos(th2)*cos(th3)*cos(th4) + 3543.2125*sin(th1)**2*sin(th5)*cos(th4) +
3405.8*sin(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5)**2 + 6385.875*sin(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5) - 579.0*sin(th1)**2*cos(th2)*cos(th3)*cos(th4) +
2000.9075*sin(th1)**2*cos(th4)*cos(th5) - 181.42*sin(th1)**2*cos(th4) - 2376*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th5) + 15403.5*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2 +
3405.6*sin(th2)*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th4) - 3691.0*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th4)*cos(th5) - 11308.125*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th4) -
3405.8*sin(th2)*sin(th3)*cos(th1)**2*cos(th4)*cos(th5)**2 - 6385.875*sin(th2)*sin(th3)*cos(th1)**2*cos(th4)*cos(th5) + 579.0*sin(th2)*sin(th3)*cos(th1)**2*cos(th4) +
6442.0*sin(th2)*sin(th4)**2*sin(th5)**2*cos(th1)**2*cos(th3) + 6811.4*sin(th2)*sin(th4)**2*sin(th5)*cos(th1)**2*cos(th3)*cos(th5) + 2751*sin(th2)*sin(th4)**2*cos(th1)**2*cos(th3)*cos(th5)**2 +
8906*sin(th2)*sin(th4)**2*cos(th1)**2*cos(th3) + 4518.36*sin(th2)*sin(th4)*cos(th1)**2*cos(th3) + 12322.8*sin(th2)*sin(th4)*cos(th1)**2 + 6442.0*sin(th2)*sin(th5)**2*cos(th1)**2*cos(th3)*cos(th4)**2 +
6811.4*sin(th2)*sin(th5)*cos(th1)**2*cos(th3)*cos(th4)**2*cos(th5) - 3317.05*sin(th2)*sin(th5)*cos(th1)**2*cos(th3)*cos(th4) - 9046.5*sin(th2)*sin(th5)*cos(th1)**2*cos(th4) +
2751*sin(th2)*cos(th1)**2*cos(th3)*cos(th4)**2*cos(th5)**2 + 8906*sin(th2)*cos(th1)**2*cos(th3)*cos(th4)**2 - 1873.19*sin(th2)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5) +
169.84*sin(th2)*cos(th1)**2*cos(th3)*cos(th4) - 5108.7*sin(th2)*cos(th1)**2*cos(th4)*cos(th5) + 463.2*sin(th2)*cos(th1)**2*cos(th4) + 6442.0*sin(th3)*sin(th4)**2*sin(th5)**2*cos(th1)**2*cos(th2) +
6811.4*sin(th3)*sin(th4)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th5) + 2751*sin(th3)*sin(th4)**2*cos(th1)**2*cos(th2)*cos(th5)**2 + 8906*sin(th3)*sin(th4)**2*cos(th1)**2*cos(th2) +
4518.36*sin(th3)*sin(th4)*cos(th1)**2*cos(th2) + 6442.0*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th4)**2 + 6811.4*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th4)**2*cos(th5) -
3317.05*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th4) + 2751*sin(th3)*cos(th1)**2*cos(th2)*cos(th4)**2*cos(th5)**2 + 8906*sin(th3)*cos(th1)**2*cos(th2)*cos(th4)**2 -
1873.19*sin(th3)*cos(th1)**2*cos(th2)*cos(th4)*cos(th5) + 169.84*sin(th3)*cos(th1)**2*cos(th2)*cos(th4) + 2376*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) -
15403.5*sin(th4)*cos(th1)**2*cos(th2)*cos(th3) - 4826.43*sin(th4)*cos(th1)**2 - 3405.6*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) +
3691.0*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5) + 11308.125*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) + 3543.2125*sin(th5)*cos(th1)**2*cos(th4) +
3405.8*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5)**2 + 6385.875*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5) - 579.0*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) +
2000.9075*cos(th1)**2*cos(th4)*cos(th5) - 181.42*cos(th1)**2*cos(th4)) + 1.0e-5*ddth5*(-9193.0*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5)**2 +
6385.875*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5) - 9193.0*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th5)**2 - 11308.125*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th5) +
2376*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th4) + 2376*sin(th1)**2*sin(th2)*sin(th4)**2*cos(th3)*cos(th5) + 1873.19*sin(th1)**2*sin(th2)*sin(th4)*sin(th5)*cos(th3) +
5108.7*sin(th1)**2*sin(th2)*sin(th4)*sin(th5) - 3317.05*sin(th1)**2*sin(th2)*sin(th4)*cos(th3)*cos(th5) - 9046.5*sin(th1)**2*sin(th2)*sin(th4)*cos(th5) +
2376*sin(th1)**2*sin(th2)*cos(th3)*cos(th4)**2*cos(th5) + 2376*sin(th1)**2*sin(th3)*sin(th4)**2*cos(th2)*cos(th5) + 1873.19*sin(th1)**2*sin(th3)*sin(th4)*sin(th5)*cos(th2) -
3317.05*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)*cos(th5) + 2376*sin(th1)**2*sin(th3)*cos(th2)*cos(th4)**2*cos(th5) + 9193.0*sin(th1)**2*sin(th4)*sin(th5)**2*cos(th2)*cos(th3) -
6385.875*sin(th1)**2*sin(th4)*sin(th5)*cos(th2)*cos(th3) - 2000.9075*sin(th1)**2*sin(th4)*sin(th5) + 9193.0*sin(th1)**2*sin(th4)*cos(th2)*cos(th3)*cos(th5)**2 +
11308.125*sin(th1)**2*sin(th4)*cos(th2)*cos(th3)*cos(th5) + 3543.2125*sin(th1)**2*sin(th4)*cos(th5) - 2376*sin(th1)**2*sin(th5)*cos(th2)*cos(th3)*cos(th4) -
9193.0*sin(th2)*sin(th3)*sin(th4)*sin(th5)**2*cos(th1)**2 + 6385.875*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2 - 9193.0*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th5)**2 -
11308.125*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th5) + 2376*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th4) + 2376*sin(th2)*sin(th4)**2*cos(th1)**2*cos(th3)*cos(th5) +
1873.19*sin(th2)*sin(th4)*sin(th5)*cos(th1)**2*cos(th3) + 5108.7*sin(th2)*sin(th4)*sin(th5)*cos(th1)**2 - 3317.05*sin(th2)*sin(th4)*cos(th1)**2*cos(th3)*cos(th5) -
9046.5*sin(th2)*sin(th4)*cos(th1)**2*cos(th5) + 2376*sin(th2)*cos(th1)**2*cos(th3)*cos(th4)**2*cos(th5) + 2376*sin(th3)*sin(th4)**2*cos(th1)**2*cos(th2)*cos(th5) +
1873.19*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2) - 3317.05*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)*cos(th5) + 2376*sin(th3)*cos(th1)**2*cos(th2)*cos(th4)**2*cos(th5) +
9193.0*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3) - 6385.875*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3) - 2000.9075*sin(th4)*sin(th5)*cos(th1)**2 +
9193.0*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5)**2 + 11308.125*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) + 3543.2125*sin(th4)*cos(th1)**2*cos(th5) -
2376*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)) + 4.174155*sin(th1)*sin(th2)*sin(th3)*sin(th5) - 7.39551375*sin(th1)*sin(th2)*sin(th3)*cos(th5) + 14.5652994*sin(th1)*sin(th2)*sin(th3) +
15.90107805*sin(th1)*sin(th2)*sin(th4)*cos(th3) + 7.39551375*sin(th1)*sin(th2)*sin(th5)*cos(th3)*cos(th4) + 4.174155*sin(th1)*sin(th2)*cos(th3)*cos(th4)*cos(th5) +
0.378666*sin(th1)*sin(th2)*cos(th3)*cos(th4) + 11.498301*sin(th1)*sin(th2)*cos(th3) + 22.42566*sin(th1)*sin(th2) + 15.90107805*sin(th1)*sin(th3)*sin(th4)*cos(th2) +
7.39551375*sin(th1)*sin(th3)*sin(th5)*cos(th2)*cos(th4) + 4.174155*sin(th1)*sin(th3)*cos(th2)*cos(th4)*cos(th5) + 0.378666*sin(th1)*sin(th3)*cos(th2)*cos(th4) + 11.498301*sin(th1)*sin(th3)*cos(th2) -
4.174155*sin(th1)*sin(th5)*cos(th2)*cos(th3) + 7.39551375*sin(th1)*cos(th2)*cos(th3)*cos(th5) - 14.5652994*sin(th1)*cos(th2)*cos(th3) - 8.3205477*sin(th1) - 19.83582*sin(th2)*cos(th1) +
7.39551375*sin(th4)*sin(th5)*cos(th1) + 4.174155*sin(th4)*cos(th1)*cos(th5) + 0.378666*sin(th4)*cos(th1) - 6.7772385*cos(th1)*cos(th2) - 15.90107805*cos(th1)*cos(th4) + 8.2634535*cos(th1)
```

```
tau2 = 1.0e-5*ddth1*(2376*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)**2*sin(th5) + 6442.0*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5)**2*cos(th4) +
    6811.4*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th4)*cos(th5) - 3317.05*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5) + 2751*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th4)*cos(th5)**2 -
    8316*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th4) - 1873.08*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th5) + 169.84*sin(th1)**2*sin(th2)*sin(th3)*sin(th4) -
    2376*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th4)**2 - 4518.36*sin(th1)**2*sin(th2)*sin(th3)*cos(th4) + 222.2*sin(th1)**2*sin(th2)*sin(th3) - 3405.8*sin(th1)**2*sin(th2)*sin(th4)*sin(th5)**2*cos(th3)
     + 3691.0*sin(th1)**2*sin(th2)*sin(th4)*sin(th5)*cos(th3)*cos(th5) + 11308.125*sin(th1)**2*sin(th2)*sin(th4)*sin(th5)*cos(th3) + 3405.6*sin(th1)**2*sin(th2)*sin(th4)*cos(th3)*cos(th5)**2 +
    6385.5*sin(th1)**2*sin(th2)*sin(th4)*cos(th3)*cos(th5) - 579.0*sin(th1)**2*sin(th2)*sin(th4)*cos(th3) - 2376*sin(th1)**2*sin(th2)*cos(th3)*cos(th4)*cos(th5) +
    15403.5*sin(th1)**2*sin(th2)*cos(th3)*cos(th4) - 3405.8*sin(th1)**2*sin(th3)*sin(th4)*sin(th5)**2*cos(th2) + 3691.0*sin(th1)**2*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th5) +
    11308.125*sin(th1)**2*sin(th3)*sin(th4)*sin(th5)*cos(th2) + 3405.6*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)*cos(th5)**2 + 6385.5*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)*cos(th5) -
    579.0*sin(th1)**2*sin(th3)*sin(th4)*cos(th2) - 2376*sin(th1)**2*sin(th3)*cos(th2)*cos(th4)*cos(th5) + 15403.5*sin(th1)**2*sin(th3)*cos(th2)*cos(th4) -
    2376*sin(th1)**2*sin(th4)**2*sin(th5)*cos(th2)*cos(th3) - 6442.0*sin(th1)**2*sin(th4)*sin(th5)**2*cos(th2)*cos(th3)*cos(th4) - 6811.4*sin(th1)**2*sin(th4)*sin(th5)*cos(th2)*cos(th3)*cos(th4)*cos(th5) +
    3317.05*sin(th1)**2*sin(th4)*sin(th5)*cos(th2)*cos(th3) + 9046.5*sin(th1)**2*sin(th4)*sin(th5)*cos(th2) - 2751*sin(th1)**2*sin(th4)*cos(th2)*cos(th3)*cos(th4)*cos(th5)**2 +
    8316*sin(th1)**2*sin(th4)*cos(th2)*cos(th3)*cos(th4) + 1873.08*sin(th1)**2*sin(th4)*cos(th2)*cos(th3)*cos(th5) - 169.84*sin(th1)**2*sin(th4)*cos(th2)*cos(th3) +
    5108.4*sin(th1)**2*sin(th4)*cos(th2)*cos(th5) - 463.2*sin(th1)**2*sin(th4)*cos(th2) + 2376*sin(th1)**2*sin(th5)*cos(th2)*cos(th3)*cos(th4)**2 + 4518.36*sin(th1)**2*cos(th2)*cos(th3)*cos(th4) -
    222.2*sin(th1)**2*cos(th2)*cos(th3) + 12322.8*sin(th1)**2*cos(th2)*cos(th4) - 1172.16*sin(th1)**2*cos(th2) + 2376*sin(th2)*sin(th3)*sin(th4)**2*sin(th5)*cos(th1)**2 +
    6442.0*sin(th2)*sin(th3)*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th4) + 6811.4*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th4)*cos(th5) - 3317.05*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2 +
    2751*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th4)*cos(th5)**2 - 8316*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th4) - 1873.08*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th5) +
    169.84*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2 - 2376*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th4)**2 - 4518.36*sin(th2)*sin(th3)*cos(th1)**2*cos(th4) + 222.2*sin(th2)*sin(th3)*cos(th1)**2 -
    3405.8*sin(th2)*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th3) + 3691.0*sin(th2)*sin(th4)*sin(th5)*cos(th1)**2*cos(th3)*cos(th5) + 11308.125*sin(th2)*sin(th4)*sin(th5)*cos(th1)**2*cos(th3) +
    3405.6*sin(th2)*sin(th4)*cos(th1)**2*cos(th3)*cos(th5)**2 + 6385.5*sin(th2)*sin(th4)*cos(th1)**2*cos(th3)*cos(th5) - 579.0*sin(th2)*sin(th4)*cos(th1)**2*cos(th3) -
    2376*sin(th2)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5) + 15403.5*sin(th2)*cos(th1)**2*cos(th3)*cos(th4) - 3405.8*sin(th3)*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th2) +
    3691.0*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th5) + 11308.125*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2) + 3405.6*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)*cos(th5)**2 +
    6385.5*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)*cos(th5) - 579.0*sin(th3)*sin(th4)*cos(th1)**2*cos(th2) - 2376*sin(th3)*cos(th1)**2*cos(th2)*cos(th4)*cos(th5) +
    15403.5*sin(th3)*cos(th1)**2*cos(th2)*cos(th4) - 2376*sin(th4)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th3) - 6442.0*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) -
    6811.4*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5) + 3317.05*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3) + 9046.5*sin(th4)*sin(th5)*cos(th1)**2*cos(th2) -
    2751*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5)**2 + 8316*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) + 1873.08*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) -
    169.84*sin(th4)*cos(th1)**2*cos(th2)*cos(th3) + 5108.4*sin(th4)*cos(th1)**2*cos(th2)*cos(th5) - 463.2*sin(th4)*cos(th1)**2*cos(th2) + 2376*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)**2 +
    4518.36*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) - 222.2*cos(th1)**2*cos(th2)*cos(th3) + 12322.8*cos(th1)**2*cos(th2)*cos(th4) - 1172.16*cos(th1)**2*cos(th2)) +
    1.0e-5*ddth2*(8611*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th4)**2 + 4752*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th4)*sin(th5)*cos(th4) + 9036.72*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th4) +
    6442.0*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th5)**2*cos(th4)**2 + 6811.4*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th4)**2*cos(th5) -
    6634.1*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th4) + 2751*sin(th1)**2*sin(th2)**2*sin(th3)**2*cos(th4)**2*cos(th5)**2 + 295*sin(th1)**2*sin(th2)**2*sin(th3)**2*cos(th4)**2 -
    3746.27*sin(th1)**2*sin(th2)**2*sin(th3)**2*cos(th4)*cos(th5) + 339.68*sin(th1)**2*sin(th2)**2*sin(th3)**2*cos(th4) + 10870.35*sin(th1)**2*sin(th2)**2*sin(th3)**2 +
    4752*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th5) - 30807.0*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*cos(th3) - 6811.4*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th3)*cos(th4) +
    7382.0*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4)*cos(th5) + 22616.25*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4) +
    3746.27*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3) + 6811.4*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th4)*cos(th5)**2 + 12771.375*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th4)*cos(th5)
     - 1158.0*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th4) - 6634.1*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th5) - 37484.832*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3) +
    2751*sin(th1)**2*sin(th2)**2*sin(th5)**2*cos(th3)**2 - 6811.4*sin(th1)**2*sin(th2)**2*sin(th5)*cos(th3)**2*cos(th5) - 12771.375*sin(th1)**2*sin(th2)**2*sin(th5)*cos(th3)**2 +
    6442.0*sin(th1)**2*sin(th2)**2*cos(th3)**2*cos(th5)**2 + 22616.25*sin(th1)**2*sin(th2)**2*cos(th3)**2*cos(th5) + 65802.4*sin(th1)**2*sin(th2)**2*cos(th3)**2 + 5632*sin(th1)**2*sin(th2)**2 +
    4752*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th5) - 30807.0*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*cos(th2) - 6811.4*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th2)*cos(th4) +
    7382.0*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4)*cos(th5) + 22616.25*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4) +
    3746.27*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2) + 6811.4*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th4)*cos(th5)**2 + 12771.375*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th4)*cos(th5)
     - 1158.0*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th4) - 6634.1*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th5) - 37484.832*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2) -
    17222*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)**2*cos(th2)*cos(th3) - 9504*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3)*cos(th4) -
    18073.44*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th2)*cos(th3) - 24645.6*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th2) -
    12884.0*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)**2*cos(th2)*cos(th3)*cos(th4)**2 + 5502*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)**2*cos(th2)*cos(th3) -
    13622.8*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4)**2*cos(th5) + 13268.2*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4) -
    13622.8*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th5) - 25542.75*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3) +
    18093.0*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th4) - 5502*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)**2*cos(th5)**2 -
    590*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)**2 + 7492.54*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)*cos(th5) -
    679.36*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4) + 12884.0*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th5)**2 + 45232.5*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th5) +
     109864.1*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3) + 10217.1*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th4)*cos(th5) - 926.4*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th4) -
    47668.8*sin(th1)**2*sin(th2)*sin(th3)*cos(th2) - 4752*sin(th1)**2*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th5) + 30807.0*sin(th1)**2*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2 +
    6811.4*sin(th1)**2*sin(th2)*sin(th5)**2*cos(th2)*cos(th3)**2*cos(th4) - 7382.0*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) -
```

```
22616.25*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4) - 3746.27*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2 - 10217.1*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3) -
6811.4*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5)**2 - 12771.375*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) + 1158.0*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)
+ 6634.1*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th5) + 37484.832*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2 + 18093.0*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)*cos(th5) +
102231.36*sin(th1)**2*sin(th2)*cos(th2)*cos(th3) - 16575.6*sin(th1)**2*sin(th2)*cos(th2) + 2751*sin(th1)**2*sin(th3)**2*sin(th5)**2*cos(th2)**2 -
6811.4*sin(th1)**2*sin(th3)**2*sin(th5)*cos(th2)**2*cos(th5) - 12771.375*sin(th1)**2*sin(th3)**2*sin(th5)*cos(th2)**2 + 6442.0*sin(th1)**2*sin(th3)**2*cos(th2)**2*cos(th5)**2 +
22616.25*sin(th1)**2*sin(th3)**2*cos(th2)**2*cos(th5) + 65802.4*sin(th1)**2*sin(th3)**2*cos(th2)**2 - 4752*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th5) +
30807.0*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)**2*cos(th3) + 6811.4*sin(th1)**2*sin(th3)*sin(th5)**2*cos(th2)**2*cos(th3)*cos(th4) -
7382.0*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) - 22616.25*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4) -
3746.27*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th3) - 10217.1*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2 - 6811.4*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5)**2 -
12771.375*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) + 1158.0*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th4) + 6634.1*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th5) +
37484.832*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3) + 18093.0*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th5) + 102231.36*sin(th1)**2*sin(th3)*cos(th2)**2 +
8611*sin(th1)**2*sin(th4)**2*cos(th2)**2*cos(th3)**2 + 4752*sin(th1)**2*sin(th4)*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th4) + 9036.72*sin(th1)**2*sin(th4)*cos(th2)**2*cos(th3)**2 +
24645.6*sin(th1)**2*sin(th4)*cos(th2)**2*cos(th3) + 6442.0*sin(th1)**2*sin(th5)**2*cos(th2)**2*cos(th3)**2*cos(th4)**2 + 6811.4*sin(th1)**2*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th4)**2*cos(th5) -
6634.1*sin(th1)**2*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th4) - 18093.0*sin(th1)**2*sin(th5)*cos(th2)**2*cos(th3)*cos(th4) + 2751*sin(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)**2*cos(th5)**2 +
295*sin(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)**2 - 3746.27*sin(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5) + 339.68*sin(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4) +
10870.35*sin(th1)**2*cos(th2)**2*cos(th3)**2 - 10217.1*sin(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) + 926.4*sin(th1)**2*cos(th2)**2*cos(th3)*cos(th4) + 47668.8*sin(th1)**2*cos(th2)**2*cos(th3) +
78713.88*sin(th1)**2*cos(th2)**2 + 8611*sin(th2)**2*sin(th3)**2*sin(th4)**2*cos(th1)**2 + 4752*sin(th2)**2*sin(th3)**2*sin(th4)*sin(th5)*cos(th1)**2*cos(th4) +
9036.72*sin(th2)**2*sin(th3)**2*sin(th4)*cos(th1)**2 + 6442.0*sin(th2)**2*sin(th3)**2*sin(th5)**2*cos(th1)**2*cos(th4)**2 + 2751*sin(th2)**2*sin(th3)**2*sin(th5)**2 +
6811.4*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th4)**2*cos(th5) - 6634.1*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th4) - 6811.4*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th5) -
12771.375*sin(th2)**2*sin(th3)**2*sin(th5) + 2751*sin(th2)**2*sin(th3)**2*cos(th1)**2*cos(th4)**2*cos(th5)**2 + 295*sin(th2)**2*sin(th3)**2*cos(th1)**2*cos(th4)**2 -
3746.27*sin(th2)**2*sin(th3)**2*cos(th1)**2*cos(th4)*cos(th5) + 339.68*sin(th2)**2*sin(th3)**2*cos(th1)**2*cos(th4) + 10870.35*sin(th2)**2*sin(th3)**2*cos(th1)**2 +
6442.0*sin(th2)**2*sin(th3)**2*cos(th5)**2 + 22616.25*sin(th2)**2*sin(th3)**2*cos(th5) + 65802.4*sin(th2)**2*sin(th3)**2 + 4752*sin(th2)**2*sin(th3)*sin(th4)*cos(th1)**2*cos(th3)*cos(th5) -
30807.0*sin(th2)**2*sin(th3)*sin(th4)*cos(th1)**2*cos(th3) - 4752*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th5) + 30807.0*sin(th2)**2*sin(th3)*sin(th4)*cos(th3) -
6811.4*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th3)*cos(th4) + 6811.4*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th3)*cos(th4) +
7382.0*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5) + 22616.25*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3)*cos(th4) +
3746.27*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3) - 7382.0*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4)*cos(th5) - 22616.25*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4) -
3746.27*sin(th2)**2*sin(th3)*sin(th5)*cos(th3) - 10217.1*sin(th2)**2*sin(th3)*sin(th5) + 6811.4*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5)**2 +
12771.375*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5) - 1158.0*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th4) - 6634.1*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th5) -
37484.832*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3) - 6811.4*sin(th2)**2*sin(th3)*cos(th3)*cos(th4)*cos(th5)**2 - 12771.375*sin(th2)**2*sin(th3)*cos(th3)*cos(th4)*cos(th5) +
1158.0*sin(th2)**2*sin(th3)*cos(th3)*cos(th4) + 6634.1*sin(th2)**2*sin(th3)*cos(th3)*cos(th5) + 37484.832*sin(th2)**2*sin(th3)*cos(th3) + 18093.0*sin(th2)**2*sin(th3)*cos(th5) +
102231.36*sin(th2)**2*sin(th3) + 8611*sin(th2)**2*sin(th4)**2*cos(th3)**2 + 4752*sin(th2)**2*sin(th4)*sin(th5)*cos(th3)**2*cos(th4) + 9036.72*sin(th2)**2*sin(th4)*cos(th3)**2 +
24645.6*sin(th2)**2*sin(th4)*cos(th3) + 2751*sin(th2)**2*sin(th5)**2*cos(th1)**2*cos(th3)**2 + 6442.0*sin(th2)**2*sin(th5)**2*cos(th3)**2*cos(th4)**2 -
6811.4*sin(th2)**2*sin(th5)*cos(th1)**2*cos(th3)**2*cos(th5) - 12771.375*sin(th2)**2*sin(th5)*cos(th1)**2*cos(th3)**2 + 6811.4*sin(th2)**2*sin(th5)*cos(th3)**2*cos(th4)**2*cos(th5) -
6634.1*sin(th2)**2*sin(th5)*cos(th3)**2*cos(th4) - 18093.0*sin(th2)**2*sin(th5)*cos(th3)*cos(th4) + 6442.0*sin(th2)**2*cos(th1)**2*cos(th3)**2*cos(th5)**2 +
22616.25*sin(th2)**2*cos(th1)**2*cos(th3)**2*cos(th5) + 65802.4*sin(th2)**2*cos(th1)**2*cos(th3)**2 + 5632*sin(th2)**2*cos(th1)**2 + 2751*sin(th2)**2*cos(th3)**2*cos(th4)**2*cos(th5)**2 +
295*sin(th2)**2*cos(th3)**2*cos(th4)**2 - 3746.27*sin(th2)**2*cos(th3)**2*cos(th4)*cos(th5) + 339.68*sin(th2)**2*cos(th3)**2*cos(th4) + 10870.35*sin(th2)**2*cos(th3)**2 -
10217.1*sin(th2)**2*cos(th3)*cos(th4)*cos(th5) + 926.4*sin(th2)**2*cos(th3)*cos(th4) + 47668.8*sin(th2)**2*cos(th3) + 78713.88*sin(th2)**2 +
4752*sin(th2)*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2)*cos(th5) - 30807.0*sin(th2)*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2) - 4752*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th5) +
30807.0*sin(th2)*sin(th3)**2*sin(th4)*cos(th2) - 6811.4*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th4) + 6811.4*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th2)*cos(th4) +
7382.0*sin(th2)*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th4)*cos(th5) + 22616.25*sin(th2)*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th4) +
3746.27*sin(th2)*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2) - 7382.0*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4)*cos(th5) - 22616.25*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4) -
3746.27*sin(th2)*sin(th3)**2*sin(th5)*cos(th2) + 6811.4*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th4)*cos(th5)**2 + 12771.375*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th4)*cos(th5) -
1158.0*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th4) - 6634.1*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th5) - 37484.832*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2) -
6811.4*sin(th2)*sin(th3)**2*cos(th2)*cos(th4)*cos(th5)**2 - 12771.375*sin(th2)*sin(th3)**2*cos(th2)*cos(th4)*cos(th5) + 1158.0*sin(th2)*sin(th3)**2*cos(th2)*cos(th4) +
6634.1*sin(th2)*sin(th3)**2*cos(th2)*cos(th5) + 37484.832*sin(th2)*sin(th3)**2*cos(th2) - 17222*sin(th2)*sin(th3)*sin(th4)**2*cos(th1)**2*cos(th2)*cos(th3) +
17222*sin(th2)*sin(th3)*sin(th4)**2*cos(th2)*cos(th3) - 9504*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) +
9504*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3)*cos(th4) - 18073.44*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3) - 24645.6*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th2) +
18073.44*sin(th2)*sin(th3)*sin(th4)*cos(th2)*cos(th3) + 24645.6*sin(th2)*sin(th3)*sin(th4)*cos(th2) - 12884.0*sin(th2)*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)**2 +
5502*sin(th2)*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3) + 12884.0*sin(th2)*sin(th3)*sin(th5)**2*cos(th2)*cos(th3)*cos(th4)**2 - 5502*sin(th2)*sin(th3)*sin(th5)**2*cos(th2)*cos(th3) -
13622.8*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)**2*cos(th5) + 13268.2*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) -
13622.8*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) - 25542.75*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3) +
18093.0*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th4) + 13622.8*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4)**2*cos(th5) -
13268.2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4) + 13622.8*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th5) + 25542.75*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3) -
18093.0*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th4) - 5502*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)**2*cos(th5)**2 - 590*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)**2 +
7492.54*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5) - 679.36*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) +
12884.0*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5)**2 + 45232.5*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) + 109864.1*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3) +
10217.1*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th4)*cos(th5) - 926.4*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th4) - 47668.8*sin(th2)*sin(th3)*cos(th1)**2*cos(th2) +
5502*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)**2*cos(th5)**2 + 590*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)**2 - 7492.54*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)*cos(th5) +
679.36*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4) - 12884.0*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th5)**2 - 45232.5*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th5) -
109864.1*sin(th2)*sin(th3)*cos(th2)*cos(th3) - 10217.1*sin(th2)*sin(th3)*cos(th2)*cos(th4)*cos(th5) + 926.4*sin(th2)*sin(th3)*cos(th2)*cos(th4) + 47668.8*sin(th2)*sin(th3)*cos(th2) -
4752*sin(th2)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th5) + 30807.0*sin(th2)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)**2 + 4752*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th5) -
30807.0*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2 + 6811.4*sin(th2)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4) - 6811.4*sin(th2)*sin(th5)**2*cos(th2)*cos(th3)**2*cos(th4) -
7382.0*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) - 22616.25*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4) -
3746.27*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2 - 10217.1*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3) + 7382.0*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) +
22616.25*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4) + 3746.27*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2 + 10217.1*sin(th2)*sin(th5)*cos(th2)*cos(th3) -
6811.4*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)*cos(th5)**2 - 12771.375*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) + 1158.0*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)
+ 6634.1*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th5) + 37484.832*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2 + 18093.0*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) +
102231.36*sin(th2)*cos(th1)**2*cos(th2)*cos(th3) - 16575.6*sin(th2)*cos(th1)**2*cos(th2) + 6811.4*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5)**2 +
```

```
12771.375*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) - 1158.0*sin(th2)*cos(th2)*cos(th3)**2*cos(th4) - 6634.1*sin(th2)*cos(th2)*cos(th3)**2*cos(th5) - 37484.832*sin(th2)*cos(th2)*cos(th3)**2 -
18093.0*sin(th2)*cos(th2)*cos(th3)*cos(th5) - 102231.36*sin(th2)*cos(th2)*cos(th3) + 16575.6*sin(th2)*cos(th2) + 8611*sin(th3)**2*sin(th4)**2*cos(th2)**2 +
4752*sin(th3)**2*sin(th4)*sin(th5)*cos(th2)**2*cos(th4) + 9036.72*sin(th3)**2*sin(th4)*cos(th2)**2 + 2751*sin(th3)**2*sin(th5)**2*cos(th1)**2*cos(th2)**2 +
6442.0*sin(th3)**2*sin(th5)**2*cos(th2)**2*cos(th4)**2 - 6811.4*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th5) - 12771.375*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)**2 +
6811.4*sin(th3)**2*sin(th5)*cos(th2)**2*cos(th4)**2*cos(th5) - 6634.1*sin(th3)**2*sin(th5)*cos(th2)**2*cos(th4) + 6442.0*sin(th3)**2*cos(th1)**2*cos(th2)**2*cos(th5)**2 +
22616.25*sin(th3)**2*cos(th1)**2*cos(th2)**2*cos(th5) + 65802.4*sin(th3)**2*cos(th1)**2*cos(th2)**2 + 2751*sin(th3)**2*cos(th2)**2*cos(th4)**2*cos(th5)**2 + 295*sin(th3)**2*cos(th2)**2*cos(th4)**2 -
3746.27*sin(th3)**2*cos(th2)**2*cos(th4)*cos(th5) + 339.68*sin(th3)**2*cos(th2)**2*cos(th4) + 10870.35*sin(th3)**2*cos(th2)**2 - 4752*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th5) +
30807.0*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3) + 4752*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th5) - 30807.0*sin(th3)*sin(th4)*cos(th2)**2*cos(th3) +
6811.4*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4) - 6811.4*sin(th3)*sin(th5)**2*cos(th2)**2*cos(th3)*cos(th4) -
7382.0*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) - 22616.25*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4) -
3746.27*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3) - 10217.1*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2 + 7382.0*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) +
22616.25*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4) + 3746.27*sin(th3)*sin(th5)*cos(th2)**2*cos(th3) - 6811.4*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5)**2 -
12771.375*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) + 1158.0*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4) + 6634.1*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th5) +
37484.832*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3) + 18093.0*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th5) + 102231.36*sin(th3)*cos(th1)**2*cos(th2)**2 +
6811.4*sin(th3)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5)**2 + 12771.375*sin(th3)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) - 1158.0*sin(th3)*cos(th2)**2*cos(th3)*cos(th4) -
6634.1*sin(th3)*cos(th2)**2*cos(th3)*cos(th5) - 37484.832*sin(th3)*cos(th2)**2*cos(th3) + 8611*sin(th4)**2*cos(th1)**2*cos(th2)**2*cos(th3)**2 +
4752*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4) + 9036.72*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3)**2 + 24645.6*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3) +
6442.0*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)**2 + 2751*sin(th5)**2*cos(th2)**2*cos(th3)**2 + 6811.4*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)**2*cos(th5) -
6634.1*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4) - 18093.0*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4) - 6811.4*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th5) -
12771.375*sin(th5)*cos(th2)**2*cos(th3)**2 + 2751*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)**2*cos(th5)**2 + 295*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)**2 -
3746.27*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5) + 339.68*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4) + 10870.35*cos(th1)**2*cos(th2)**2*cos(th3)**2 -
10217.1*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) + 926.4*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4) + 47668.8*cos(th1)**2*cos(th2)**2*cos(th3) + 78713.88*cos(th1)**2*cos(th2)**2 +
6442.0*cos(th2)**2*cos(th3)**2*cos(th5)**2 + 22616.25*cos(th2)**2*cos(th3)**2*cos(th5) + 65802.4*cos(th2)**2*cos(th3)**2 + 5632*cos(th2)**2) + 4.0e-11*ddth2 +
1.0e-5*ddth3*(8611*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th4)**2 + 4752*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th4)*sin(th5)*cos(th4) + 9036.72*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th4) +
6442.0*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th5)**2*cos(th4)**2 + 6811.4*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th4)**2*cos(th5) -
6634.1*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th4) + 2751*sin(th1)**2*sin(th2)**2*sin(th3)**2*cos(th4)**2*cos(th5)**2 + 295*sin(th1)**2*sin(th2)**2*sin(th3)**2*cos(th4)**2 -
3746.27*sin(th1)**2*sin(th2)**2*sin(th3)**2*cos(th4)*cos(th5) + 339.68*sin(th1)**2*sin(th2)**2*sin(th3)**2*cos(th4) + 10870.35*sin(th1)**2*sin(th2)**2*sin(th3)**2 +
4752*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th5) - 30807.0*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*cos(th3) - 6811.4*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th3)*cos(th4) +
7382.0*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4)*cos(th5) + 22616.25*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4) +
3746.27*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3) + 6811.4*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th4)*cos(th5)**2 + 12771.375*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th4)*cos(th5)
 - 1158.0*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th4) - 6634.1*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th5) - 37484.832*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3) +
2751*sin(th1)**2*sin(th2)**2*sin(th5)**2*cos(th3)**2 - 6811.4*sin(th1)**2*sin(th2)**2*sin(th5)*cos(th3)**2*cos(th5) - 12771.375*sin(th1)**2*sin(th2)**2*sin(th5)*cos(th3)**2 +
6442.0*sin(th1)**2*sin(th2)**2*cos(th3)**2*cos(th5)**2 + 22616.25*sin(th1)**2*sin(th2)**2*cos(th3)**2*cos(th5) + 65802.4*sin(th1)**2*sin(th2)**2*cos(th3)**2 +
4752*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th5) - 30807.0*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*cos(th2) - 6811.4*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th2)*cos(th4) +
7382.0*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4)*cos(th5) + 22616.25*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4) +
3746.27*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2) + 6811.4*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th4)*cos(th5)**2 + 12771.375*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th4)*cos(th5)
 - 1158.0*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th4) - 6634.1*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th5) - 37484.832*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2) -
17222*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)**2*cos(th2)*cos(th3) - 9504*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3)*cos(th4) -
18073.44*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th2)*cos(th3) - 12322.8*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th2) -
12884.0*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)**2*cos(th2)*cos(th3)*cos(th4)**2 + 5502*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)**2*cos(th2)*cos(th3) -
13622.8*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4)**2*cos(th5) + 13268.2*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4) -
13622.8*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th5) - 25542.75*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3) +
9046.5*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th4) - 5502*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)**2*cos(th5)**2 -
590*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)**2 + 7492.54*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)*cos(th5) -
679.36*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4) + 12884.0*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th5)**2 + 45232.5*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th5) +
 109864.1*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3) + 5108.7*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th4)*cos(th5) - 463.2*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th4) -
23834.4*sin(th1)**2*sin(th2)*sin(th3)*cos(th2) - 4752*sin(th1)**2*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th5) + 30807.0*sin(th1)**2*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2 +
6811.4*sin(th1)**2*sin(th2)*sin(th5)**2*cos(th2)*cos(th3)**2*cos(th4) - 7382.0*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) -
22616.25*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4) - 3746.27*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2 - 5108.7*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3) -
6811.4*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5)**2 - 12771.375*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) + 1158.0*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)
+ 6634.1*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th5) + 37484.832*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2 + 9046.5*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)*cos(th5) +
51115.68*sin(th1)**2*sin(th2)*cos(th2)*cos(th3) + 2751*sin(th1)**2*sin(th3)**2*sin(th5)**2*cos(th2)**2 - 6811.4*sin(th1)**2*sin(th3)**2*sin(th5)*cos(th2)**2*cos(th5) -
12771.375*sin(th1)**2*sin(th3)**2*sin(th5)*cos(th2)**2 + 6442.0*sin(th1)**2*sin(th3)**2*cos(th2)**2*cos(th5)**2 + 22616.25*sin(th1)**2*sin(th3)**2*cos(th2)**2*cos(th5) +
65802.4*sin(th1)**2*sin(th3)**2*cos(th2)**2 - 4752*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th5) + 30807.0*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)**2*cos(th3) +
6811.4*sin(th1)**2*sin(th3)*sin(th5)**2*cos(th2)**2*cos(th3)*cos(th4) - 7382.0*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) -
22616.25*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4) - 3746.27*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th3) - 5108.7*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2 -
6811.4*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5)**2 - 12771.375*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th4)*cos(th5) + 1158.0*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th4)
+ 6634.1*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th5) + 37484.832*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3) + 9046.5*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th5) +
51115.68*sin(th1)**2*sin(th3)*cos(th2)**2 + 8611*sin(th1)**2*sin(th4)**2*cos(th2)**2*cos(th3)**2 + 4752*sin(th1)**2*sin(th4)*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th4) +
9036.72*sin(th1)**2*sin(th4)*cos(th2)**2*cos(th3)**2 + 12322.8*sin(th1)**2*sin(th4)*cos(th2)**2*cos(th3) + 6442.0*sin(th1)**2*sin(th5)**2*cos(th2)**2*cos(th3)**2*cos(th4)**2 +
6811.4*sin(th1)**2*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th4)**2*cos(th5) - 6634.1*sin(th1)**2*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th4) - 9046.5*sin(th1)**2*sin(th5)*cos(th2)**2*cos(th3)*cos(th4) +
2751*sin(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)**2*cos(th5)**2 + 295*sin(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)**2 - 3746.27*sin(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5) +
339.68*sin(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4) + 10870.35*sin(th1)**2*cos(th2)**2*cos(th3)**2 - 5108.7*sin(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) +
463.2*sin(th1)**2*cos(th2)**2*cos(th3)*cos(th4) + 23834.4*sin(th1)**2*cos(th2)**2*cos(th3) + 8611*sin(th2)**2*sin(th3)**2*sin(th4)**2*cos(th1)**2 +
4752*sin(th2)**2*sin(th3)**2*sin(th4)*sin(th5)*cos(th1)**2*cos(th4) + 9036.72*sin(th2)**2*sin(th3)**2*sin(th4)*cos(th1)**2 + 6442.0*sin(th2)**2*sin(th3)**2*sin(th5)**2*cos(th1)**2*cos(th4)**2 +
2751*sin(th2)**2*sin(th3)**2*sin(th5)**2 + 6811.4*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th4)**2*cos(th5) - 6634.1*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th4) -
6811.4*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th5) - 12771.375*sin(th2)**2*sin(th3)**2*sin(th5) + 2751*sin(th2)**2*sin(th3)**2*cos(th1)**2*cos(th4)**2*cos(th5)**2 +
295*sin(th2)**2*sin(th3)**2*cos(th1)**2*cos(th4)**2 - 3746.27*sin(th2)**2*sin(th3)**2*cos(th1)**2*cos(th4)*cos(th5) + 339.68*sin(th2)**2*sin(th3)**2*cos(th1)**2*cos(th4) +
```

```
10870.35*sin(th2)**2*sin(th3)**2*cos(th1)**2 + 6442.0*sin(th2)**2*sin(th3)**2*cos(th5)**2 + 22616.25*sin(th2)**2*sin(th3)**2*cos(th5) + 65802.4*sin(th2)**2*sin(th3)**2 +
4752*sin(th2)**2*sin(th3)*sin(th4)*cos(th1)**2*cos(th3)*cos(th5) - 30807.0*sin(th2)**2*sin(th3)*sin(th4)*cos(th1)**2*cos(th3) - 4752*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th5) +
30807.0*sin(th2)**2*sin(th3)*sin(th4)*cos(th3) - 6811.4*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th3)*cos(th4) + 6811.4*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th3)*cos(th4) +
7382.0*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5) + 22616.25*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3)*cos(th4) +
3746.27*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3) - 7382.0*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4)*cos(th5) - 22616.25*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4) -
3746.27*sin(th2)**2*sin(th3)*sin(th5)*cos(th3) - 5108.7*sin(th2)**2*sin(th3)*sin(th5) + 6811.4*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5)**2 +
12771.375*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5) - 1158.0*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th4) - 6634.1*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th5) -
37484.832*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3) - 6811.4*sin(th2)**2*sin(th3)*cos(th3)*cos(th4)*cos(th5)**2 - 12771.375*sin(th2)**2*sin(th3)*cos(th3)*cos(th4)*cos(th5) +
1158.0*sin(th2)**2*sin(th3)*cos(th3)*cos(th4) + 6634.1*sin(th2)**2*sin(th3)*cos(th3)*cos(th5) + 37484.832*sin(th2)**2*sin(th3)*cos(th3) + 9046.5*sin(th2)**2*sin(th3)*cos(th5) +
51115.68*sin(th2)**2*sin(th3) + 8611*sin(th2)**2*sin(th4)**2*cos(th3)**2 + 4752*sin(th2)**2*sin(th4)*sin(th5)*cos(th3)**2*cos(th4) + 9036.72*sin(th2)**2*sin(th4)*cos(th3)**2 +
12322.8*sin(th2)**2*sin(th4)*cos(th3) + 2751*sin(th2)**2*sin(th5)**2*cos(th1)**2*cos(th3)**2 + 6442.0*sin(th2)**2*sin(th5)**2*cos(th3)**2*cos(th4)**2 -
6811.4*sin(th2)**2*sin(th5)*cos(th1)**2*cos(th3)**2*cos(th5) - 12771.375*sin(th2)**2*sin(th5)*cos(th1)**2*cos(th3)**2 + 6811.4*sin(th2)**2*sin(th5)*cos(th3)**2*cos(th4)**2*cos(th5) -
6634.1*sin(th2)**2*sin(th5)*cos(th3)**2*cos(th4) - 9046.5*sin(th2)**2*sin(th5)*cos(th3)*cos(th4) + 6442.0*sin(th2)**2*cos(th1)**2*cos(th3)**2*cos(th5)**2 +
22616.25*sin(th2)**2*cos(th1)**2*cos(th3)**2*cos(th5) + 65802.4*sin(th2)**2*cos(th1)**2*cos(th3)**2 + 2751*sin(th2)**2*cos(th3)**2*cos(th4)**2*cos(th5)**2 + 295*sin(th2)**2*cos(th3)**2*cos(th4)**2 -
3746.27*sin(th2)**2*cos(th3)**2*cos(th4)*cos(th5) + 339.68*sin(th2)**2*cos(th3)**2*cos(th4) + 10870.35*sin(th2)**2*cos(th3)**2 - 5108.7*sin(th2)**2*cos(th3)*cos(th4)*cos(th5) +
463.2*sin(th2)**2*cos(th3)*cos(th4) + 23834.4*sin(th2)**2*cos(th3) + 4752*sin(th2)*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2)*cos(th5) - 30807.0*sin(th2)*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2) -
4752*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th5) + 30807.0*sin(th2)*sin(th3)**2*sin(th4)*cos(th2) - 6811.4*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th4) +
6811.4*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th2)*cos(th4) + 7382.0*sin(th2)*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th4)*cos(th5) +
22616.25*sin(th2)*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th4) + 3746.27*sin(th2)*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2) - 7382.0*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4)*cos(th5) -
22616.25*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4) - 3746.27*sin(th2)*sin(th3)**2*sin(th5)*cos(th2) + 6811.4*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th4)*cos(th5)**2 +
12771.375*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th4)*cos(th5) - 1158.0*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th4) - 6634.1*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th5) -
37484.832*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2) - 6811.4*sin(th2)*sin(th3)**2*cos(th2)*cos(th4)*cos(th5)**2 - 12771.375*sin(th2)*sin(th3)**2*cos(th2)*cos(th4)*cos(th5) +
1158.0*sin(th2)*sin(th3)**2*cos(th2)*cos(th4) + 6634.1*sin(th2)*sin(th3)**2*cos(th2)*cos(th5) + 37484.832*sin(th2)*sin(th3)**2*cos(th2) -
17222*sin(th2)*sin(th3)*sin(th4)**2*cos(th1)**2*cos(th2)*cos(th3) + 17222*sin(th2)*sin(th3)*sin(th4)**2*cos(th2)*cos(th3) -
9504*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) + 9504*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3)*cos(th4) -
18073.44*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3) - 12322.8*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th2) + 18073.44*sin(th2)*sin(th3)*sin(th4)*cos(th2)*cos(th3) +
12322.8*sin(th2)*sin(th3)*sin(th4)*cos(th2) - 12884.0*sin(th2)*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)**2 + 5502*sin(th2)*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3) +
12884.0*sin(th2)*sin(th3)*sin(th5)**2*cos(th2)*cos(th3)*cos(th4)**2 - 5502*sin(th2)*sin(th3)*sin(th5)**2*cos(th2)*cos(th3) -
13622.8*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)**2*cos(th5) + 13268.2*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) -
13622.8*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) - 25542.75*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3) +
9046.5*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th4) + 13622.8*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4)**2*cos(th5) -
13268.2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4) + 13622.8*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th5) + 25542.75*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3) -
9046.5*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th4) - 5502*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)**2*cos(th5)**2 - 590*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)**2 +
7492.54*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5) - 679.36*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) +
12884.0*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5)**2 + 45232.5*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) + 109864.1*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3) +
5108.7*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th4)*cos(th5) - 463.2*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th4) - 23834.4*sin(th2)*sin(th3)*cos(th1)**2*cos(th2) +
5502*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)**2*cos(th5)**2 + 590*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)**2 - 7492.54*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)*cos(th5) +
679.36*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4) - 12884.0*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th5)**2 - 45232.5*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th5) -
109864.1*sin(th2)*sin(th3)*cos(th2)*cos(th3) - 5108.7*sin(th2)*sin(th3)*cos(th2)*cos(th4)*cos(th5) + 463.2*sin(th2)*sin(th3)*cos(th2)*cos(th4) + 23834.4*sin(th2)*sin(th3)*cos(th2) -
4752*sin(th2)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th5) + 30807.0*sin(th2)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)**2 + 4752*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th5) -
30807.0*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2 + 6811.4*sin(th2)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4) - 6811.4*sin(th2)*sin(th5)**2*cos(th2)*cos(th3)**2*cos(th4) -
7382.0*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) - 22616.25*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4) -
3746.27*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2 - 5108.7*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3) + 7382.0*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) +
22616.25*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4) + 3746.27*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2 + 5108.7*sin(th2)*sin(th5)*cos(th2)*cos(th3) -
6811.4*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)*cos(th5)**2 - 12771.375*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) + 1158.0*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)
+ 6634.1*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th5) + 37484.832*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2 + 9046.5*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) +
51115.68*sin(th2)*cos(th1)**2*cos(th2)*cos(th3) + 6811.4*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5)**2 + 12771.375*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) -
1158.0*sin(th2)*cos(th2)*cos(th3)**2*cos(th4) - 6634.1*sin(th2)*cos(th2)*cos(th3)**2*cos(th5) - 37484.832*sin(th2)*cos(th2)*cos(th3)**2 - 9046.5*sin(th2)*cos(th2)*cos(th3)*cos(th5) -
51115.68*sin(th2)*cos(th2)*cos(th3) + 8611*sin(th3)**2*sin(th4)**2*cos(th2)**2 + 4752*sin(th3)**2*sin(th4)*sin(th5)*cos(th2)**2*cos(th4) + 9036.72*sin(th3)**2*sin(th4)*cos(th2)**2 +
2751*sin(th3)**2*sin(th5)**2*cos(th1)**2*cos(th2)**2 + 6442.0*sin(th3)**2*sin(th5)**2*cos(th2)**2*cos(th4)**2 - 6811.4*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th5) -
12771.375*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)**2 + 6811.4*sin(th3)**2*sin(th5)*cos(th2)**2*cos(th4)**2*cos(th5) - 6634.1*sin(th3)**2*sin(th5)*cos(th2)**2*cos(th4) +
6442.0*sin(th3)**2*cos(th1)**2*cos(th2)**2*cos(th5)**2 + 22616.25*sin(th3)**2*cos(th1)**2*cos(th2)**2*cos(th5) + 65802.4*sin(th3)**2*cos(th1)**2*cos(th2)**2 +
2751*sin(th3)**2*cos(th2)**2*cos(th4)**2*cos(th5)**2 + 295*sin(th3)**2*cos(th2)**2*cos(th4)**2 - 3746.27*sin(th3)**2*cos(th2)**2*cos(th4)*cos(th5) + 339.68*sin(th3)**2*cos(th2)**2*cos(th4) +
10870.35*sin(th3)**2*cos(th2)**2 - 4752*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th5) + 30807.0*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3) +
4752*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th5) - 30807.0*sin(th3)*sin(th4)*cos(th2)**2*cos(th3) + 6811.4*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4) -
6811.4*sin(th3)*sin(th5)**2*cos(th2)**2*cos(th3)*cos(th4) - 7382.0*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) -
22616.25*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4) - 3746.27*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3) - 5108.7*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2 +
7382.0*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) + 22616.25*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4) + 3746.27*sin(th3)*sin(th5)*cos(th2)**2*cos(th3) -
6811.4*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5)**2 - 12771.375*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) + 1158.0*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)
+ 6634.1*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th5) + 37484.832*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3) + 9046.5*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th5) +
51115.68*sin(th3)*cos(th1)**2*cos(th2)**2 + 6811.4*sin(th3)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5)**2 + 12771.375*sin(th3)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) -
1158.0*sin(th3)*cos(th2)**2*cos(th3)*cos(th4) - 6634.1*sin(th3)*cos(th2)**2*cos(th3)*cos(th5) - 37484.832*sin(th3)*cos(th2)**2*cos(th3) + 8611*sin(th4)**2*cos(th1)**2*cos(th2)**2*cos(th3)**2 +
4752*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4) + 9036.72*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3)**2 + 12322.8*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3) +
6442.0*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)**2 + 2751*sin(th5)**2*cos(th2)**2*cos(th3)**2 + 6811.4*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)**2*cos(th5) -
6634.1*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4) - 9046.5*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4) - 6811.4*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th5) -
12771.375*sin(th5)*cos(th2)**2*cos(th3)**2 + 2751*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)**2*cos(th5)**2 + 295*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)**2 -
3746.27*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5) + 339.68*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4) + 10870.35*cos(th1)**2*cos(th2)**2*cos(th3)**2 -
5108.7*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) + 463.2*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4) + 23834.4*cos(th1)**2*cos(th2)**2*cos(th3) + 6442.0*cos(th2)**2*cos(th3)**2*cos(th5)**2 +
```

```
22616.25*cos(th2)**2*cos(th3)**2*cos(th5) + 65802.4*cos(th2)**2*cos(th3)**2) + 1.0e-5*ddth4*(2376*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)**2*sin(th5)*cos(th3) +
6442.0*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)**2*cos(th3)*cos(th4) + 6811.4*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)*cos(th3)*cos(th4)*cos(th5) -
3317.05*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)*cos(th3) + 2751*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th4)*cos(th5)**2 -
8316*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th4) - 1873.19*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th5) + 169.84*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*cos(th3) -
2376*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4)**2 - 4518.36*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th4) - 3405.6*sin(th1)**2*sin(th2)**2*sin(th4)*sin(th5)**2*cos(th3)**2 +
3691.0*sin(th1)**2*sin(th2)**2*sin(th4)*sin(th5)*cos(th3)**2*cos(th5) + 11308.125*sin(th1)**2*sin(th2)**2*sin(th4)*sin(th5)*cos(th3)**2 + 3405.8*sin(th1)**2*sin(th2)**2*sin(th4)*cos(th3)**2*cos(th5)**2
+ 6385.875*sin(th1)**2*sin(th2)**2*sin(th4)*cos(th3)**2*cos(th5) - 579.0*sin(th1)**2*sin(th2)**2*sin(th4)*cos(th3)**2 - 2376*sin(th1)**2*sin(th2)**2*cos(th3)**2*cos(th4)*cos(th5) +
15403.5*sin(th1)**2*sin(th2)**2*cos(th3)**2*cos(th4) + 2376*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)**2*sin(th5)*cos(th2) +
6442.0*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)**2*cos(th2)*cos(th4) + 6811.4*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)*cos(th2)*cos(th4)*cos(th5) -
3317.05*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)*cos(th2) + 2751*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th4)*cos(th5)**2 -
8316*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th4) - 1873.19*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th5) + 169.84*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*cos(th2) -
2376*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4)**2 - 4518.36*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th4) -
6811.2*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5)**2*cos(th2)*cos(th3) + 7382.0*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3)*cos(th5) +
22616.25*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3) + 6811.6*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th2)*cos(th3)*cos(th5)**2 +
12771.75*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th2)*cos(th3)*cos(th5) - 1158.0*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th2)*cos(th3) -
4752*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)*cos(th5) + 30807.0*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4) -
2376*sin(th1)**2*sin(th2)*sin(th4)**2*sin(th5)*cos(th2)*cos(th3)**2 - 6442.0*sin(th1)**2*sin(th2)*sin(th4)*sin(th5)**2*cos(th2)*cos(th3)**2*cos(th4) -
6811.4*sin(th1)**2*sin(th2)*sin(th4)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4) + 3317.05*sin(th1)**2*sin(th2)*sin(th4)*sin(th5)*cos(th2)*cos(th3)**2 +
9046.5*sin(th1)**2*sin(th2)*sin(th4)*sin(th5)*cos(th2)*cos(th3) - 2751*sin(th1)**2*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5)**2 +
8316*sin(th1)**2*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th4) + 1873.19*sin(th1)**2*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th5) - 169.84*sin(th1)**2*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2 +
5108.7*sin(th1)**2*sin(th2)*sin(th4)*cos(th2)*cos(th3)*cos(th5) - 463.2*sin(th1)**2*sin(th2)*sin(th4)*cos(th2)*cos(th3) + 2376*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4)**2 +
4518.36*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th4) + 12322.8*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)*cos(th4) - 3405.6*sin(th1)**2*sin(th3)**2*sin(th4)*sin(th5)**2*cos(th2)**2 +
3691.0*sin(th1)**2*sin(th3)**2*sin(th4)*sin(th5)*cos(th2)**2*cos(th5) + 11308.125*sin(th1)**2*sin(th3)**2*sin(th4)*sin(th5)*cos(th2)**2 + 3405.8*sin(th1)**2*sin(th3)**2*sin(th4)*cos(th2)**2*cos(th5)**2
+ 6385.875*sin(th1)**2*sin(th3)**2*sin(th4)*cos(th2)**2*cos(th5) - 579.0*sin(th1)**2*sin(th3)**2*sin(th4)*cos(th2)**2 - 2376*sin(th1)**2*sin(th3)**2*cos(th2)**2*cos(th4)*cos(th5) +
15403.5*sin(th1)**2*sin(th3)**2*cos(th2)**2*cos(th4) - 2376*sin(th1)**2*sin(th3)*sin(th4)**2*sin(th5)*cos(th2)**2*cos(th3) -
6442.0*sin(th1)**2*sin(th3)*sin(th4)*sin(th5)**2*cos(th2)**2*cos(th3)*cos(th4) - 6811.4*sin(th1)**2*sin(th3)*sin(th4)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) +
3317.05*sin(th1)**2*sin(th3)*sin(th4)*sin(th5)*cos(th2)**2*cos(th3) + 9046.5*sin(th1)**2*sin(th3)*sin(th4)*sin(th5)*cos(th2)**2 -
2751*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5)**2 + 8316*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th4) +
1873.19*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th5) - 169.84*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)**2*cos(th3) + 5108.7*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)**2*cos(th5) -
463.2*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)**2 + 2376*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4)**2 + 4518.36*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th4) +
12322.8*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th4) - 3405.6*sin(th2)**2*sin(th3)**2*sin(th4)*sin(th5)**2 + 3691.0*sin(th2)**2*sin(th3)**2*sin(th4)*sin(th5)*cos(th5) +
11308.125*sin(th2)**2*sin(th3)**2*sin(th4)*sin(th5) + 3405.8*sin(th2)**2*sin(th3)**2*sin(th4)*cos(th5)**2 + 6385.875*sin(th2)**2*sin(th3)**2*sin(th4)*cos(th5) - 579.0*sin(th2)**2*sin(th3)**2*sin(th4) -
2376*sin(th2)**2*sin(th3)**2*cos(th4)*cos(th5) + 15403.5*sin(th2)**2*sin(th3)**2*cos(th4) + 2376*sin(th2)**2*sin(th3)*sin(th4)**2*sin(th5)*cos(th1)**2*cos(th3) -
2376*sin(th2)**2*sin(th3)*sin(th4)**2*sin(th5)*cos(th3) + 6442.0*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th3)*cos(th4) -
6442.0*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)**2*cos(th3)*cos(th4) + 6811.4*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5) -
3317.05*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th3) - 6811.4*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)*cos(th3)*cos(th4)*cos(th5) + 3317.05*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)*cos(th3) +
 9046.5*sin(th2)**2*sin(th3)*sin(th4)*sin(th5) + 2751*sin(th2)**2*sin(th3)*sin(th4)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5)**2 - 8316*sin(th2)**2*sin(th3)*sin(th4)*cos(th1)**2*cos(th3)*cos(th4) -
1873.19*sin(th2)**2*sin(th3)*sin(th4)*cos(th1)**2*cos(th3)*cos(th5) + 169.84*sin(th2)**2*sin(th3)*sin(th4)*cos(th1)**2*cos(th3) - 2751*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th4)*cos(th5)**2 +
8316*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th4) + 1873.19*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th5) - 169.84*sin(th2)**2*sin(th3)*sin(th4)*cos(th3) +
5108.7*sin(th2)**2*sin(th3)*sin(th4)*cos(th5) - 463.2*sin(th2)**2*sin(th3)*sin(th4) - 2376*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3)*cos(th4)**2 +
2376*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4)**2 - 4518.36*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th3)*cos(th4) + 4518.36*sin(th2)**2*sin(th3)*cos(th3)*cos(th4) +
12322.8*sin(th2)**2*sin(th3)*cos(th4) - 3405.6*sin(th2)**2*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th3)**2 + 3691.0*sin(th2)**2*sin(th4)*sin(th5)*cos(th1)**2*cos(th3)**2*cos(th5) +
11308.125*sin(th2)**2*sin(th4)*sin(th5)*cos(th1)**2*cos(th3)**2 + 3405.8*sin(th2)**2*sin(th4)*cos(th1)**2*cos(th3)**2*cos(th5)**2 + 6385.875*sin(th2)**2*sin(th4)*cos(th1)**2*cos(th3)**2*cos(th5) -
579.0*sin(th2)**2*sin(th4)*cos(th1)**2*cos(th3)**2 - 2376*sin(th2)**2*cos(th1)**2*cos(th3)**2*cos(th4)*cos(th5) + 15403.5*sin(th2)**2*cos(th1)**2*cos(th3)**2*cos(th4) +
2376*sin(th2)*sin(th3)**2*sin(th4)**2*sin(th5)*cos(th1)**2*cos(th2) - 2376*sin(th2)*sin(th3)**2*sin(th4)**2*sin(th5)*cos(th2) +
6442.0*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th4) - 6442.0*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)**2*cos(th2)*cos(th4) +
6811.4*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th4)*cos(th5) - 3317.05*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)*cos(th1)**2*cos(th2) -
6811.4*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)*cos(th2)*cos(th4)*cos(th5) + 3317.05*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)*cos(th2) +
2751*sin(th2)*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2)*cos(th4)*cos(th5)**2 - 8316*sin(th2)*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2)*cos(th4) -
1873.19*sin(th2)*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2)*cos(th5) + 169.84*sin(th2)*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2) - 2751*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th4)*cos(th5)**2 +
8316*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th4) + 1873.19*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th5) - 169.84*sin(th2)*sin(th3)**2*sin(th4)*cos(th2) -
2376*sin(th2)*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th4)**2 + 2376*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4)**2 - 4518.36*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th4) +
4518.36*sin(th2)*sin(th3)**2*cos(th2)*cos(th4) - 6811.2*sin(th2)*sin(th3)*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3) + 6811.2*sin(th2)*sin(th3)*sin(th4)*sin(th5)**2*cos(th2)*cos(th3) +
7382.0*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) + 22616.25*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3) -
7382.0*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3)*cos(th5) - 22616.25*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3) +
6811.6*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5)**2 + 12771.75*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) -
1158.0*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3) - 6811.6*sin(th2)*sin(th3)*sin(th4)*cos(th2)*cos(th3)*cos(th5)**2 - 12771.75*sin(th2)*sin(th3)*sin(th4)*cos(th2)*cos(th3)*cos(th5) +
1158.0*sin(th2)*sin(th3)*sin(th4)*cos(th2)*cos(th3) - 4752*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5) + 30807.0*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) +
4752*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)*cos(th5) - 30807.0*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4) - 2376*sin(th2)*sin(th4)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2 +
2376*sin(th2)*sin(th4)**2*sin(th5)*cos(th2)*cos(th3)**2 - 6442.0*sin(th2)*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4) +
6442.0*sin(th2)*sin(th4)*sin(th5)**2*cos(th2)*cos(th3)**2*cos(th4) - 6811.4*sin(th2)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) +
3317.05*sin(th2)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2 + 9046.5*sin(th2)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3) +
6811.4*sin(th2)*sin(th4)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) - 3317.05*sin(th2)*sin(th4)*sin(th5)*cos(th2)*cos(th3)**2 - 9046.5*sin(th2)*sin(th4)*sin(th5)*cos(th2)*cos(th3) -
2751*sin(th2)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)*cos(th5)**2 + 8316*sin(th2)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4) +
1873.19*sin(th2)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th5) - 169.84*sin(th2)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)**2 + 5108.7*sin(th2)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) -
463.2*sin(th2)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3) + 2751*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5)**2 - 8316*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th4) -
1873.19*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th5) + 169.84*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2 - 5108.7*sin(th2)*sin(th4)*cos(th2)*cos(th3)*cos(th5) + 463.2*sin(th2)*sin(th4)*cos(th2)*cos(th3)
+ 2376*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)**2 - 2376*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4)**2 + 4518.36*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4) +
```

```
12322.8*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) - 4518.36*sin(th2)*cos(th2)*cos(th3)**2*cos(th4) - 12322.8*sin(th2)*cos(th2)*cos(th3)*cos(th4) -
3405.6*sin(th3)**2*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th2)**2 + 3691.0*sin(th3)**2*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th5) + 11308.125*sin(th3)**2*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)**2
+ 3405.8*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th5)**2 + 6385.875*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th5) - 579.0*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2)**2 -
2376*sin(th3)**2*cos(th1)**2*cos(th2)**2*cos(th4)*cos(th5) + 15403.5*sin(th3)**2*cos(th1)**2*cos(th2)**2*cos(th4) - 2376*sin(th3)*sin(th4)**2*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3) +
2376*sin(th3)*sin(th4)**2*sin(th5)*cos(th2)**2*cos(th3) - 6442.0*sin(th3)*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4) +
6442.0*sin(th3)*sin(th4)*sin(th5)**2*cos(th2)**2*cos(th3)*cos(th4) - 6811.4*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) +
3317.05*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3) + 9046.5*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)**2 + 6811.4*sin(th3)*sin(th4)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5)
 - 3317.05*sin(th3)*sin(th4)*sin(th5)*cos(th2)**2*cos(th3) - 2751*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5)**2 +
8316*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4) + 1873.19*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th5) - 169.84*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3) +
5108.7*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th5) - 463.2*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)**2 + 2751*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5)**2 -
8316*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th4) - 1873.19*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th5) + 169.84*sin(th3)*sin(th4)*cos(th2)**2*cos(th3) +
2376*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)**2 - 2376*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4)**2 + 4518.36*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4) +
12322.8*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th4) - 4518.36*sin(th3)*cos(th2)**2*cos(th3)*cos(th4) - 3405.6*sin(th4)*sin(th5)**2*cos(th2)**2*cos(th3)**2 +
3691.0*sin(th4)*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th5) + 11308.125*sin(th4)*sin(th5)*cos(th2)**2*cos(th3)**2 + 3405.8*sin(th4)*cos(th2)**2*cos(th3)**2*cos(th5)**2 +
6385.875*sin(th4)*cos(th2)**2*cos(th3)**2*cos(th5) - 579.0*sin(th4)*cos(th2)**2*cos(th3)**2 - 2376*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5) + 15403.5*cos(th2)**2*cos(th3)**2*cos(th4)) +
1.0e-5*ddth5*(-2376*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th4)*sin(th5) - 6442.0*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th5)**2*cos(th4) -
6811.4*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th4)*cos(th5) + 3317.05*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th5) - 2751*sin(th1)**2*sin(th2)**2*sin(th3)**2*cos(th4)*cos(th5)**2 +
1873.19*sin(th1)**2*sin(th2)**2*sin(th3)**2*cos(th5) - 2376*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th4)*cos(th5) +
3405.8*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th3)*cos(th4)**2 + 3405.6*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th3) -
3691.0*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4)**2*cos(th5) - 1873.19*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4) -
3691.0*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th5) - 11308.125*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3) -
3405.6*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th4)**2*cos(th5)**2 + 3317.05*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th4)*cos(th5) -
3405.8*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th5)**2 - 6385.875*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th5) - 2751*sin(th1)**2*sin(th2)**2*sin(th5)**2*cos(th3)**2*cos(th4) +
6811.4*sin(th1)**2*sin(th2)**2*sin(th5)*cos(th3)**2*cos(th4)*cos(th5) + 6385.875*sin(th1)**2*sin(th2)**2*sin(th5)*cos(th3)**2*cos(th4) - 6442.0*sin(th1)**2*sin(th2)**2*cos(th3)**2*cos(th4)*cos(th5)**2 -
 11308.125*sin(th1)**2*sin(th2)**2*cos(th3)**2*cos(th4)*cos(th5) - 2376*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th4)*cos(th5) +
3405.8*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th2)*cos(th4)**2 + 3405.6*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th2) -
3691.0*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4)**2*cos(th5) - 1873.19*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4) -
3691.0*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th5) - 11308.125*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2) -
3405.6*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th4)**2*cos(th5)**2 + 3317.05*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th4)*cos(th5) -
3405.8*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th5)**2 - 6385.875*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th5) + 4752*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3) +
7382.0*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)**2*cos(th2)*cos(th3)*cos(th4) + 27245.6*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4)*cos(th5) +
12771.75*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4) - 6634.1*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3) - 9046.5*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2) -
7382.0*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)*cos(th5)**2 - 22616.25*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)*cos(th5) -
3746.38*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th5) - 5108.7*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th5) + 2376*sin(th1)**2*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) -
3405.8*sin(th1)**2*sin(th2)*sin(th5)**2*cos(th2)*cos(th3)**2*cos(th4)**2 - 3405.6*sin(th1)**2*sin(th2)*sin(th5)**2*cos(th2)*cos(th3)**2 +
3691.0*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4)**2*cos(th5) + 1873.19*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4) +
3691.0*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th5) + 11308.125*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2 + 5108.7*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)*cos(th4) +
3405.6*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)**2*cos(th5)**2 - 3317.05*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) +
3405.8*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th5)**2 + 6385.875*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th5) - 9046.5*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)*cos(th4)*cos(th5) -
2751*sin(th1)**2*sin(th3)**2*sin(th5)**2*cos(th2)**2*cos(th4) + 6811.4*sin(th1)**2*sin(th3)**2*sin(th5)*cos(th2)**2*cos(th4)*cos(th5) + 6385.875*sin(th1)**2*sin(th3)**2*sin(th5)*cos(th2)**2*cos(th4) -
6442.0*sin(th1)**2*sin(th3)**2*cos(th2)**2*cos(th4)*cos(th5)**2 - 11308.125*sin(th1)**2*sin(th3)**2*cos(th2)**2*cos(th4)*cos(th5) +
2376*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) - 3405.8*sin(th1)**2*sin(th3)*sin(th5)**2*cos(th2)**2*cos(th3)*cos(th4)**2 -
3405.6*sin(th1)**2*sin(th3)*sin(th5)**2*cos(th2)**2*cos(th3) + 3691.0*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4)**2*cos(th5) +
1873.19*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4) + 3691.0*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th5) + 11308.125*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th3) +
5108.7*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th4) + 3405.6*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th4)**2*cos(th5)**2 - 3317.05*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5)
 + 3405.8*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th5)**2 + 6385.875*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th5) - 9046.5*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th4)*cos(th5) -
2376*sin(th1)**2*sin(th4)*sin(th5)*cos(th2)**2*cos(th3)**2 - 6442.0*sin(th1)**2*sin(th5)**2*cos(th2)**2*cos(th3)**2*cos(th4) - 6811.4*sin(th1)**2*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5) +
3317.05*sin(th1)**2*sin(th5)*cos(th2)**2*cos(th3)**2 + 9046.5*sin(th1)**2*sin(th5)*cos(th2)**2*cos(th3) - 2751*sin(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5)**2 +
1873.19*sin(th1)**2*cos(th2)**2*cos(th3)**2*cos(th5) + 5108.7*sin(th1)**2*cos(th2)**2*cos(th3)*cos(th5) - 2376*sin(th2)**2*sin(th3)**2*sin(th4)*sin(th5)*cos(th1)**2 -
6442.0*sin(th2)**2*sin(th3)**2*sin(th5)**2*cos(th1)**2*cos(th4) - 2751*sin(th2)**2*sin(th3)**2*sin(th5)**2*cos(th4) - 6811.4*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th4)*cos(th5) +
3317.05*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th1)**2 + 6811.4*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th4)*cos(th5) + 6385.875*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th4) -
2751*sin(th2)**2*sin(th3)**2*cos(th1)**2*cos(th4)*cos(th5)**2 + 1873.19*sin(th2)**2*sin(th3)**2*cos(th1)**2*cos(th5) - 6442.0*sin(th2)**2*sin(th3)**2*cos(th4)*cos(th5)**2 -
11308.125*sin(th2)**2*sin(th3)**2*cos(th4)*cos(th5) - 2376*sin(th2)**2*sin(th3)*sin(th4)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5) + 2376*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th4)*cos(th5) +
3405.8*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th3)*cos(th4)**2 + 3405.6*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th3) - 3405.8*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th3)*cos(th4)**2 -
3405.6*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th3) - 3691.0*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3)*cos(th4)**2*cos(th5) - 1873.19*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3)*cos(th4) -
3691.0*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3)*cos(th5) - 11308.125*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3) + 3691.0*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4)**2*cos(th5) +
1873.19*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4) + 3691.0*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th5) + 11308.125*sin(th2)**2*sin(th3)*sin(th5)*cos(th3) +
5108.7*sin(th2)**2*sin(th3)*sin(th5)*cos(th4) - 3405.6*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th4)**2*cos(th5)**2 + 3317.05*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5) -
3405.8*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th5)**2 - 6385.875*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th5) + 3405.6*sin(th2)**2*sin(th3)*cos(th3)*cos(th4)**2*cos(th5)**2 -
3317.05*sin(th2)**2*sin(th3)*cos(th3)*cos(th4)*cos(th5) + 3405.8*sin(th2)**2*sin(th3)*cos(th3)*cos(th5)**2 + 6385.875*sin(th2)**2*sin(th3)*cos(th3)*cos(th5) -
9046.5*sin(th2)**2*sin(th3)*cos(th4)*cos(th5) - 2376*sin(th2)**2*sin(th4)*sin(th5)*cos(th3)**2 - 2751*sin(th2)**2*sin(th5)**2*cos(th1)**2*cos(th3)**2*cos(th4) -
6442.0*sin(th2)**2*sin(th5)**2*cos(th3)**2*cos(th4) + 6811.4*sin(th2)**2*sin(th5)*cos(th1)**2*cos(th3)**2*cos(th4)*cos(th5) + 6385.875*sin(th2)**2*sin(th5)*cos(th1)**2*cos(th3)**2*cos(th4) -
6811.4*sin(th2)**2*sin(th5)*cos(th3)**2*cos(th4)*cos(th5) + 3317.05*sin(th2)**2*sin(th5)*cos(th3)**2 + 9046.5*sin(th2)**2*sin(th5)*cos(th3) -
6442.0*sin(th2)**2*cos(th1)**2*cos(th3)**2*cos(th4)*cos(th5)**2 - 11308.125*sin(th2)**2*cos(th1)**2*cos(th3)**2*cos(th4)*cos(th5) - 2751*sin(th2)**2*cos(th3)**2*cos(th4)*cos(th5)**2 +
1873.19*sin(th2)**2*cos(th3)**2*cos(th5) + 5108.7*sin(th2)**2*cos(th3)*cos(th5) - 2376*sin(th2)*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2)*cos(th4)*cos(th5) +
2376*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th4)*cos(th5) + 3405.8*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th4)**2 + 3405.6*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th1)**2*cos(th2) -
3405.8*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th2)*cos(th4)**2 - 3405.6*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th2) - 3691.0*sin(th2)*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th4)**2*cos(th5) -
1873.19*sin(th2)*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th4) - 3691.0*sin(th2)*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th5) - 11308.125*sin(th2)*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2) +
```

```
3691.0*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4)**2*cos(th5) + 1873.19*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4) + 3691.0*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th5) +
11308.125*sin(th2)*sin(th3)**2*sin(th5)*cos(th2) - 3405.6*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th4)**2*cos(th5)**2 + 3317.05*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th4)*cos(th5) -
3405.8*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th5)**2 - 6385.875*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th5) + 3405.6*sin(th2)*sin(th3)**2*cos(th2)*cos(th4)**2*cos(th5)**2 -
3317.05*sin(th2)*sin(th3)**2*cos(th2)*cos(th4)*cos(th5) + 3405.8*sin(th2)*sin(th3)**2*cos(th2)*cos(th5)**2 + 6385.875*sin(th2)*sin(th3)**2*cos(th2)*cos(th5) +
4752*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3) - 4752*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3) +
7382.0*sin(th2)*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) - 7382.0*sin(th2)*sin(th3)*sin(th5)**2*cos(th2)*cos(th3)*cos(th4) +
27245.6*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5) + 12771.75*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) -
6634.1*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3) - 9046.5*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2) - 27245.6*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4)*cos(th5) -
12771.75*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4) + 6634.1*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3) + 9046.5*sin(th2)*sin(th3)*sin(th5)*cos(th2) -
7382.0*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5)**2 - 22616.25*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5) -
3746.38*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) - 5108.7*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th5) + 7382.0*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)*cos(th5)**2 +
22616.25*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)*cos(th5) + 3746.38*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th5) + 5108.7*sin(th2)*sin(th3)*cos(th2)*cos(th5) +
2376*sin(th2)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) - 2376*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) -
3405.8*sin(th2)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)**2 - 3405.6*sin(th2)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3)**2 + 3405.8*sin(th2)*sin(th5)**2*cos(th2)*cos(th3)**2*cos(th4)**2 +
3405.6*sin(th2)*sin(th5)**2*cos(th2)*cos(th3)**2 + 3691.0*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)**2*cos(th5) + 1873.19*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4) +
3691.0*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th5) + 11308.125*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2 + 5108.7*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) -
3691.0*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4)**2*cos(th5) - 1873.19*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4) - 3691.0*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th5) -
11308.125*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2 - 5108.7*sin(th2)*sin(th5)*cos(th2)*cos(th3)*cos(th4) + 3405.6*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)**2*cos(th5)**2 -
3317.05*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) + 3405.8*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th5)**2 + 6385.875*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th5) -
9046.5*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5) - 3405.6*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)**2*cos(th5)**2 + 3317.05*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) -
3405.8*sin(th2)*cos(th2)*cos(th3)**2*cos(th5)**2 - 6385.875*sin(th2)*cos(th2)*cos(th3)**2*cos(th5) + 9046.5*sin(th2)*cos(th2)*cos(th3)*cos(th4)*cos(th5) - 2376*sin(th3)**2*sin(th4)*sin(th5)*cos(th2)**2
- 2751*sin(th3)**2*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th4) - 6442.0*sin(th3)**2*sin(th5)**2*cos(th2)**2*cos(th4) + 6811.4*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th4)*cos(th5) +
6385.875*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th4) - 6811.4*sin(th3)**2*sin(th5)*cos(th2)**2*cos(th4)*cos(th5) + 3317.05*sin(th3)**2*sin(th5)*cos(th2)**2 -
6442.0*sin(th3)**2*cos(th1)**2*cos(th2)**2*cos(th4)*cos(th5)**2 - 11308.125*sin(th3)**2*cos(th1)**2*cos(th2)**2*cos(th4)*cos(th5) - 2751*sin(th3)**2*cos(th2)**2*cos(th4)*cos(th5)**2 +
1873.19*sin(th3)**2*cos(th2)**2*cos(th5) + 2376*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) - 2376*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) -
3405.8*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)**2 - 3405.6*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th3) + 3405.8*sin(th3)*sin(th5)**2*cos(th2)**2*cos(th3)*cos(th4)**2 +
3405.6*sin(th3)*sin(th5)**2*cos(th2)**2*cos(th3) + 3691.0*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)**2*cos(th5) + 1873.19*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4) +
3691.0*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th5) + 11308.125*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3) + 5108.7*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th4) -
3691.0*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4)**2*cos(th5) - 1873.19*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4) - 3691.0*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th5) -
11308.125*sin(th3)*sin(th5)*cos(th2)**2*cos(th3) + 3405.6*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)**2*cos(th5)**2 - 3317.05*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) +
3405.8*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th5)**2 + 6385.875*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th5) - 9046.5*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th4)*cos(th5) -
3405.6*sin(th3)*cos(th2)**2*cos(th3)*cos(th4)**2*cos(th5)**2 + 3317.05*sin(th3)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) - 3405.8*sin(th3)*cos(th2)**2*cos(th3)*cos(th5)**2 -
6385.875*sin(th3)*cos(th2)**2*cos(th3)*cos(th5) - 2376*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)**2 - 6442.0*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4) -
2751*sin(th5)**2*cos(th2)**2*cos(th3)**2*cos(th4) - 6811.4*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5) + 3317.05*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)**2 +
9046.5*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3) + 6811.4*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5) + 6385.875*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th4) -
2751*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5)**2 + 1873.19*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th5) + 5108.7*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th5) -
6442.0*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5)**2 - 11308.125*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5)) + 6.7772385*sin(th1)*sin(th2) - 19.83582*sin(th1)*cos(th2) +
15.90107805*sin(th2)*sin(th3)*sin(th4)*cos(th1) + 7.39551375*sin(th2)*sin(th3)*sin(th5)*cos(th1)*cos(th4) + 4.174155*sin(th2)*sin(th3)*cos(th1)*cos(th4)*cos(th5) +
0.378666*sin(th2)*sin(th3)*cos(th1)*cos(th4) + 11.498301*sin(th2)*sin(th3)*cos(th1) - 4.174155*sin(th2)*sin(th5)*cos(th1)*cos(th3) + 7.39551375*sin(th2)*cos(th1)*cos(th3)*cos(th5) -
14.5652994*sin(th2)*cos(th1)*cos(th3) - 4.174155*sin(th3)*sin(th5)*cos(th1)*cos(th2) + 7.39551375*sin(th3)*cos(th1)*cos(th2)*cos(th5) - 14.5652994*sin(th3)*cos(th1)*cos(th2) -
15.90107805*sin(th4)*cos(th1)*cos(th2)*cos(th3) - 7.39551375*sin(th5)*cos(th1)*cos(th2)*cos(th3)*cos(th4) - 4.174155*cos(th1)*cos(th2)*cos(th3)*cos(th4)*cos(th5) -
0.378666*cos(th1)*cos(th2)*cos(th3)*cos(th4) - 11.498301*cos(th1)*cos(th2)*cos(th3) - 22.42566*cos(th1)*cos(th2)
```

```
tau3 = 1.0e-5*ddth1*(2376*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)**2*sin(th5) + 6442.0*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5)**2*cos(th4) +
    6811.4*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th4)*cos(th5) - 3317.05*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5) + 2751*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th4)*cos(th5)**2 -
    8316*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th4) - 1873.08*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th5) + 169.84*sin(th1)**2*sin(th2)*sin(th3)*sin(th4) -
    2376*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th4)**2 - 4518.36*sin(th1)**2*sin(th2)*sin(th3)*cos(th4) + 222.2*sin(th1)**2*sin(th2)*sin(th3) - 3405.8*sin(th1)**2*sin(th2)*sin(th4)*sin(th5)**2*cos(th3)
     + 3691.0*sin(th1)**2*sin(th2)*sin(th4)*sin(th5)*cos(th3)*cos(th5) + 11308.125*sin(th1)**2*sin(th2)*sin(th4)*sin(th5)*cos(th3) + 3405.6*sin(th1)**2*sin(th2)*sin(th4)*cos(th3)*cos(th5)**2 +
    6385.5*sin(th1)**2*sin(th2)*sin(th4)*cos(th3)*cos(th5) - 579.0*sin(th1)**2*sin(th2)*sin(th4)*cos(th3) - 2376*sin(th1)**2*sin(th2)*cos(th3)*cos(th4)*cos(th5) +
    15403.5*sin(th1)**2*sin(th2)*cos(th3)*cos(th4) - 3405.8*sin(th1)**2*sin(th3)*sin(th4)*sin(th5)**2*cos(th2) + 3691.0*sin(th1)**2*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th5) +
    11308.125*sin(th1)**2*sin(th3)*sin(th4)*sin(th5)*cos(th2) + 3405.6*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)*cos(th5)**2 + 6385.5*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)*cos(th5) -
    579.0*sin(th1)**2*sin(th3)*sin(th4)*cos(th2) - 2376*sin(th1)**2*sin(th3)*cos(th2)*cos(th4)*cos(th5) + 15403.5*sin(th1)**2*sin(th3)*cos(th2)*cos(th4) -
    2376*sin(th1)**2*sin(th4)**2*sin(th5)*cos(th2)*cos(th3) - 6442.0*sin(th1)**2*sin(th4)*sin(th5)**2*cos(th2)*cos(th3)*cos(th4) - 6811.4*sin(th1)**2*sin(th4)*sin(th5)*cos(th2)*cos(th3)*cos(th4)*cos(th5) +
    3317.05*sin(th1)**2*sin(th4)*sin(th5)*cos(th2)*cos(th3) - 2751*sin(th1)**2*sin(th4)*cos(th2)*cos(th3)*cos(th4)*cos(th5)**2 + 8316*sin(th1)**2*sin(th4)*cos(th2)*cos(th3)*cos(th4) +
    1873.08*sin(th1)**2*sin(th4)*cos(th2)*cos(th3)*cos(th5) - 169.84*sin(th1)**2*sin(th4)*cos(th2)*cos(th3) + 2376*sin(th1)**2*sin(th5)*cos(th2)*cos(th3)*cos(th4)**2 +
    4518.36*sin(th1)**2*cos(th2)*cos(th3)*cos(th4) - 222.2*sin(th1)**2*cos(th2)*cos(th3) + 2376*sin(th2)*sin(th3)*sin(th4)**2*sin(th5)*cos(th1)**2 +
    6442.0*sin(th2)*sin(th3)*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th4) + 6811.4*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th4)*cos(th5) - 3317.05*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2 +
    2751*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th4)*cos(th5)**2 - 8316*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th4) - 1873.08*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th5) +
    169.84*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2 - 2376*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th4)**2 - 4518.36*sin(th2)*sin(th3)*cos(th1)**2*cos(th4) + 222.2*sin(th2)*sin(th3)*cos(th1)**2 -
    3405.8*sin(th2)*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th3) + 3691.0*sin(th2)*sin(th4)*sin(th5)*cos(th1)**2*cos(th3)*cos(th5) + 11308.125*sin(th2)*sin(th4)*sin(th5)*cos(th1)**2*cos(th3) +
    3405.6*sin(th2)*sin(th4)*cos(th1)**2*cos(th3)*cos(th5)**2 + 6385.5*sin(th2)*sin(th4)*cos(th1)**2*cos(th3)*cos(th5) - 579.0*sin(th2)*sin(th4)*cos(th1)**2*cos(th3) -
    2376*sin(th2)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5) + 15403.5*sin(th2)*cos(th1)**2*cos(th3)*cos(th4) - 3405.8*sin(th3)*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th2) +
    3691.0*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th5) + 11308.125*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2) + 3405.6*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)*cos(th5)**2 +
    6385.5*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)*cos(th5) - 579.0*sin(th3)*sin(th4)*cos(th1)**2*cos(th2) - 2376*sin(th3)*cos(th1)**2*cos(th2)*cos(th4)*cos(th5) +
    15403.5*sin(th3)*cos(th1)**2*cos(th2)*cos(th4) - 2376*sin(th4)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) - 6442.0*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) -
    6811.4*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5) + 3317.05*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3) - 2751*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5)**2 +
    8316*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) + 1873.08*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) - 169.84*sin(th4)*cos(th1)**2*cos(th2)*cos(th3) +
    2376*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)**2 + 4518.36*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) - 222.2*cos(th1)**2*cos(th2)*cos(th3)) +
    1.0e-5*ddth2*(8611*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th4)**2 + 4752*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th4)*sin(th5)*cos(th4) + 9036.72*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th4) +
    6442.0*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th5)**2*cos(th4)**2 + 6811.4*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th4)**2*cos(th5) -
    6634.1*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th4) + 2751*sin(th1)**2*sin(th2)**2*sin(th3)**2*cos(th4)**2*cos(th5)**2 + 295*sin(th1)**2*sin(th2)**2*sin(th3)**2*cos(th4)**2 -
    3746.27*sin(th1)**2*sin(th2)**2*sin(th3)**2*cos(th4)*cos(th5) + 339.68*sin(th1)**2*sin(th2)**2*sin(th3)**2*cos(th4) + 10870.35*sin(th1)**2*sin(th2)**2*sin(th3)**2 +
    4752*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th5) - 30807.0*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*cos(th3) - 6811.4*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th3)*cos(th4) +
    7382.0*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4)*cos(th5) + 22616.25*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4) +
    3746.27*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3) + 6811.4*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th4)*cos(th5)**2 + 12771.375*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th4)*cos(th5)
     - 1158.0*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th4) - 6634.1*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th5) - 37484.832*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3) +
    2751*sin(th1)**2*sin(th2)**2*sin(th5)**2*cos(th3)**2 - 6811.4*sin(th1)**2*sin(th2)**2*sin(th5)*cos(th3)**2*cos(th5) - 12771.375*sin(th1)**2*sin(th2)**2*sin(th5)*cos(th3)**2 +
    6442.0*sin(th1)**2*sin(th2)**2*cos(th3)**2*cos(th5)**2 + 22616.25*sin(th1)**2*sin(th2)**2*cos(th3)**2*cos(th5) + 65802.4*sin(th1)**2*sin(th2)**2*cos(th3)**2 +
    4752*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th5) - 30807.0*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*cos(th2) - 6811.4*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th2)*cos(th4) +
    7382.0*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4)*cos(th5) + 22616.25*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4) +
    3746.27*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2) + 6811.4*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th4)*cos(th5)**2 + 12771.375*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th4)*cos(th5)
     - 1158.0*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th4) - 6634.1*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th5) - 37484.832*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2) -
    17222*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)**2*cos(th2)*cos(th3) - 9504*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3)*cos(th4) -
    18073.44*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th2)*cos(th3) - 12322.8*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th2) -
    12884.0*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)**2*cos(th2)*cos(th3)*cos(th4)**2 + 5502*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)**2*cos(th2)*cos(th3) -
    13622.8*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4)**2*cos(th5) + 13268.2*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4) -
    13622.8*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th5) - 25542.75*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3) +
    9046.5*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th4) - 5502*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)**2*cos(th5)**2 -
    590*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)**2 + 7492.54*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)*cos(th5) -
    679.36*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4) + 12884.0*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th5)**2 + 45232.5*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th5) +
     109864.1*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3) + 5108.7*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th4)*cos(th5) - 463.2*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th4) -
    23834.4*sin(th1)**2*sin(th2)*sin(th3)*cos(th2) - 4752*sin(th1)**2*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th5) + 30807.0*sin(th1)**2*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2 +
    6811.4*sin(th1)**2*sin(th2)*sin(th5)**2*cos(th2)*cos(th3)**2*cos(th4) - 7382.0*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) -
    22616.25*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4) - 3746.27*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2 - 5108.7*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3) -
    6811.4*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5)**2 - 12771.375*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) + 1158.0*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)
    + 6634.1*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th5) + 37484.832*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2 + 9046.5*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)*cos(th5) +
    51115.68*sin(th1)**2*sin(th2)*cos(th2)*cos(th3) + 2751*sin(th1)**2*sin(th3)**2*sin(th5)**2*cos(th2)**2 - 6811.4*sin(th1)**2*sin(th3)**2*sin(th5)*cos(th2)**2*cos(th5) -
    12771.375*sin(th1)**2*sin(th3)**2*sin(th5)*cos(th2)**2 + 6442.0*sin(th1)**2*sin(th3)**2*cos(th2)**2*cos(th5)**2 + 22616.25*sin(th1)**2*sin(th3)**2*cos(th2)**2*cos(th5) +
    65802.4*sin(th1)**2*sin(th3)**2*cos(th2)**2 - 4752*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th5) + 30807.0*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)**2*cos(th3) +
    6811.4*sin(th1)**2*sin(th3)*sin(th5)**2*cos(th2)**2*cos(th3)*cos(th4) - 7382.0*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) -
    22616.25*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4) - 3746.27*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th3) - 5108.7*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2 -
    6811.4*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5)**2 - 12771.375*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) + 1158.0*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th4)
    + 6634.1*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th5) + 37484.832*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3) + 9046.5*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th5) +
    51115.68*sin(th1)**2*sin(th3)*cos(th2)**2 + 8611*sin(th1)**2*sin(th4)**2*cos(th2)**2*cos(th3)**2 + 4752*sin(th1)**2*sin(th4)*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th4) +
    9036.72*sin(th1)**2*sin(th4)*cos(th2)**2*cos(th3)**2 + 12322.8*sin(th1)**2*sin(th4)*cos(th2)**2*cos(th3) + 6442.0*sin(th1)**2*sin(th5)**2*cos(th2)**2*cos(th3)**2*cos(th4)**2 +
    6811.4*sin(th1)**2*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th4)**2*cos(th5) - 6634.1*sin(th1)**2*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th4) - 9046.5*sin(th1)**2*sin(th5)*cos(th2)**2*cos(th3)*cos(th4) +
    2751*sin(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)**2*cos(th5)**2 + 295*sin(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)**2 - 3746.27*sin(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5) +
    339.68*sin(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4) + 10870.35*sin(th1)**2*cos(th2)**2*cos(th3)**2 - 5108.7*sin(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) +
```

```
463.2*sin(th1)**2*cos(th2)**2*cos(th3)*cos(th4) + 23834.4*sin(th1)**2*cos(th2)**2*cos(th3) + 8611*sin(th2)**2*sin(th3)**2*sin(th4)**2*cos(th1)**2 +
4752*sin(th2)**2*sin(th3)**2*sin(th4)*sin(th5)*cos(th1)**2*cos(th4) + 9036.72*sin(th2)**2*sin(th3)**2*sin(th4)*cos(th1)**2 + 6442.0*sin(th2)**2*sin(th3)**2*sin(th5)**2*cos(th1)**2*cos(th4)**2 +
2751*sin(th2)**2*sin(th3)**2*sin(th5)**2 + 6811.4*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th4)**2*cos(th5) - 6634.1*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th4) -
6811.4*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th5) - 12771.375*sin(th2)**2*sin(th3)**2*sin(th5) + 2751*sin(th2)**2*sin(th3)**2*cos(th1)**2*cos(th4)**2*cos(th5)**2 +
295*sin(th2)**2*sin(th3)**2*cos(th1)**2*cos(th4)**2 - 3746.27*sin(th2)**2*sin(th3)**2*cos(th1)**2*cos(th4)*cos(th5) + 339.68*sin(th2)**2*sin(th3)**2*cos(th1)**2*cos(th4) +
10870.35*sin(th2)**2*sin(th3)**2*cos(th1)**2 + 6442.0*sin(th2)**2*sin(th3)**2*cos(th5)**2 + 22616.25*sin(th2)**2*sin(th3)**2*cos(th5) + 65802.4*sin(th2)**2*sin(th3)**2 +
4752*sin(th2)**2*sin(th3)*sin(th4)*cos(th1)**2*cos(th3)*cos(th5) - 30807.0*sin(th2)**2*sin(th3)*sin(th4)*cos(th1)**2*cos(th3) - 4752*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th5) +
30807.0*sin(th2)**2*sin(th3)*sin(th4)*cos(th3) - 6811.4*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th3)*cos(th4) + 6811.4*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th3)*cos(th4) +
7382.0*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5) + 22616.25*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3)*cos(th4) +
3746.27*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3) - 7382.0*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4)*cos(th5) - 22616.25*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4) -
3746.27*sin(th2)**2*sin(th3)*sin(th5)*cos(th3) - 5108.7*sin(th2)**2*sin(th3)*sin(th5) + 6811.4*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5)**2 +
12771.375*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5) - 1158.0*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th4) - 6634.1*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th5) -
37484.832*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3) - 6811.4*sin(th2)**2*sin(th3)*cos(th3)*cos(th4)*cos(th5)**2 - 12771.375*sin(th2)**2*sin(th3)*cos(th3)*cos(th4)*cos(th5) +
1158.0*sin(th2)**2*sin(th3)*cos(th3)*cos(th4) + 6634.1*sin(th2)**2*sin(th3)*cos(th3)*cos(th5) + 37484.832*sin(th2)**2*sin(th3)*cos(th3) + 9046.5*sin(th2)**2*sin(th3)*cos(th5) +
51115.68*sin(th2)**2*sin(th3) + 8611*sin(th2)**2*sin(th4)**2*cos(th3)**2 + 4752*sin(th2)**2*sin(th4)*sin(th5)*cos(th3)**2*cos(th4) + 9036.72*sin(th2)**2*sin(th4)*cos(th3)**2 +
12322.8*sin(th2)**2*sin(th4)*cos(th3) + 2751*sin(th2)**2*sin(th5)**2*cos(th1)**2*cos(th3)**2 + 6442.0*sin(th2)**2*sin(th5)**2*cos(th3)**2*cos(th4)**2 -
6811.4*sin(th2)**2*sin(th5)*cos(th1)**2*cos(th3)**2*cos(th5) - 12771.375*sin(th2)**2*sin(th5)*cos(th1)**2*cos(th3)**2 + 6811.4*sin(th2)**2*sin(th5)*cos(th3)**2*cos(th4)**2*cos(th5) -
6634.1*sin(th2)**2*sin(th5)*cos(th3)**2*cos(th4) - 9046.5*sin(th2)**2*sin(th5)*cos(th3)*cos(th4) + 6442.0*sin(th2)**2*cos(th1)**2*cos(th3)**2*cos(th5)**2 +
22616.25*sin(th2)**2*cos(th1)**2*cos(th3)**2*cos(th5) + 65802.4*sin(th2)**2*cos(th1)**2*cos(th3)**2 + 2751*sin(th2)**2*cos(th3)**2*cos(th4)**2*cos(th5)**2 + 295*sin(th2)**2*cos(th3)**2*cos(th4)**2 -
3746.27*sin(th2)**2*cos(th3)**2*cos(th4)*cos(th5) + 339.68*sin(th2)**2*cos(th3)**2*cos(th4) + 10870.35*sin(th2)**2*cos(th3)**2 - 5108.7*sin(th2)**2*cos(th3)*cos(th4)*cos(th5) +
463.2*sin(th2)**2*cos(th3)*cos(th4) + 23834.4*sin(th2)**2*cos(th3) + 4752*sin(th2)*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2)*cos(th5) - 30807.0*sin(th2)*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2) -
4752*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th5) + 30807.0*sin(th2)*sin(th3)**2*sin(th4)*cos(th2) - 6811.4*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th4) +
6811.4*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th2)*cos(th4) + 7382.0*sin(th2)*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th4)*cos(th5) +
22616.25*sin(th2)*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th4) + 3746.27*sin(th2)*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2) - 7382.0*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4)*cos(th5) -
22616.25*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4) - 3746.27*sin(th2)*sin(th3)**2*sin(th5)*cos(th2) + 6811.4*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th4)*cos(th5)**2 +
12771.375*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th4)*cos(th5) - 1158.0*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th4) - 6634.1*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th5) -
37484.832*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2) - 6811.4*sin(th2)*sin(th3)**2*cos(th2)*cos(th4)*cos(th5)**2 - 12771.375*sin(th2)*sin(th3)**2*cos(th2)*cos(th4)*cos(th5) +
1158.0*sin(th2)*sin(th3)**2*cos(th2)*cos(th4) + 6634.1*sin(th2)*sin(th3)**2*cos(th2)*cos(th5) + 37484.832*sin(th2)*sin(th3)**2*cos(th2) -
17222*sin(th2)*sin(th3)*sin(th4)**2*cos(th1)**2*cos(th2)*cos(th3) + 17222*sin(th2)*sin(th3)*sin(th4)**2*cos(th2)*cos(th3) -
9504*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) + 9504*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3)*cos(th4) -
18073.44*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3) - 12322.8*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th2) + 18073.44*sin(th2)*sin(th3)*sin(th4)*cos(th2)*cos(th3) +
12322.8*sin(th2)*sin(th3)*sin(th4)*cos(th2) - 12884.0*sin(th2)*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)**2 + 5502*sin(th2)*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3) +
12884.0*sin(th2)*sin(th3)*sin(th5)**2*cos(th2)*cos(th3)*cos(th4)**2 - 5502*sin(th2)*sin(th3)*sin(th5)**2*cos(th2)*cos(th3) -
13622.8*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)**2*cos(th5) + 13268.2*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) -
13622.8*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) - 25542.75*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3) +
9046.5*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th4) + 13622.8*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4)**2*cos(th5) -
13268.2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4) + 13622.8*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th5) + 25542.75*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3) -
9046.5*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th4) - 5502*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)**2*cos(th5)**2 - 590*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)**2 +
7492.54*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5) - 679.36*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) +
12884.0*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5)**2 + 45232.5*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) + 109864.1*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3) +
5108.7*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th4)*cos(th5) - 463.2*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th4) - 23834.4*sin(th2)*sin(th3)*cos(th1)**2*cos(th2) +
5502*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)**2*cos(th5)**2 + 590*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)**2 - 7492.54*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)*cos(th5) +
679.36*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4) - 12884.0*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th5)**2 - 45232.5*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th5) -
109864.1*sin(th2)*sin(th3)*cos(th2)*cos(th3) - 5108.7*sin(th2)*sin(th3)*cos(th2)*cos(th4)*cos(th5) + 463.2*sin(th2)*sin(th3)*cos(th2)*cos(th4) + 23834.4*sin(th2)*sin(th3)*cos(th2) -
4752*sin(th2)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th5) + 30807.0*sin(th2)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)**2 + 4752*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th5) -
30807.0*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2 + 6811.4*sin(th2)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) - 6811.4*sin(th2)*sin(th5)**2*cos(th2)*cos(th3)**2*cos(th4) -
7382.0*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) - 22616.25*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4) -
3746.27*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2 - 5108.7*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3) + 7382.0*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) +
22616.25*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4) + 3746.27*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2 + 5108.7*sin(th2)*sin(th5)*cos(th2)*cos(th3) -
6811.4*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)*cos(th5)**2 - 12771.375*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) + 1158.0*sin(th2)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)
+ 6634.1*sin(th2)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th5) + 37484.832*sin(th2)*cos(th1)**2*cos(th2)**2*cos(th3) + 9046.5*sin(th2)*cos(th1)**2*cos(th2)**2*cos(th5) +
51115.68*sin(th2)*cos(th1)**2*cos(th2)**2 + 6811.4*sin(th2)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5)**2 + 12771.375*sin(th2)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) -
1158.0*sin(th2)*cos(th2)**2*cos(th3)*cos(th4) - 6634.1*sin(th2)*cos(th2)**2*cos(th3)*cos(th5) - 37484.832*sin(th2)*cos(th2)**2*cos(th3) + 8611*sin(th4)**2*cos(th1)**2*cos(th2)**2*cos(th3)**2 +
4752*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4) + 9036.72*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3)**2 + 12322.8*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3) +
```

```
6442.0*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)**2 + 2751*sin(th5)**2*cos(th2)**2*cos(th3)**2 + 6811.4*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)**2*cos(th5) -
6634.1*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4) - 9046.5*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4) - 6811.4*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th5) -
12771.375*sin(th5)*cos(th2)**2*cos(th3)**2 + 2751*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)**2*cos(th5)**2 + 295*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)**2 -
3746.27*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5) + 339.68*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4) + 10870.35*cos(th1)**2*cos(th2)**2*cos(th3)**2 -
5108.7*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) + 463.2*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4) + 23834.4*cos(th1)**2*cos(th2)**2*cos(th3) + 6442.0*cos(th2)**2*cos(th3)**2*cos(th5)**2 +
22616.25*cos(th2)**2*cos(th3)**2*cos(th5) + 65802.4*cos(th2)**2*cos(th3)**2) + 1.0e-5*ddth3*(8611*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th4)**2 +
4752*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th4)*sin(th5)*cos(th4) + 9036.72*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th4) + 6442.0*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th5)**2*cos(th4)**2 +
6811.4*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th4)**2*cos(th5) - 6634.1*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th4) + 2751*sin(th1)**2*sin(th2)**2*sin(th3)**2*cos(th4)**2*cos(th5)**2
 + 295*sin(th1)**2*sin(th2)**2*sin(th3)**2*cos(th4)**2 - 3746.27*sin(th1)**2*sin(th2)**2*sin(th3)**2*cos(th4)*cos(th5) + 339.68*sin(th1)**2*sin(th2)**2*sin(th3)**2*cos(th4) +
10870.35*sin(th1)**2*sin(th2)**2*sin(th3)**2 + 4752*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th5) - 30807.0*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*cos(th3) -
6811.4*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th3)*cos(th4) + 7382.0*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4)*cos(th5) +
22616.25*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4) + 3746.27*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3) + 6811.4*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th4)*cos(th5)**2
+ 12771.375*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th4)*cos(th5) - 1158.0*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th4) - 6634.1*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th5) -
37484.832*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3) + 2751*sin(th1)**2*sin(th2)**2*sin(th5)**2*cos(th3)**2 - 6811.4*sin(th1)**2*sin(th2)**2*sin(th5)*cos(th3)**2*cos(th5) -
12771.375*sin(th1)**2*sin(th2)**2*sin(th5)*cos(th3)**2 + 6442.0*sin(th1)**2*sin(th2)**2*cos(th3)**2*cos(th5)**2 + 22616.25*sin(th1)**2*sin(th2)**2*cos(th3)**2*cos(th5) +
65802.4*sin(th1)**2*sin(th2)**2*cos(th3)**2 + 4752*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th5) - 30807.0*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*cos(th2) -
6811.4*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th2)*cos(th4) + 7382.0*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4)*cos(th5) +
22616.25*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4) + 3746.27*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2) + 6811.4*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th4)*cos(th5)**2
+ 12771.375*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th4)*cos(th5) - 1158.0*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th4) - 6634.1*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th5) -
37484.832*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2) - 17222*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)**2*cos(th2)*cos(th3) - 9504*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3)*cos(th4)
- 18073.44*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th2)*cos(th3) - 12884.0*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)**2*cos(th2)*cos(th3)*cos(th4)**2 +
5502*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)**2*cos(th2)*cos(th3) - 13622.8*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4)**2*cos(th5) +
13268.2*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4) - 13622.8*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th5) -
25542.75*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3) - 5502*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)**2*cos(th5)**2 -
590*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)**2 + 7492.54*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)*cos(th5) -
679.36*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4) + 12884.0*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th5)**2 + 45232.5*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th5) +
 109864.1*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3) - 4752*sin(th1)**2*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th5) + 30807.0*sin(th1)**2*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2 +
6811.4*sin(th1)**2*sin(th2)*sin(th5)**2*cos(th2)*cos(th3)**2*cos(th4) - 7382.0*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) -
22616.25*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4) - 3746.27*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2 - 6811.4*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5)**2
- 12771.375*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) + 1158.0*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th4) + 6634.1*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th5) +
37484.832*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2 + 2751*sin(th1)**2*sin(th3)**2*sin(th5)**2*cos(th2)**2 - 6811.4*sin(th1)**2*sin(th3)**2*sin(th5)*cos(th2)**2*cos(th5) -
12771.375*sin(th1)**2*sin(th3)**2*sin(th5)*cos(th2)**2 + 6442.0*sin(th1)**2*sin(th3)**2*cos(th2)**2*cos(th5)**2 + 22616.25*sin(th1)**2*sin(th3)**2*cos(th2)**2*cos(th5) +
65802.4*sin(th1)**2*sin(th3)**2*cos(th2)**2 - 4752*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th5) + 30807.0*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)**2*cos(th3) +
6811.4*sin(th1)**2*sin(th3)*sin(th5)**2*cos(th2)**2*cos(th3)*cos(th4) - 7382.0*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) -
22616.25*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4) - 3746.27*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th3) - 6811.4*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5)**2
- 12771.375*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) + 1158.0*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th4) + 6634.1*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th5) +
37484.832*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3) + 8611*sin(th1)**2*sin(th4)**2*cos(th2)**2*cos(th3)**2 + 4752*sin(th1)**2*sin(th4)*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th4) +
9036.72*sin(th1)**2*sin(th4)*cos(th2)**2*cos(th3)**2 + 6442.0*sin(th1)**2*sin(th5)**2*cos(th2)**2*cos(th3)**2*cos(th4)**2 + 6811.4*sin(th1)**2*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th4)**2*cos(th5) -
6634.1*sin(th1)**2*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th4) + 2751*sin(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)**2*cos(th5)**2 + 295*sin(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)**2 -
3746.27*sin(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5) + 339.68*sin(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4) + 10870.35*sin(th1)**2*cos(th2)**2*cos(th3)**2 +
8611*sin(th2)**2*sin(th3)**2*sin(th4)**2*cos(th1)**2 + 4752*sin(th2)**2*sin(th3)**2*sin(th4)*sin(th5)*cos(th1)**2*cos(th4) + 9036.72*sin(th2)**2*sin(th3)**2*sin(th4)*cos(th1)**2 +
6442.0*sin(th2)**2*sin(th3)**2*sin(th5)**2*cos(th1)**2*cos(th4)**2 + 2751*sin(th2)**2*sin(th3)**2*sin(th5)**2 + 6811.4*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th4)**2*cos(th5) -
6634.1*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th4) - 6811.4*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th5) - 12771.375*sin(th2)**2*sin(th3)**2*sin(th5) +
2751*sin(th2)**2*sin(th3)**2*cos(th1)**2*cos(th4)**2*cos(th5)**2 + 295*sin(th2)**2*sin(th3)**2*cos(th1)**2*cos(th4)**2 - 3746.27*sin(th2)**2*sin(th3)**2*cos(th1)**2*cos(th4)*cos(th5) +
339.68*sin(th2)**2*sin(th3)**2*cos(th1)**2*cos(th4) + 10870.35*sin(th2)**2*sin(th3)**2*cos(th1)**2 + 6442.0*sin(th2)**2*sin(th3)**2*cos(th5)**2 + 22616.25*sin(th2)**2*sin(th3)**2*cos(th5) +
65802.4*sin(th2)**2*sin(th3)**2 + 4752*sin(th2)**2*sin(th3)*sin(th4)*cos(th1)**2*cos(th3)*cos(th5) - 30807.0*sin(th2)**2*sin(th3)*sin(th4)*cos(th1)**2*cos(th3) -
4752*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th5) + 30807.0*sin(th2)**2*sin(th3)*sin(th4)*cos(th3) - 6811.4*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th3)*cos(th4) +
6811.4*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th3)*cos(th4) + 7382.0*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5) +
22616.25*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3)*cos(th4) + 3746.27*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3) - 7382.0*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4)*cos(th5) -
22616.25*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4) - 3746.27*sin(th2)**2*sin(th3)*sin(th5)*cos(th3) + 6811.4*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5)**2 +
12771.375*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5) - 1158.0*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th4) - 6634.1*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th5) -
37484.832*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3) - 6811.4*sin(th2)**2*sin(th3)*cos(th3)*cos(th4)*cos(th5)**2 - 12771.375*sin(th2)**2*sin(th3)*cos(th3)*cos(th4)*cos(th5) +
1158.0*sin(th2)**2*sin(th3)*cos(th3)*cos(th4) + 6634.1*sin(th2)**2*sin(th3)*cos(th3)*cos(th5) + 37484.832*sin(th2)**2*sin(th3)*cos(th3) + 8611*sin(th2)**2*sin(th4)**2*cos(th3)**2 +
4752*sin(th2)**2*sin(th4)*sin(th5)*cos(th3)**2*cos(th4) + 9036.72*sin(th2)**2*sin(th4)*cos(th3)**2 + 2751*sin(th2)**2*sin(th5)**2*cos(th1)**2*cos(th3)**2 +
6442.0*sin(th2)**2*sin(th5)**2*cos(th3)**2*cos(th4)**2 - 6811.4*sin(th2)**2*sin(th5)*cos(th1)**2*cos(th3)**2*cos(th5) - 12771.375*sin(th2)**2*sin(th5)*cos(th1)**2*cos(th3)**2 +
6811.4*sin(th2)**2*sin(th5)*cos(th3)**2*cos(th4)**2*cos(th5) - 6634.1*sin(th2)**2*sin(th5)*cos(th3)**2*cos(th4) + 6442.0*sin(th2)**2*cos(th1)**2*cos(th3)**2*cos(th5)**2 +
22616.25*sin(th2)**2*cos(th1)**2*cos(th3)**2*cos(th5) + 65802.4*sin(th2)**2*cos(th1)**2*cos(th3)**2 + 2751*sin(th2)**2*cos(th3)**2*cos(th4)**2*cos(th5)**2 + 295*sin(th2)**2*cos(th3)**2*cos(th4)**2 -
3746.27*sin(th2)**2*cos(th3)**2*cos(th4)*cos(th5) + 339.68*sin(th2)**2*cos(th3)**2*cos(th4) + 10870.35*sin(th2)**2*cos(th3)**2 + 4752*sin(th2)*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2)*cos(th5) -
30807.0*sin(th2)*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2) - 4752*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th5) + 30807.0*sin(th2)*sin(th3)**2*sin(th4)*cos(th2) -
6811.4*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th4) + 6811.4*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th2)*cos(th4) +
7382.0*sin(th2)*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th4)*cos(th5) + 22616.25*sin(th2)*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th4) +
3746.27*sin(th2)*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2) - 7382.0*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4)*cos(th5) - 22616.25*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4) -
3746.27*sin(th2)*sin(th3)**2*sin(th5)*cos(th2) + 6811.4*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th4)*cos(th5)**2 + 12771.375*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th4)*cos(th5) -
1158.0*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th4) - 6634.1*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th5) - 37484.832*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2) -
6811.4*sin(th2)*sin(th3)**2*cos(th2)*cos(th4)*cos(th5)**2 - 12771.375*sin(th2)*sin(th3)**2*cos(th2)*cos(th4)*cos(th5) + 1158.0*sin(th2)*sin(th3)**2*cos(th2)*cos(th4) +
6634.1*sin(th2)*sin(th3)**2*cos(th2)*cos(th5) + 37484.832*sin(th2)*sin(th3)**2*cos(th2) - 17222*sin(th2)*sin(th3)*sin(th4)**2*cos(th1)**2*cos(th2)*cos(th3) +
17222*sin(th2)*sin(th3)*sin(th4)**2*cos(th2)*cos(th3) - 9504*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) +
9504*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3)*cos(th4) - 18073.44*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3) + 18073.44*sin(th2)*sin(th3)*sin(th4)*cos(th2)*cos(th3) -
```

```
12884.0*sin(th2)*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)**2 + 5502*sin(th2)*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3) +
12884.0*sin(th2)*sin(th3)*sin(th5)**2*cos(th2)*cos(th3)*cos(th4)**2 - 5502*sin(th2)*sin(th3)*sin(th5)**2*cos(th2)*cos(th3) -
13622.8*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)**2*cos(th5) + 13268.2*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) -
13622.8*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) - 25542.75*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3) +
13622.8*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4)**2*cos(th5) - 13268.2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4) + 13622.8*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th5)
+ 25542.75*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3) - 5502*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)**2*cos(th5)**2 - 590*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)**2 +
 7492.54*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5) - 679.36*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) +
12884.0*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5)**2 + 45232.5*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) + 109864.1*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3) +
5502*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)**2*cos(th5)**2 + 590*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)**2 - 7492.54*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)*cos(th5) +
679.36*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4) - 12884.0*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th5)**2 - 45232.5*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th5) -
109864.1*sin(th2)*sin(th3)*cos(th2)*cos(th3) - 4752*sin(th2)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th5) + 30807.0*sin(th2)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)**2 +
4752*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th5) - 30807.0*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2 + 6811.4*sin(th2)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4) -
6811.4*sin(th2)*sin(th5)**2*cos(th2)*cos(th3)**2*cos(th4) - 7382.0*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) -
22616.25*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4) - 3746.27*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2 + 7382.0*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) +
22616.25*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4) + 3746.27*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2 - 6811.4*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)*cos(th5)**2 -
12771.375*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) + 1158.0*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4) + 6634.1*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th5) +
37484.832*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2 + 6811.4*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5)**2 + 12771.375*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) -
1158.0*sin(th2)*cos(th2)*cos(th3)**2*cos(th4) - 6634.1*sin(th2)*cos(th2)*cos(th3)**2*cos(th5) - 37484.832*sin(th2)*cos(th2)*cos(th3)**2 + 8611*sin(th3)**2*sin(th4)**2*cos(th2)**2 +
4752*sin(th3)**2*sin(th4)*sin(th5)*cos(th2)**2*cos(th4) + 9036.72*sin(th3)**2*sin(th4)*cos(th2)**2 + 2751*sin(th3)**2*sin(th5)**2*cos(th1)**2*cos(th2)**2 +
6442.0*sin(th3)**2*sin(th5)**2*cos(th2)**2*cos(th4)**2 - 6811.4*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th5) - 12771.375*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)**2 +
6811.4*sin(th3)**2*sin(th5)*cos(th2)**2*cos(th4)**2*cos(th5) - 6634.1*sin(th3)**2*sin(th5)*cos(th2)**2*cos(th4) + 6442.0*sin(th3)**2*cos(th1)**2*cos(th2)**2*cos(th5)**2 +
22616.25*sin(th3)**2*cos(th1)**2*cos(th2)**2*cos(th5) + 65802.4*sin(th3)**2*cos(th1)**2*cos(th2)**2 + 2751*sin(th3)**2*cos(th2)**2*cos(th4)**2*cos(th5)**2 + 295*sin(th3)**2*cos(th2)**2*cos(th4)**2 -
3746.27*sin(th3)**2*cos(th2)**2*cos(th4)*cos(th5) + 339.68*sin(th3)**2*cos(th2)**2*cos(th4) + 10870.35*sin(th3)**2*cos(th2)**2 - 4752*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th5) +
30807.0*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3) + 4752*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th5) - 30807.0*sin(th3)*sin(th4)*cos(th2)**2*cos(th3) +
6811.4*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4) - 6811.4*sin(th3)*sin(th5)**2*cos(th2)**2*cos(th3)*cos(th4) -
7382.0*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) - 22616.25*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4) -
3746.27*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3) + 7382.0*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) + 22616.25*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4) +
3746.27*sin(th3)*sin(th5)*cos(th2)**2*cos(th3) - 6811.4*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5)**2 - 12771.375*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) +
1158.0*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4) + 6634.1*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th5) + 37484.832*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3) +
6811.4*sin(th3)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5)**2 + 12771.375*sin(th3)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) - 1158.0*sin(th3)*cos(th2)**2*cos(th3)*cos(th4) -
6634.1*sin(th3)*cos(th2)**2*cos(th3)*cos(th5) - 37484.832*sin(th3)*cos(th2)**2*cos(th3) + 8611*sin(th4)**2*cos(th1)**2*cos(th2)**2*cos(th3)**2 +
4752*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4) + 9036.72*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3)**2 + 6442.0*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)**2 +
2751*sin(th5)**2*cos(th2)**2*cos(th3)**2 + 6811.4*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)**2*cos(th5) - 6634.1*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4) -
6811.4*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th5) - 12771.375*sin(th5)*cos(th2)**2*cos(th3)**2 + 2751*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)**2*cos(th5)**2 +
295*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)**2 - 3746.27*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5) + 339.68*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4) +
10870.35*cos(th1)**2*cos(th2)**2*cos(th3)**2 + 6442.0*cos(th2)**2*cos(th3)**2*cos(th5)**2 + 22616.25*cos(th2)**2*cos(th3)**2*cos(th5) + 65802.4*cos(th2)**2*cos(th3)**2) + 4.0e-11*ddth3 +
1.0e-5*ddth4*(2376*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)**2*sin(th5)*cos(th3) + 6442.0*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)**2*cos(th3)*cos(th4) +
6811.4*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)*cos(th3)*cos(th4)*cos(th5) - 3317.05*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)*cos(th3) +
2751*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th4)*cos(th5)**2 - 8316*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th4) -
1873.19*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th5) + 169.84*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*cos(th3) - 2376*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4)**2 -
4518.36*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th4) - 3405.6*sin(th1)**2*sin(th2)**2*sin(th4)*sin(th5)**2*cos(th3)**2 + 3691.0*sin(th1)**2*sin(th2)**2*sin(th4)*sin(th5)*cos(th3)**2*cos(th5) +
11308.125*sin(th1)**2*sin(th2)**2*sin(th4)*sin(th5)*cos(th3)**2 + 3405.8*sin(th1)**2*sin(th2)**2*sin(th4)*cos(th3)**2*cos(th5)**2 + 6385.875*sin(th1)**2*sin(th2)**2*sin(th4)*cos(th3)**2*cos(th5) -
579.0*sin(th1)**2*sin(th2)**2*sin(th4)*cos(th3)**2 - 2376*sin(th1)**2*sin(th2)**2*cos(th3)**2*cos(th4)*cos(th5) + 15403.5*sin(th1)**2*sin(th2)**2*cos(th3)**2*cos(th4) +
2376*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)**2*sin(th5)*cos(th2) + 6442.0*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)**2*cos(th2)*cos(th4) +
6811.4*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)*cos(th2)*cos(th4)*cos(th5) - 3317.05*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)*cos(th2) +
2751*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th4)*cos(th5)**2 - 8316*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th4) -
1873.19*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th5) + 169.84*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*cos(th2) - 2376*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4)**2 -
4518.36*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th4) - 6811.2*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5)**2*cos(th2)*cos(th3) +
7382.0*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3)*cos(th5) + 22616.25*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3) +
6811.6*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th2)*cos(th3)*cos(th5)**2 + 12771.75*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th2)*cos(th3)*cos(th5) -
1158.0*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th2)*cos(th3) - 4752*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)*cos(th5) +
30807.0*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4) - 2376*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)**2*sin(th5)*cos(th2)*cos(th3)**2 -
6442.0*sin(th1)**2*sin(th2)*sin(th4)*sin(th5)**2*cos(th2)*cos(th3)**2*cos(th4) - 6811.4*sin(th1)**2*sin(th2)*sin(th4)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) +
3317.05*sin(th1)**2*sin(th2)*sin(th4)*sin(th5)*cos(th2)*cos(th3)**2 - 2751*sin(th1)**2*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5)**2 +
8316*sin(th1)**2*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th4) + 1873.19*sin(th1)**2*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th5) - 169.84*sin(th1)**2*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2 +
2376*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4)**2 + 4518.36*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th4) - 3405.6*sin(th1)**2*sin(th3)**2*sin(th4)*sin(th5)**2*cos(th2)**2 +
3691.0*sin(th1)**2*sin(th3)**2*sin(th4)*sin(th5)*cos(th2)**2*cos(th5) + 11308.125*sin(th1)**2*sin(th3)**2*sin(th4)*sin(th5)*cos(th2)**2 + 3405.8*sin(th1)**2*sin(th3)**2*sin(th4)*cos(th2)**2*cos(th5)**2
+ 6385.875*sin(th1)**2*sin(th3)**2*sin(th4)*cos(th2)**2*cos(th5) - 579.0*sin(th1)**2*sin(th3)**2*sin(th4)*cos(th2)**2 - 2376*sin(th1)**2*sin(th3)**2*cos(th2)**2*cos(th4)*cos(th5) +
15403.5*sin(th1)**2*sin(th3)**2*cos(th2)**2*cos(th4) - 2376*sin(th1)**2*sin(th3)*sin(th4)**2*sin(th5)*cos(th2)**2*cos(th3) -
6442.0*sin(th1)**2*sin(th3)*sin(th4)*sin(th5)**2*cos(th2)**2*cos(th3)*cos(th4) - 6811.4*sin(th1)**2*sin(th3)*sin(th4)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) +
3317.05*sin(th1)**2*sin(th3)*sin(th4)*sin(th5)*cos(th2)**2*cos(th3) - 2751*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5)**2 +
8316*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th4) + 1873.19*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th5) - 169.84*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)**2*cos(th3) +
2376*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4)**2 + 4518.36*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th4) - 3405.6*sin(th2)**2*sin(th3)**2*sin(th4)*sin(th5)**2 +
3691.0*sin(th2)**2*sin(th3)**2*sin(th4)*sin(th5)*cos(th5) + 11308.125*sin(th2)**2*sin(th3)**2*sin(th4)*sin(th5) + 3405.8*sin(th2)**2*sin(th3)**2*sin(th4)*cos(th5)**2 +
6385.875*sin(th2)**2*sin(th3)**2*sin(th4)*cos(th5) - 579.0*sin(th2)**2*sin(th3)**2*sin(th4) - 2376*sin(th2)**2*sin(th3)**2*cos(th4)*cos(th5) + 15403.5*sin(th2)**2*sin(th3)**2*cos(th4) +
2376*sin(th2)**2*sin(th3)*sin(th4)**2*sin(th5)*cos(th1)**2*cos(th3) - 2376*sin(th2)**2*sin(th3)*sin(th4)**2*sin(th5)*cos(th3) +
6442.0*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th3)*cos(th4) - 6442.0*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)**2*cos(th3)*cos(th4) +
6811.4*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5) - 3317.05*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th3) -
```

```
6811.4*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)*cos(th3)*cos(th4)*cos(th5) + 3317.05*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)*cos(th3) +
2751*sin(th2)**2*sin(th3)*sin(th4)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5)**2 - 8316*sin(th2)**2*sin(th3)*sin(th4)*cos(th1)**2*cos(th3)*cos(th4) -
1873.19*sin(th2)**2*sin(th3)*sin(th4)*cos(th1)**2*cos(th3)*cos(th5) + 169.84*sin(th2)**2*sin(th3)*sin(th4)*cos(th1)**2*cos(th3) - 2751*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th4)*cos(th5)**2 +
8316*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th4) + 1873.19*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th5) - 169.84*sin(th2)**2*sin(th3)*sin(th4)*cos(th3) -
2376*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3)*cos(th4)**2 + 2376*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4)**2 - 4518.36*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th4) +
4518.36*sin(th2)**2*sin(th3)*cos(th3)*cos(th4) - 3405.6*sin(th2)**2*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th3)**2 + 3691.0*sin(th2)**2*sin(th4)*sin(th5)*cos(th1)**2*cos(th3)**2*cos(th5) +
11308.125*sin(th2)**2*sin(th4)*sin(th5)*cos(th1)**2*cos(th3)**2 + 3405.8*sin(th2)**2*sin(th4)*cos(th1)**2*cos(th3)**2*cos(th5)**2 + 6385.875*sin(th2)**2*sin(th4)*cos(th1)**2*cos(th3)**2*cos(th5) -
579.0*sin(th2)**2*sin(th4)*cos(th1)**2*cos(th3)**2 - 2376*sin(th2)**2*cos(th1)**2*cos(th3)**2*cos(th4)*cos(th5) + 15403.5*sin(th2)**2*cos(th1)**2*cos(th3)**2*cos(th4) +
2376*sin(th2)*sin(th3)**2*sin(th4)**2*sin(th5)*cos(th1)**2*cos(th2) - 2376*sin(th2)*sin(th3)**2*sin(th4)**2*sin(th5)*cos(th2) +
6442.0*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th4) - 6442.0*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)**2*cos(th2)*cos(th4) +
6811.4*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th4)*cos(th5) - 3317.05*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)*cos(th1)**2*cos(th2) -
6811.4*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)*cos(th2)*cos(th4)*cos(th5) + 3317.05*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)*cos(th2) +
2751*sin(th2)*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2)*cos(th4)*cos(th5)**2 - 8316*sin(th2)*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2)*cos(th4) -
1873.19*sin(th2)*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2)*cos(th5) + 169.84*sin(th2)*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2) - 2751*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th4)*cos(th5)**2 +
8316*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th4) + 1873.19*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th5) - 169.84*sin(th2)*sin(th3)**2*sin(th4)*cos(th2) -
2376*sin(th2)*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th4)**2 + 2376*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4)**2 - 4518.36*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th4) +
4518.36*sin(th2)*sin(th3)**2*cos(th2)*cos(th4) - 6811.2*sin(th2)*sin(th3)*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3) + 6811.2*sin(th2)*sin(th3)*sin(th4)*sin(th5)**2*cos(th2)*cos(th3) +
7382.0*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) + 22616.25*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3) -
7382.0*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3)*cos(th5) - 22616.25*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3) +
6811.6*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5)**2 + 12771.75*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) -
1158.0*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3) - 6811.6*sin(th2)*sin(th3)*sin(th4)*cos(th2)*cos(th3)*cos(th5)**2 - 12771.75*sin(th2)*sin(th3)*sin(th4)*cos(th2)*cos(th3)*cos(th5) +
1158.0*sin(th2)*sin(th3)*sin(th4)*cos(th2)*cos(th3) - 4752*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5) + 30807.0*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) +
4752*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)*cos(th5) - 30807.0*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4) - 2376*sin(th2)*sin(th4)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2 +
2376*sin(th2)*sin(th4)**2*sin(th5)*cos(th2)*cos(th3)**2 - 6442.0*sin(th2)*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4) +
6442.0*sin(th2)*sin(th4)*sin(th5)**2*cos(th2)*cos(th3)**2*cos(th4) - 6811.4*sin(th2)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) +
3317.05*sin(th2)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2 + 6811.4*sin(th2)*sin(th4)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) - 3317.05*sin(th2)*sin(th4)*sin(th5)*cos(th2)*cos(th3)**2 -
 2751*sin(th2)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)*cos(th5)**2 + 8316*sin(th2)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4) +
1873.19*sin(th2)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th5) - 169.84*sin(th2)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)**2 + 2751*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5)**2 -
8316*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th4) - 1873.19*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th5) + 169.84*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2 +
2376*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)**2 - 2376*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4)**2 + 4518.36*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4) -
4518.36*sin(th2)*cos(th2)*cos(th3)**2*cos(th4) - 3405.6*sin(th3)**2*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th2)**2 + 3691.0*sin(th3)**2*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th5) +
11308.125*sin(th3)**2*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)**2 + 3405.8*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th5)**2 + 6385.875*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th5) -
579.0*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2)**2 - 2376*sin(th3)**2*cos(th1)**2*cos(th2)**2*cos(th4)*cos(th5) + 15403.5*sin(th3)**2*cos(th1)**2*cos(th2)**2*cos(th4) -
2376*sin(th3)*sin(th4)**2*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3) + 2376*sin(th3)*sin(th4)**2*sin(th5)*cos(th2)**2*cos(th3) -
6442.0*sin(th3)*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4) + 6442.0*sin(th3)*sin(th4)*sin(th5)**2*cos(th2)**2*cos(th3)*cos(th4) -
6811.4*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) + 3317.05*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3) +
6811.4*sin(th3)*sin(th4)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) - 3317.05*sin(th3)*sin(th4)*sin(th5)*cos(th2)**2*cos(th3) -
2751*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5)**2 + 8316*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4) +
1873.19*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th5) - 169.84*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3) + 2751*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5)**2 -
8316*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th4) - 1873.19*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th5) + 169.84*sin(th3)*sin(th4)*cos(th2)**2*cos(th3) +
2376*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)**2 - 2376*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4)**2 + 4518.36*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4) -
4518.36*sin(th3)*cos(th2)**2*cos(th3)*cos(th4) - 3405.6*sin(th4)*sin(th5)**2*cos(th2)**2*cos(th3)**2 + 3691.0*sin(th4)*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th5) +
11308.125*sin(th4)*sin(th5)*cos(th2)**2*cos(th3)**2 + 3405.8*sin(th4)*cos(th2)**2*cos(th3)**2*cos(th5)**2 + 6385.875*sin(th4)*cos(th2)**2*cos(th3)**2*cos(th5) - 579.0*sin(th4)*cos(th2)**2*cos(th3)**2 -
2376*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5) + 15403.5*cos(th2)**2*cos(th3)**2*cos(th4)) + 1.0e-5*ddth5*(-2376*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th4)*sin(th5) -
6442.0*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th5)**2*cos(th4) - 6811.4*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th4)*cos(th5) + 3317.05*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th5) -
2751*sin(th1)**2*sin(th2)**2*sin(th3)**2*cos(th4)*cos(th5)**2 + 1873.19*sin(th1)**2*sin(th2)**2*sin(th3)**2*cos(th5) - 2376*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th4)*cos(th5) +
3405.8*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th3)*cos(th4)**2 + 3405.6*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th3) -
3691.0*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4)**2*cos(th5) - 1873.19*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4) -
3691.0*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th5) - 11308.125*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3) -
3405.6*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th4)**2*cos(th5)**2 + 3317.05*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th4)*cos(th5) -
3405.8*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th5)**2 - 6385.875*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th5) - 2751*sin(th1)**2*sin(th2)**2*sin(th5)**2*cos(th3)**2*cos(th4) +
6811.4*sin(th1)**2*sin(th2)**2*sin(th5)*cos(th3)**2*cos(th4)*cos(th5) + 6385.875*sin(th1)**2*sin(th2)**2*sin(th5)*cos(th3)**2*cos(th4) - 6442.0*sin(th1)**2*sin(th2)**2*cos(th3)**2*cos(th4)*cos(th5)**2 -
 11308.125*sin(th1)**2*sin(th2)**2*cos(th3)**2*cos(th4)*cos(th5) - 2376*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th4)*cos(th5) +
3405.8*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th2)*cos(th4)**2 + 3405.6*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th2) -
3691.0*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4)**2*cos(th5) - 1873.19*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4) -
3691.0*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th5) - 11308.125*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2) -
3405.6*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th4)**2*cos(th5)**2 + 3317.05*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th4)*cos(th5) -
3405.8*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th5)**2 - 6385.875*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th5) + 4752*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3) +
7382.0*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)**2*cos(th2)*cos(th3)*cos(th4) + 27245.6*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4)*cos(th5) +
12771.75*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4) - 6634.1*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3) -
7382.0*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)*cos(th5)**2 - 22616.25*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)*cos(th5) -
3746.38*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th5) + 2376*sin(th1)**2*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) -
3405.8*sin(th1)**2*sin(th2)*sin(th5)**2*cos(th2)*cos(th3)**2*cos(th4)**2 - 3405.6*sin(th1)**2*sin(th2)*sin(th5)**2*cos(th2)*cos(th3)**2 +
3691.0*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4)**2*cos(th5) + 1873.19*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4) +
3691.0*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th5) + 11308.125*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2 +
3405.6*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)**2*cos(th5)**2 - 3317.05*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) +
3405.8*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th5)**2 + 6385.875*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th5) - 2751*sin(th1)**2*sin(th3)**2*sin(th5)**2*cos(th2)**2*cos(th4) +
6811.4*sin(th1)**2*sin(th3)**2*sin(th5)*cos(th2)**2*cos(th4)*cos(th5) + 6385.875*sin(th1)**2*sin(th3)**2*sin(th5)*cos(th2)**2*cos(th4) - 6442.0*sin(th1)**2*sin(th3)**2*cos(th2)**2*cos(th4)*cos(th5)**2 -
```

```
 11308.125*sin(th1)**2*sin(th3)**2*cos(th2)**2*cos(th4)*cos(th5) + 2376*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) -
3405.8*sin(th1)**2*sin(th3)*sin(th5)**2*cos(th2)**2*cos(th3)*cos(th4)**2 - 3405.6*sin(th1)**2*sin(th3)*sin(th5)**2*cos(th2)**2*cos(th3) +
3691.0*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4)**2*cos(th5) + 1873.19*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4) +
3691.0*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th5) + 11308.125*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th3) +
3405.6*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th4)**2*cos(th5)**2 - 3317.05*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) +
3405.8*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th5)**2 + 6385.875*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th5) - 2376*sin(th1)**2*sin(th4)*sin(th5)*cos(th2)**2*cos(th3)**2 -
6442.0*sin(th1)**2*sin(th5)**2*cos(th2)**2*cos(th3)**2*cos(th4) - 6811.4*sin(th1)**2*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5) + 3317.05*sin(th1)**2*sin(th5)*cos(th2)**2*cos(th3)**2 -
2751*sin(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5)**2 + 1873.19*sin(th1)**2*cos(th2)**2*cos(th3)**2*cos(th5) - 2376*sin(th2)**2*sin(th3)**2*sin(th4)*sin(th5)*cos(th1)**2 -
6442.0*sin(th2)**2*sin(th3)**2*sin(th5)**2*cos(th1)**2*cos(th4) - 2751*sin(th2)**2*sin(th3)**2*sin(th5)**2*cos(th4) - 6811.4*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th4)*cos(th5) +
3317.05*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th1)**2 + 6811.4*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th4)*cos(th5) + 6385.875*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th4) -
2751*sin(th2)**2*sin(th3)**2*cos(th1)**2*cos(th4)*cos(th5)**2 + 1873.19*sin(th2)**2*sin(th3)**2*cos(th1)**2*cos(th5) - 6442.0*sin(th2)**2*sin(th3)**2*cos(th4)*cos(th5)**2 -
11308.125*sin(th2)**2*sin(th3)**2*cos(th4)*cos(th5) - 2376*sin(th2)**2*sin(th3)*sin(th4)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5) + 2376*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th4)*cos(th5) +
3405.8*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th3)*cos(th4)**2 + 3405.6*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th3) - 3405.8*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th3)*cos(th4)**2 -
3405.6*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th3) - 3691.0*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3)*cos(th4)**2*cos(th5) - 1873.19*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3)*cos(th4) -
3691.0*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3)*cos(th5) - 11308.125*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3) + 3691.0*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4)**2*cos(th5) +
1873.19*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4) + 3691.0*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th5) + 11308.125*sin(th2)**2*sin(th3)*sin(th5)*cos(th3) -
3405.6*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th4)**2*cos(th5)**2 + 3317.05*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5) -
3405.8*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th5)**2 - 6385.875*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th5) + 3405.6*sin(th2)**2*sin(th3)*cos(th3)*cos(th4)**2*cos(th5)**2 -
3317.05*sin(th2)**2*sin(th3)*cos(th3)*cos(th4)*cos(th5) + 3405.8*sin(th2)**2*sin(th3)*cos(th3)*cos(th5)**2 + 6385.875*sin(th2)**2*sin(th3)*cos(th3)*cos(th5) -
2376*sin(th2)**2*sin(th4)*sin(th5)*cos(th3)**2 - 2751*sin(th2)**2*sin(th5)**2*cos(th1)**2*cos(th3)**2*cos(th4) - 6442.0*sin(th2)**2*sin(th5)**2*cos(th3)**2*cos(th4) +
6811.4*sin(th2)**2*sin(th5)*cos(th1)**2*cos(th3)**2*cos(th4)*cos(th5) + 6385.875*sin(th2)**2*sin(th5)*cos(th1)**2*cos(th3)**2*cos(th4) - 6811.4*sin(th2)**2*sin(th5)*cos(th3)**2*cos(th4)*cos(th5) +
3317.05*sin(th2)**2*sin(th5)*cos(th3)**2 - 6442.0*sin(th2)**2*cos(th1)**2*cos(th3)**2*cos(th4)*cos(th5)**2 - 11308.125*sin(th2)**2*cos(th1)**2*cos(th3)**2*cos(th4)*cos(th5) -
2751*sin(th2)**2*cos(th3)**2*cos(th4)*cos(th5)**2 + 1873.19*sin(th2)**2*cos(th3)**2*cos(th5) - 2376*sin(th2)*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2)*cos(th4)*cos(th5) +
2376*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th4)*cos(th5) + 3405.8*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th4)**2 + 3405.6*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th1)**2*cos(th2) -
3405.8*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th2)*cos(th4)**2 - 3405.6*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th2) - 3691.0*sin(th2)*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th4)**2*cos(th5) -
1873.19*sin(th2)*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th4) - 3691.0*sin(th2)*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th5) - 11308.125*sin(th2)*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2) +
3691.0*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4)**2*cos(th5) + 1873.19*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4) + 3691.0*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th5) +
11308.125*sin(th2)*sin(th3)**2*sin(th5)*cos(th2) - 3405.6*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th4)**2*cos(th5)**2 + 3317.05*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th4)*cos(th5) -
3405.8*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th5)**2 - 6385.875*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th5) + 3405.6*sin(th2)*sin(th3)**2*cos(th2)*cos(th4)**2*cos(th5)**2 -
3317.05*sin(th2)*sin(th3)**2*cos(th2)*cos(th4)*cos(th5) + 3405.8*sin(th2)*sin(th3)**2*cos(th2)*cos(th5)**2 + 6385.875*sin(th2)*sin(th3)**2*cos(th2)*cos(th5) +
4752*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3) - 4752*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3) +
7382.0*sin(th2)*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) - 7382.0*sin(th2)*sin(th3)*sin(th5)**2*cos(th2)*cos(th3)*cos(th4) +
27245.6*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5) + 12771.75*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) -
6634.1*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3) - 27245.6*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4)*cos(th5) - 12771.75*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4)
+ 6634.1*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3) - 7382.0*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5)**2 -
22616.25*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5) - 3746.38*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) +
7382.0*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)*cos(th5)**2 + 22616.25*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)*cos(th5) + 3746.38*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th5) +
2376*sin(th2)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) - 2376*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) -
3405.8*sin(th2)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)**2 - 3405.6*sin(th2)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3)**2 + 3405.8*sin(th2)*sin(th5)**2*cos(th2)*cos(th3)**2*cos(th4)**2 +
3405.6*sin(th2)*sin(th5)**2*cos(th2)*cos(th3)**2 + 3691.0*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4) + 1873.19*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) +
3691.0*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th5) + 11308.125*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2 - 3691.0*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4)**2*cos(th5) -
1873.19*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4) - 3691.0*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th5) - 11308.125*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2 +
3405.6*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)**2*cos(th5)**2 - 3317.05*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) +
3405.8*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th5)**2 + 6385.875*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th5) - 3405.6*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)**2*cos(th5)**2 +
3317.05*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) - 3405.8*sin(th2)*cos(th2)*cos(th3)**2*cos(th5)**2 - 6385.875*sin(th2)*cos(th2)*cos(th3)**2*cos(th5) -
2376*sin(th3)**2*sin(th4)*sin(th5)*cos(th2)**2 - 2751*sin(th3)**2*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th4) - 6442.0*sin(th3)**2*sin(th5)**2*cos(th2)**2*cos(th4) +
6811.4*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th4)*cos(th5) + 6385.875*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th4) - 6811.4*sin(th3)**2*sin(th5)*cos(th2)**2*cos(th4)*cos(th5) +
3317.05*sin(th3)**2*sin(th5)*cos(th2)**2 - 6442.0*sin(th3)**2*cos(th1)**2*cos(th2)**2*cos(th4)*cos(th5)**2 - 11308.125*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) -
2751*sin(th3)**2*cos(th2)**2*cos(th4)*cos(th5)**2 + 1873.19*sin(th3)**2*cos(th2)**2*cos(th5) + 2376*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) -
2376*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) - 3405.8*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)**2 - 3405.6*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th3) +
3405.8*sin(th3)*sin(th5)**2*cos(th2)**2*cos(th3)*cos(th4)**2 + 3405.6*sin(th3)*sin(th5)**2*cos(th2)**2*cos(th3) + 3691.0*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)**2*cos(th5) +
1873.19*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4) + 3691.0*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th5) + 11308.125*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3) -
3691.0*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4)**2*cos(th5) - 1873.19*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4) - 3691.0*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th5) -
11308.125*sin(th3)*sin(th5)*cos(th2)**2*cos(th3) + 3405.6*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)**2*cos(th5)**2 - 3317.05*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) +
3405.8*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th5)**2 + 6385.875*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th5) - 3405.6*sin(th3)*cos(th2)**2*cos(th3)*cos(th4)**2*cos(th5)**2 +
3317.05*sin(th3)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) - 3405.8*sin(th3)*cos(th2)**2*cos(th3)*cos(th5)**2 - 6385.875*sin(th3)*cos(th2)**2*cos(th3)*cos(th5) -
2376*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)**2 - 6442.0*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4) - 2751*sin(th5)**2*cos(th2)**2*cos(th3)**2*cos(th4) -
6811.4*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5) + 3317.05*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)**2 + 6811.4*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5) +
6385.875*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th4) - 2751*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5)**2 + 1873.19*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th5) -
6442.0*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5)**2 - 11308.125*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5)) + 15.90107805*sin(th2)*sin(th3)*sin(th4)*cos(th1) +
7.39551375*sin(th2)*sin(th3)*sin(th5)*cos(th1)*cos(th4) + 4.174155*sin(th2)*sin(th3)*cos(th1)*cos(th4)*cos(th5) + 0.378666*sin(th2)*sin(th3)*cos(th1)*cos(th4) + 11.498301*sin(th2)*sin(th3)*cos(th1) -
4.174155*sin(th2)*sin(th5)*cos(th1)*cos(th3) + 7.39551375*sin(th2)*cos(th1)*cos(th3)*cos(th5) - 14.5652994*sin(th2)*cos(th1)*cos(th3) - 4.174155*sin(th3)*sin(th5)*cos(th1)*cos(th2) +
7.39551375*sin(th3)*cos(th1)*cos(th2)*cos(th5) - 14.5652994*sin(th3)*cos(th1)*cos(th2) - 15.90107805*sin(th4)*cos(th1)*cos(th2)*cos(th3) - 7.39551375*sin(th5)*cos(th1)*cos(th2)*cos(th3)*cos(th4) -
4.174155*cos(th1)*cos(th2)*cos(th3)*cos(th4)*cos(th5) - 0.378666*cos(th1)*cos(th2)*cos(th3)*cos(th4) - 11.498301*cos(th1)*cos(th2)*cos(th3)
```

```
tau4 = 1.0e-5*ddth1*(-2376*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th5) + 15403.5*sin(th1)**2*sin(th2)*sin(th3)*sin(th4) + 3405.6*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)**2*cos(th4) -
    3691.0*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th4)*cos(th5) - 11308.125*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th4) - 3405.8*sin(th1)**2*sin(th2)*sin(th3)*cos(th4)*cos(th5)**2 -
    6385.875*sin(th1)**2*sin(th2)*sin(th3)*cos(th4)*cos(th5) + 579.0*sin(th1)**2*sin(th2)*sin(th3)*cos(th4) + 6442.0*sin(th1)**2*sin(th2)*sin(th4)**2*sin(th5)**2*cos(th3) +
    6811.4*sin(th1)**2*sin(th2)*sin(th4)**2*sin(th5)*cos(th3)*cos(th5) + 2751*sin(th1)**2*sin(th2)*sin(th4)**2*cos(th3)*cos(th5)**2 + 8906*sin(th1)**2*sin(th2)*sin(th4)**2*cos(th3) +
    4518.36*sin(th1)**2*sin(th2)*sin(th4)*cos(th3) + 12322.8*sin(th1)**2*sin(th2)*sin(th4) + 6442.0*sin(th1)**2*sin(th2)*sin(th5)**2*cos(th3)*cos(th4)**2 +
    6811.4*sin(th1)**2*sin(th2)*sin(th5)*cos(th3)*cos(th4)**2*cos(th5) - 3317.05*sin(th1)**2*sin(th2)*sin(th5)*cos(th3)*cos(th4) - 9046.5*sin(th1)**2*sin(th2)*sin(th5)*cos(th4) +
    2751*sin(th1)**2*sin(th2)*cos(th3)*cos(th4)**2*cos(th5)**2 + 8906*sin(th1)**2*sin(th2)*cos(th3)*cos(th4)**2 - 1873.19*sin(th1)**2*sin(th2)*cos(th3)*cos(th4)*cos(th5) +
    169.84*sin(th1)**2*sin(th2)*cos(th3)*cos(th4) - 5108.7*sin(th1)**2*sin(th2)*cos(th4)*cos(th5) + 463.2*sin(th1)**2*sin(th2)*cos(th4) + 6442.0*sin(th1)**2*sin(th3)*sin(th4)**2*sin(th5)**2*cos(th2) +
    6811.4*sin(th1)**2*sin(th3)*sin(th4)**2*sin(th5)*cos(th2)*cos(th5) + 2751*sin(th1)**2*sin(th3)*sin(th4)**2*cos(th2)*cos(th5)**2 + 8906*sin(th1)**2*sin(th3)*sin(th4)**2*cos(th2) +
    4518.36*sin(th1)**2*sin(th3)*sin(th4)*cos(th2) + 6442.0*sin(th1)**2*sin(th3)*sin(th5)**2*cos(th2)*cos(th4)**2 + 6811.4*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)*cos(th4)**2*cos(th5) -
    3317.05*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)*cos(th4) + 2751*sin(th1)**2*sin(th3)*cos(th2)*cos(th4)**2*cos(th5)**2 + 8906*sin(th1)**2*sin(th3)*cos(th2)*cos(th4)**2 -
    1873.19*sin(th1)**2*sin(th3)*cos(th2)*cos(th4)*cos(th5) + 169.84*sin(th1)**2*sin(th3)*cos(th2)*cos(th4) + 2376*sin(th1)**2*sin(th4)*cos(th2)*cos(th3)*cos(th5) -
    15403.5*sin(th1)**2*sin(th4)*cos(th2)*cos(th3) - 4826.43*sin(th1)**2*sin(th4) - 3405.6*sin(th1)**2*sin(th5)**2*cos(th2)*cos(th3)*cos(th4) +
    3691.0*sin(th1)**2*sin(th5)*cos(th2)*cos(th3)*cos(th4)*cos(th5) + 11308.125*sin(th1)**2*sin(th5)*cos(th2)*cos(th3)*cos(th4) + 3543.2125*sin(th1)**2*sin(th5)*cos(th4) +
    3405.8*sin(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5)**2 + 6385.875*sin(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5) - 579.0*sin(th1)**2*cos(th2)*cos(th3)*cos(th4) +
    2000.9075*sin(th1)**2*cos(th4)*cos(th5) - 181.42*sin(th1)**2*cos(th4) - 2376*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th5) + 15403.5*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2 +
    3405.6*sin(th2)*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th4) - 3691.0*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th4)*cos(th5) - 11308.125*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th4) -
    3405.8*sin(th2)*sin(th3)*cos(th1)**2*cos(th4)*cos(th5)**2 - 6385.875*sin(th2)*sin(th3)*cos(th1)**2*cos(th4)*cos(th5) + 579.0*sin(th2)*sin(th3)*cos(th1)**2*cos(th4) +
    6442.0*sin(th2)*sin(th4)**2*sin(th5)**2*cos(th1)**2*cos(th3) + 6811.4*sin(th2)*sin(th4)**2*sin(th5)*cos(th1)**2*cos(th3)*cos(th5) + 2751*sin(th2)*sin(th4)**2*cos(th1)**2*cos(th3)*cos(th5)**2 +
    8906*sin(th2)*sin(th4)**2*cos(th1)**2*cos(th3) + 4518.36*sin(th2)*sin(th4)*cos(th1)**2*cos(th3) + 12322.8*sin(th2)*sin(th4)*cos(th1)**2 + 6442.0*sin(th2)*sin(th5)**2*cos(th1)**2*cos(th3)*cos(th4)**2 +
    6811.4*sin(th2)*sin(th5)*cos(th1)**2*cos(th3)*cos(th4)**2*cos(th5) - 3317.05*sin(th2)*sin(th5)*cos(th1)**2*cos(th3)*cos(th4) - 9046.5*sin(th2)*sin(th5)*cos(th1)**2*cos(th4) +
    2751*sin(th2)*cos(th1)**2*cos(th3)*cos(th4)**2*cos(th5)**2 + 8906*sin(th2)*cos(th1)**2*cos(th3)*cos(th4)**2 - 1873.19*sin(th2)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5) +
    169.84*sin(th2)*cos(th1)**2*cos(th3)*cos(th4) - 5108.7*sin(th2)*cos(th1)**2*cos(th4)*cos(th5) + 463.2*sin(th2)*cos(th1)**2*cos(th4) + 6442.0*sin(th3)*sin(th4)**2*sin(th5)**2*cos(th1)**2*cos(th2) +
    6811.4*sin(th3)*sin(th4)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th5) + 2751*sin(th3)*sin(th4)**2*cos(th1)**2*cos(th2)*cos(th5)**2 + 8906*sin(th3)*sin(th4)**2*cos(th1)**2*cos(th2) +
    4518.36*sin(th3)*sin(th4)*cos(th1)**2*cos(th2) + 6442.0*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th4)**2 + 6811.4*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th4)**2*cos(th5) -
    3317.05*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th4) + 2751*sin(th3)*cos(th1)**2*cos(th2)*cos(th4)**2*cos(th5)**2 + 8906*sin(th3)*cos(th1)**2*cos(th2)*cos(th4)**2 -
    1873.19*sin(th3)*cos(th1)**2*cos(th2)*cos(th4)*cos(th5) + 169.84*sin(th3)*cos(th1)**2*cos(th2)*cos(th4) + 2376*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) -
    15403.5*sin(th4)*cos(th1)**2*cos(th2)*cos(th3) - 4826.43*sin(th4)*cos(th1)**2 - 3405.6*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) +
    3691.0*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5) + 11308.125*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) + 3543.2125*sin(th5)*cos(th1)**2*cos(th4) +
    3405.8*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5)**2 + 6385.875*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5) - 579.0*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) +
    2000.9075*cos(th1)**2*cos(th4)*cos(th5) - 181.42*cos(th1)**2*cos(th4)) + 1.0e-5*ddth2*(2376*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)**2*sin(th5)*cos(th3) +
    6442.0*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)**2*cos(th3)*cos(th4) + 6811.4*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)*cos(th3)*cos(th4)*cos(th5) -
    3317.05*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)*cos(th3) + 2751*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th4)*cos(th5)**2 -
    8316*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th4) - 1873.19*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th5) + 169.84*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*cos(th3) -
    2376*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4)**2 - 4518.36*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th4) - 3405.6*sin(th1)**2*sin(th2)**2*sin(th4)*sin(th5)**2*cos(th3)**2 +
    3691.0*sin(th1)**2*sin(th2)**2*sin(th4)*sin(th5)*cos(th3)**2*cos(th5) + 11308.125*sin(th1)**2*sin(th2)**2*sin(th4)*sin(th5)*cos(th3)**2 + 3405.8*sin(th1)**2*sin(th2)**2*sin(th4)*cos(th3)**2*cos(th5)**2
    + 6385.875*sin(th1)**2*sin(th2)**2*sin(th4)*cos(th3)**2*cos(th5) - 579.0*sin(th1)**2*sin(th2)**2*sin(th4)*cos(th3)**2 - 2376*sin(th1)**2*sin(th2)**2*cos(th3)**2*cos(th4)*cos(th5) +
    15403.5*sin(th1)**2*sin(th2)**2*cos(th3)**2*cos(th4) + 2376*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)**2*sin(th5)*cos(th2) +
    6442.0*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)**2*cos(th2)*cos(th4) + 6811.4*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)*cos(th2)*cos(th4)*cos(th5) -
    3317.05*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)*cos(th2) + 2751*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th4)*cos(th5)**2 -
    8316*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th4) - 1873.19*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th5) + 169.84*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*cos(th2) -
    2376*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4)**2 - 4518.36*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th4) -
    6811.2*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5)**2*cos(th2)*cos(th3) + 7382.0*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3)*cos(th5) +
    22616.25*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3) + 6811.6*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th2)*cos(th3)*cos(th5)**2 +
```

```
12771.75*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th2)*cos(th3)*cos(th5) - 1158.0*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th2)*cos(th3) -
4752*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)*cos(th5) + 30807.0*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4) -
2376*sin(th1)**2*sin(th2)*sin(th4)**2*sin(th5)*cos(th2)*cos(th3)**2 - 6442.0*sin(th1)**2*sin(th2)*sin(th4)*sin(th5)**2*cos(th2)*cos(th3)**2*cos(th4) -
6811.4*sin(th1)**2*sin(th2)*sin(th4)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) + 3317.05*sin(th1)**2*sin(th2)*sin(th4)*sin(th5)*cos(th2)*cos(th3)**2 +
9046.5*sin(th1)**2*sin(th2)*sin(th4)*sin(th5)*cos(th2)*cos(th3) - 2751*sin(th1)**2*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5)**2 +
8316*sin(th1)**2*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th4) + 1873.19*sin(th1)**2*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th5) - 169.84*sin(th1)**2*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2 +
5108.7*sin(th1)**2*sin(th2)*sin(th4)*cos(th2)*cos(th3)*cos(th5) - 463.2*sin(th1)**2*sin(th2)*sin(th4)*cos(th2)*cos(th3) + 2376*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4)**2 +
4518.36*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th4) + 12322.8*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)*cos(th4) - 3405.6*sin(th1)**2*sin(th3)**2*sin(th4)*sin(th5)**2*cos(th2)**2 +
3691.0*sin(th1)**2*sin(th3)**2*sin(th4)*sin(th5)*cos(th2)**2*cos(th5) + 11308.125*sin(th1)**2*sin(th3)**2*sin(th4)*sin(th5)*cos(th2)**2 + 3405.8*sin(th1)**2*sin(th3)**2*sin(th4)*cos(th2)**2*cos(th5)**2
+ 6385.875*sin(th1)**2*sin(th3)**2*sin(th4)*cos(th2)**2*cos(th5) - 579.0*sin(th1)**2*sin(th3)**2*sin(th4)*cos(th2)**2 - 2376*sin(th1)**2*sin(th3)**2*cos(th2)**2*cos(th4)*cos(th5) +
15403.5*sin(th1)**2*sin(th3)**2*cos(th2)**2*cos(th4) - 2376*sin(th1)**2*sin(th3)*sin(th4)**2*sin(th5)*cos(th2)**2*cos(th3) -
6442.0*sin(th1)**2*sin(th3)*sin(th4)*sin(th5)**2*cos(th2)**2*cos(th3)*cos(th4) - 6811.4*sin(th1)**2*sin(th3)*sin(th4)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) +
3317.05*sin(th1)**2*sin(th3)*sin(th4)*sin(th5)*cos(th2)**2*cos(th3) + 9046.5*sin(th1)**2*sin(th3)*sin(th4)*sin(th5)*cos(th2)**2 -
2751*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5)**2 + 8316*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th4) +
1873.19*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th5) - 169.84*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)**2*cos(th3) + 5108.7*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)**2*cos(th5) -
463.2*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)**2 + 2376*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4)**2 + 4518.36*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th4) +
12322.8*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th4) - 3405.6*sin(th2)**2*sin(th3)**2*sin(th4)*sin(th5)**2 + 3691.0*sin(th2)**2*sin(th3)**2*sin(th4)*sin(th5)*cos(th5) +
11308.125*sin(th2)**2*sin(th3)**2*sin(th4)*sin(th5) + 3405.8*sin(th2)**2*sin(th3)**2*sin(th4)*cos(th5)**2 + 6385.875*sin(th2)**2*sin(th3)**2*sin(th4)*cos(th5) - 579.0*sin(th2)**2*sin(th3)**2*sin(th4) -
2376*sin(th2)**2*sin(th3)**2*cos(th4)*cos(th5) + 15403.5*sin(th2)**2*sin(th3)**2*cos(th4) + 2376*sin(th2)**2*sin(th3)*sin(th4)**2*sin(th5)*cos(th1)**2*cos(th3) -
2376*sin(th2)**2*sin(th3)*sin(th4)**2*sin(th5)*cos(th3) + 6442.0*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th3)*cos(th4) -
6442.0*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)**2*cos(th3)*cos(th4) + 6811.4*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5) -
3317.05*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th3) - 6811.4*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)*cos(th3)*cos(th4)*cos(th5) + 3317.05*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)*cos(th3) +
 9046.5*sin(th2)**2*sin(th3)*sin(th4)*sin(th5) + 2751*sin(th2)**2*sin(th3)*sin(th4)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5)**2 - 8316*sin(th2)**2*sin(th3)*sin(th4)*cos(th1)**2*cos(th3)*cos(th4) -
1873.19*sin(th2)**2*sin(th3)*sin(th4)*cos(th1)**2*cos(th3)*cos(th5) + 169.84*sin(th2)**2*sin(th3)*sin(th4)*cos(th1)**2*cos(th3) - 2751*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th4)*cos(th5)**2 +
8316*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th4) + 1873.19*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th5) - 169.84*sin(th2)**2*sin(th3)*sin(th4)*cos(th3) +
5108.7*sin(th2)**2*sin(th3)*sin(th4)*cos(th5) - 463.2*sin(th2)**2*sin(th3)*sin(th4) - 2376*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3)*cos(th4)**2 +
2376*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4)**2 - 4518.36*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th4) + 4518.36*sin(th2)**2*sin(th3)*cos(th3)*cos(th4) +
12322.8*sin(th2)**2*sin(th3)*cos(th4) - 3405.6*sin(th2)**2*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th3)**2 + 3691.0*sin(th2)**2*sin(th4)*sin(th5)*cos(th1)**2*cos(th3)**2*cos(th5) +
11308.125*sin(th2)**2*sin(th4)*sin(th5)*cos(th1)**2*cos(th3)**2 + 3405.8*sin(th2)**2*sin(th4)*cos(th1)**2*cos(th3)**2*cos(th5)**2 + 6385.875*sin(th2)**2*sin(th4)*cos(th1)**2*cos(th3)**2*cos(th5) -
579.0*sin(th2)**2*sin(th4)*cos(th1)**2*cos(th3)**2 - 2376*sin(th2)**2*cos(th1)**2*cos(th3)**2*cos(th4)*cos(th5) + 15403.5*sin(th2)**2*cos(th1)**2*cos(th3)**2*cos(th4) +
2376*sin(th2)*sin(th3)**2*sin(th4)**2*sin(th5)*cos(th1)**2*cos(th2) - 2376*sin(th2)*sin(th3)**2*sin(th4)**2*sin(th5)*cos(th2) +
6442.0*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th4) - 6442.0*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)**2*cos(th2)*cos(th4) +
6811.4*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th4)*cos(th5) - 3317.05*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)*cos(th1)**2*cos(th2) -
6811.4*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)*cos(th2)*cos(th4)*cos(th5) + 3317.05*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)*cos(th2) +
2751*sin(th2)*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2)*cos(th4)*cos(th5)**2 - 8316*sin(th2)*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2)*cos(th4) -
1873.19*sin(th2)*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2)*cos(th5) + 169.84*sin(th2)*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2) - 2751*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th4)*cos(th5)**2 +
8316*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th4) + 1873.19*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th5) - 169.84*sin(th2)*sin(th3)**2*sin(th4)*cos(th2) -
2376*sin(th2)*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th4)**2 + 2376*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4)**2 - 4518.36*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th4) +
4518.36*sin(th2)*sin(th3)**2*cos(th2)*cos(th4) - 6811.2*sin(th2)*sin(th3)*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3) + 6811.2*sin(th2)*sin(th3)*sin(th4)*sin(th5)**2*cos(th2)*cos(th3) +
7382.0*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) + 22616.25*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3) -
7382.0*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3)*cos(th5) - 22616.25*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3) +
6811.6*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5)**2 + 12771.75*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) -
1158.0*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3) - 6811.6*sin(th2)*sin(th3)*sin(th4)*cos(th2)*cos(th3)*cos(th5)**2 - 12771.75*sin(th2)*sin(th3)*sin(th4)*cos(th2)*cos(th3)*cos(th5) +
1158.0*sin(th2)*sin(th3)*sin(th4)*cos(th2)*cos(th3) - 4752*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5) + 30807.0*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) +
4752*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)*cos(th5) - 30807.0*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4) - 2376*sin(th2)*sin(th4)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2 +
2376*sin(th2)*sin(th4)**2*sin(th5)*cos(th2)*cos(th3)**2 - 6442.0*sin(th2)*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4) +
6442.0*sin(th2)*sin(th4)*sin(th5)**2*cos(th2)*cos(th3)**2*cos(th4) - 6811.4*sin(th2)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) +
3317.05*sin(th2)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2 + 9046.5*sin(th2)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3) +
6811.4*sin(th2)*sin(th4)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) - 3317.05*sin(th2)*sin(th4)*sin(th5)*cos(th2)*cos(th3)**2 - 9046.5*sin(th2)*sin(th4)*sin(th5)*cos(th2)*cos(th3) -
2751*sin(th2)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)*cos(th5)**2 + 8316*sin(th2)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4) +
1873.19*sin(th2)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th5) - 169.84*sin(th2)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)**2 + 5108.7*sin(th2)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) -
463.2*sin(th2)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3) + 2751*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5)**2 - 8316*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th4) -
1873.19*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th5) + 169.84*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2 - 5108.7*sin(th2)*sin(th4)*cos(th2)*cos(th3)*cos(th5) + 463.2*sin(th2)*sin(th4)*cos(th2)*cos(th3)
+ 2376*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)**2 - 2376*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4)**2 + 4518.36*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4) +
12322.8*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) - 4518.36*sin(th2)*cos(th2)*cos(th3)**2*cos(th4) - 12322.8*sin(th2)*cos(th2)*cos(th3)*cos(th4) -
3405.6*sin(th3)**2*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th2)**2 + 3691.0*sin(th3)**2*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th5) + 11308.125*sin(th3)**2*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)**2
+ 3405.8*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th5)**2 + 6385.875*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th5) - 579.0*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2)**2 -
2376*sin(th3)**2*cos(th1)**2*cos(th2)**2*cos(th4)*cos(th5) + 15403.5*sin(th3)**2*cos(th1)**2*cos(th2)**2*cos(th4) - 2376*sin(th3)*sin(th4)**2*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3) +
2376*sin(th3)*sin(th4)**2*sin(th5)*cos(th2)**2*cos(th3) - 6442.0*sin(th3)*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4) +
6442.0*sin(th3)*sin(th4)*sin(th5)**2*cos(th2)**2*cos(th3)*cos(th4) - 6811.4*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) +
3317.05*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3) + 9046.5*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)**2 + 6811.4*sin(th3)*sin(th4)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5)
 - 3317.05*sin(th3)*sin(th4)*sin(th5)*cos(th2)**2*cos(th3) - 2751*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5)**2 +
8316*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4) + 1873.19*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th5) - 169.84*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3) +
5108.7*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th5) - 463.2*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)**2 + 2751*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5)**2 -
8316*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th4) - 1873.19*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th5) + 169.84*sin(th3)*sin(th4)*cos(th2)**2*cos(th3) +
2376*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)**2 - 2376*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4)**2 + 4518.36*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4) +
12322.8*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th4) - 4518.36*sin(th3)*cos(th2)**2*cos(th3)*cos(th4) - 3405.6*sin(th4)*sin(th5)**2*cos(th2)**2*cos(th3)**2 +
3691.0*sin(th4)*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th5) + 11308.125*sin(th4)*sin(th5)*cos(th2)**2*cos(th3)**2 + 3405.8*sin(th4)*cos(th2)**2*cos(th3)**2*cos(th5)**2 +
```

```
6385.875*sin(th4)*cos(th2)**2*cos(th3)**2*cos(th5) - 579.0*sin(th4)*cos(th2)**2*cos(th3)**2 - 2376*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5) + 15403.5*cos(th2)**2*cos(th3)**2*cos(th4)) +
1.0e-5*ddth3*(2376*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)**2*sin(th5)*cos(th3) + 6442.0*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)**2*cos(th3)*cos(th4) +
6811.4*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)*cos(th3)*cos(th4)*cos(th5) - 3317.05*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)*cos(th3) +
2751*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th4)*cos(th5)**2 - 8316*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th4) -
1873.19*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th5) + 169.84*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*cos(th3) - 2376*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4)**2 -
4518.36*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th4) - 3405.6*sin(th1)**2*sin(th2)**2*sin(th4)*sin(th5)**2*cos(th3)**2 + 3691.0*sin(th1)**2*sin(th2)**2*sin(th4)*sin(th5)*cos(th3)**2*cos(th5) +
11308.125*sin(th1)**2*sin(th2)**2*sin(th4)*sin(th5)*cos(th3)**2 + 3405.8*sin(th1)**2*sin(th2)**2*sin(th4)*cos(th3)**2*cos(th5)**2 + 6385.875*sin(th1)**2*sin(th2)**2*sin(th4)*cos(th3)**2*cos(th5) -
579.0*sin(th1)**2*sin(th2)**2*sin(th4)*cos(th3)**2 - 2376*sin(th1)**2*sin(th2)**2*cos(th3)**2*cos(th4) - 6811.4*sin(th1)**2*sin(th2)**2*cos(th3)**2*cos(th4) +
2376*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)**2*sin(th5)*cos(th2) + 6442.0*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)**2*cos(th2)*cos(th4) +
6811.4*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)*cos(th2)*cos(th4)*cos(th5) - 3317.05*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)*cos(th2) +
2751*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th4)*cos(th5)**2 - 8316*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th4) -
1873.19*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th5) + 169.84*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*cos(th2) - 2376*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4)**2 -
4518.36*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th4) - 6811.2*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5)**2*cos(th2)*cos(th3) +
7382.0*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3)*cos(th5) + 22616.25*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3) +
6811.6*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th2)*cos(th3)*cos(th5)**2 + 12771.75*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th2)*cos(th3)*cos(th5) -
1158.0*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th2)*cos(th3) - 4752*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)*cos(th5) +
30807.0*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4) - 2376*sin(th1)**2*sin(th2)*sin(th4)**2*sin(th5)*cos(th2)*cos(th3)**2 -
6442.0*sin(th1)**2*sin(th2)*sin(th4)*sin(th5)**2*cos(th2)*cos(th3)**2*cos(th4) - 6811.4*sin(th1)**2*sin(th2)*sin(th4)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) +
3317.05*sin(th1)**2*sin(th2)*sin(th4)*sin(th5)*cos(th2)*cos(th3)**2 - 2751*sin(th1)**2*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5)**2 +
8316*sin(th1)**2*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th4) + 1873.19*sin(th1)**2*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th5) - 169.84*sin(th1)**2*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2 +
2376*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4)**2 + 4518.36*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th4) - 3405.6*sin(th1)**2*sin(th3)**2*sin(th4)*sin(th5)**2*cos(th2)**2 +
3691.0*sin(th1)**2*sin(th3)**2*sin(th4)*sin(th5)*cos(th2)**2*cos(th5) + 11308.125*sin(th1)**2*sin(th3)**2*sin(th4)*sin(th5)*cos(th2)**2 + 3405.8*sin(th1)**2*sin(th3)**2*sin(th4)*cos(th2)**2*cos(th5)**2
+ 6385.875*sin(th1)**2*sin(th3)**2*sin(th4)*cos(th2)**2*cos(th5) - 579.0*sin(th1)**2*sin(th3)**2*sin(th4)*cos(th2)**2 - 2376*sin(th1)**2*sin(th3)**2*cos(th2)**2*cos(th4)*cos(th5) +
15403.5*sin(th1)**2*sin(th3)**2*cos(th2)**2*cos(th4) - 2376*sin(th1)**2*sin(th3)*sin(th4)**2*sin(th5)*cos(th2)**2*cos(th3) -
6442.0*sin(th1)**2*sin(th3)*sin(th4)*sin(th5)**2*cos(th2)**2*cos(th3)*cos(th4) - 6811.4*sin(th1)**2*sin(th3)*sin(th4)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) +
3317.05*sin(th1)**2*sin(th3)*sin(th4)*sin(th5)*cos(th2)**2*cos(th3) - 2751*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5)**2 +
8316*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th4) + 1873.19*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th5) - 169.84*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)**2*cos(th3) +
2376*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4)**2 + 4518.36*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th4) - 3405.6*sin(th2)**2*sin(th3)**2*sin(th4)*sin(th5)**2 +
3691.0*sin(th2)**2*sin(th3)**2*sin(th4)*sin(th5)*cos(th5) + 11308.125*sin(th2)**2*sin(th3)**2*sin(th4)*sin(th5) + 3405.8*sin(th2)**2*sin(th3)**2*sin(th4)*cos(th5)**2 +
6385.875*sin(th2)**2*sin(th3)**2*sin(th4)*cos(th5) - 579.0*sin(th2)**2*sin(th3)**2*sin(th4) - 2376*sin(th2)**2*sin(th3)**2*cos(th4)*cos(th5) + 15403.5*sin(th2)**2*sin(th3)**2*cos(th4) +
2376*sin(th2)**2*sin(th3)*sin(th4)**2*sin(th5)*cos(th1)**2*cos(th3) - 2376*sin(th2)**2*sin(th3)*sin(th4)**2*sin(th5)*cos(th3) +
6442.0*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th3)*cos(th4) - 6442.0*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)**2*cos(th3)*cos(th4) +
6811.4*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5) - 3317.05*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th3) -
6811.4*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)*cos(th3)*cos(th4)*cos(th5) + 3317.05*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)*cos(th3) +
2751*sin(th2)**2*sin(th3)*sin(th4)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5)**2 - 8316*sin(th2)**2*sin(th3)*sin(th4)*cos(th1)**2*cos(th3)*cos(th4) -
1873.19*sin(th2)**2*sin(th3)*sin(th4)*cos(th1)**2*cos(th3)*cos(th5) + 169.84*sin(th2)**2*sin(th3)*sin(th4)*cos(th1)**2*cos(th3) - 2751*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th4)*cos(th5)**2 +
8316*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th4) + 1873.19*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th5) - 169.84*sin(th2)**2*sin(th3)*sin(th4)*cos(th3) -
2376*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3)*cos(th4)**2 + 2376*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4)**2 - 4518.36*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th4) +
4518.36*sin(th2)**2*sin(th3)*cos(th3)*cos(th4) - 3405.6*sin(th2)**2*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th3)**2 + 3691.0*sin(th2)**2*sin(th4)*sin(th5)*cos(th1)**2*cos(th3)**2*cos(th5) +
11308.125*sin(th2)**2*sin(th4)*sin(th5)*cos(th1)**2*cos(th3)**2 + 3405.8*sin(th2)**2*sin(th4)*cos(th1)**2*cos(th3)**2*cos(th5)**2 + 6385.875*sin(th2)**2*sin(th4)*cos(th1)**2*cos(th3)**2*cos(th5) -
579.0*sin(th2)**2*sin(th4)*cos(th1)**2*cos(th3)**2 - 2376*sin(th2)**2*cos(th1)**2*cos(th3)**2*cos(th4)*cos(th5) + 15403.5*sin(th2)**2*cos(th1)**2*cos(th3)**2*cos(th4) +
2376*sin(th2)*sin(th3)**2*sin(th4)**2*sin(th5)*cos(th1)**2*cos(th2) - 2376*sin(th2)*sin(th3)**2*sin(th4)**2*sin(th5)*cos(th2) +
6442.0*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th4) - 6442.0*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)**2*cos(th2)*cos(th4) +
6811.4*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th4)*cos(th5) - 3317.05*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)*cos(th1)**2*cos(th2) -
6811.4*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)*cos(th2)*cos(th4)*cos(th5) + 3317.05*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)*cos(th2) +
2751*sin(th2)*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2)*cos(th4)*cos(th5)**2 - 8316*sin(th2)*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2)*cos(th4) -
1873.19*sin(th2)*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2)*cos(th5) + 169.84*sin(th2)*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2) - 2751*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th4)*cos(th5)**2 +
8316*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th4) + 1873.19*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th5) - 169.84*sin(th2)*sin(th3)**2*sin(th4)*cos(th2) -
2376*sin(th2)*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th4)**2 + 2376*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4)**2 - 4518.36*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th4) +
4518.36*sin(th2)*sin(th3)**2*cos(th2)*cos(th4) - 6811.2*sin(th2)*sin(th3)*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3) + 6811.2*sin(th2)*sin(th3)*sin(th4)*sin(th5)**2*cos(th2)*cos(th3) +
7382.0*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) + 22616.25*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3) -
7382.0*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3)*cos(th5) - 22616.25*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3) +
6811.6*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5)**2 + 12771.75*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) -
1158.0*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3) - 6811.6*sin(th2)*sin(th3)*sin(th4)*cos(th2)*cos(th3)*cos(th5)**2 - 12771.75*sin(th2)*sin(th3)*sin(th4)*cos(th2)*cos(th3)*cos(th5) +
1158.0*sin(th2)*sin(th3)*sin(th4)*cos(th2)*cos(th3) - 4752*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5) + 30807.0*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) +
4752*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)*cos(th5) - 30807.0*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4) - 2376*sin(th2)*sin(th4)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2 +
2376*sin(th2)*sin(th4)**2*sin(th5)*cos(th2)*cos(th3)**2 - 6442.0*sin(th2)*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4) +
6442.0*sin(th2)*sin(th4)*sin(th5)**2*cos(th2)*cos(th3)**2*cos(th4) - 6811.4*sin(th2)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) +
3317.05*sin(th2)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2 + 6811.4*sin(th2)*sin(th4)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) - 3317.05*sin(th2)*sin(th4)*sin(th5)*cos(th2)*cos(th3)**2 -
 2751*sin(th2)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)*cos(th5)**2 + 8316*sin(th2)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4) +
1873.19*sin(th2)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th5) - 169.84*sin(th2)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)**2 + 2751*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5)**2 -
8316*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th4) - 1873.19*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th5) + 169.84*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2 +
2376*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)**2 - 2376*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4)**2 + 4518.36*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4) -
4518.36*sin(th2)*cos(th2)*cos(th3)**2*cos(th4) - 3405.6*sin(th3)**2*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th2)**2 + 3691.0*sin(th3)**2*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th5) +
11308.125*sin(th3)**2*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)**2 + 3405.8*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th5)**2 + 6385.875*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th5) -
579.0*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2)**2 - 2376*sin(th3)**2*cos(th1)**2*cos(th2)**2*cos(th4)*cos(th5) + 15403.5*sin(th3)**2*cos(th1)**2*cos(th2)**2*cos(th4) -
2376*sin(th3)*sin(th4)**2*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3) + 2376*sin(th3)*sin(th4)**2*sin(th5)*cos(th2)**2*cos(th3) -
6442.0*sin(th3)*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4) + 6442.0*sin(th3)*sin(th4)*sin(th5)**2*cos(th2)**2*cos(th3)*cos(th4) -
```

```
6811.4*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) + 3317.05*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3) +
6811.4*sin(th3)*sin(th4)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) - 3317.05*sin(th3)*sin(th4)*sin(th5)*cos(th2)**2*cos(th3) -
2751*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5)**2 + 8316*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4) +
1873.19*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th5) - 169.84*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3) + 2751*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5)**2 -
8316*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th4) - 1873.19*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th5) + 169.84*sin(th3)*sin(th4)*cos(th2)**2*cos(th3) +
2376*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)**2 - 2376*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4)**2 + 4518.36*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4) -
4518.36*sin(th3)*cos(th2)**2*cos(th3)*cos(th4) - 3405.6*sin(th4)*sin(th5)**2*cos(th2)**2*cos(th3)**2 + 3691.0*sin(th4)*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th5) +
11308.125*sin(th4)*sin(th5)*cos(th2)**2*cos(th3)**2 + 3405.8*sin(th4)*cos(th2)**2*cos(th3)**2*cos(th5)**2 + 6385.875*sin(th4)*cos(th2)**2*cos(th3)**2*cos(th5) - 579.0*sin(th4)*cos(th2)**2*cos(th3)**2 -
2376*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5) + 15403.5*cos(th2)**2*cos(th3)**2*cos(th4)) + 1.0e-5*ddth4*(6442.0*sin(th1)**2*sin(th2)**2*sin(th4)**2*sin(th5)**2*cos(th3)**2 +
6811.4*sin(th1)**2*sin(th2)**2*sin(th4)**2*sin(th5)*cos(th3)**2*cos(th5) + 2751*sin(th1)**2*sin(th2)**2*sin(th4)**2*cos(th3)**2*cos(th5)**2 + 295*sin(th1)**2*sin(th2)**2*sin(th4)**2*cos(th3)**2 -
4752*sin(th1)**2*sin(th2)**2*sin(th4)*sin(th5)*cos(th3)**2*cos(th4) + 8611*sin(th1)**2*sin(th2)**2*cos(th3)**2*cos(th4)**2 +
12884.0*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)**2*sin(th5)**2*cos(th2)*cos(th3) + 13622.8*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)**2*sin(th5)*cos(th2)*cos(th3)*cos(th5) +
5502*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)**2*cos(th2)*cos(th3)*cos(th5)**2 + 590*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)**2*cos(th2)*cos(th3) -
9504*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3)*cos(th4) + 17222*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)**2 +
6442.0*sin(th1)**2*sin(th3)**2*sin(th4)**2*sin(th5)**2*cos(th2)**2 + 6811.4*sin(th1)**2*sin(th3)**2*sin(th4)**2*sin(th5)*cos(th2)**2*cos(th5) +
2751*sin(th1)**2*sin(th3)**2*sin(th4)**2*cos(th2)**2*cos(th5)**2 + 295*sin(th1)**2*sin(th3)**2*sin(th4)**2*cos(th2)**2 - 4752*sin(th1)**2*sin(th3)**2*sin(th4)*sin(th5)*cos(th2)**2*cos(th4) +
8611*sin(th1)**2*sin(th3)**2*cos(th2)**2*cos(th4)**2 + 8611*sin(th1)**2*sin(th4)**2 + 4752*sin(th1)**2*sin(th4)*sin(th5)*cos(th4) + 6442.0*sin(th1)**2*sin(th5)**2*cos(th4)**2 +
6811.4*sin(th1)**2*sin(th5)*cos(th4)**2*cos(th5) + 2751*sin(th1)**2*cos(th4)**2*cos(th5)**2 + 295*sin(th1)**2*cos(th4)**2 + 6442.0*sin(th2)**2*sin(th3)**2*sin(th4)**2*sin(th5)**2 +
6811.4*sin(th2)**2*sin(th3)**2*sin(th4)**2*sin(th5)*cos(th5) + 2751*sin(th2)**2*sin(th3)**2*sin(th4)**2*cos(th5)**2 + 295*sin(th2)**2*sin(th3)**2*sin(th4)**2 -
4752*sin(th2)**2*sin(th3)**2*sin(th4)*sin(th5)*cos(th4) + 8611*sin(th2)**2*sin(th3)**2*cos(th4)**2 + 6442.0*sin(th2)**2*sin(th4)**2*sin(th5)**2*cos(th1)**2*cos(th3)**2 +
6811.4*sin(th2)**2*sin(th4)**2*sin(th5)*cos(th1)**2*cos(th3)**2*cos(th5) + 2751*sin(th2)**2*sin(th4)**2*cos(th1)**2*cos(th3)**2*cos(th5)**2 + 295*sin(th2)**2*sin(th4)**2*cos(th1)**2*cos(th3)**2 -
4752*sin(th2)**2*sin(th4)*sin(th5)*cos(th1)**2*cos(th3)**2*cos(th4) + 8611*sin(th2)**2*cos(th1)**2*cos(th3)**2*cos(th4)**2 +
12884.0*sin(th2)*sin(th3)*sin(th4)**2*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3) - 12884.0*sin(th2)*sin(th3)*sin(th4)**2*sin(th5)**2*cos(th2)*cos(th3) +
13622.8*sin(th2)*sin(th3)*sin(th4)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) - 13622.8*sin(th2)*sin(th3)*sin(th4)**2*sin(th5)*cos(th2)*cos(th3)*cos(th5) +
5502*sin(th2)*sin(th3)*sin(th4)**2*cos(th1)**2*cos(th2)*cos(th3)*cos(th5)**2 + 590*sin(th2)*sin(th3)*sin(th4)**2*cos(th1)**2*cos(th2)*cos(th3) -
5502*sin(th2)*sin(th3)*sin(th4)**2*cos(th2)*cos(th3)*cos(th5)**2 - 590*sin(th2)*sin(th3)*sin(th4)**2*cos(th2)*cos(th3) - 9504*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) +
 9504*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3)*cos(th4) + 17222*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)**2 - 17222*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)**2 +
6442.0*sin(th3)**2*sin(th4)**2*sin(th5)**2*cos(th1)**2*cos(th2)**2 + 6811.4*sin(th3)**2*sin(th4)**2*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th5) +
2751*sin(th3)**2*sin(th4)**2*cos(th1)**2*cos(th2)**2*cos(th5)**2 + 295*sin(th3)**2*sin(th4)**2*cos(th1)**2*cos(th2)**2 - 4752*sin(th3)**2*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th4) +
8611*sin(th3)**2*cos(th1)**2*cos(th2)**2*cos(th4)**2 + 6442.0*sin(th4)**2*sin(th5)**2*cos(th2)**2*cos(th3)**2 + 6811.4*sin(th4)**2*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th5) +
8611*sin(th4)**2*cos(th1)**2 + 2751*sin(th4)**2*cos(th2)**2*cos(th3)**2*cos(th5)**2 + 295*sin(th4)**2*cos(th2)**2*cos(th3)**2 + 4752*sin(th4)*sin(th5)*cos(th1)**2*cos(th4) -
4752*sin(th4)*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th4) + 6442.0*sin(th5)**2*cos(th1)**2*cos(th4)**2 + 6811.4*sin(th5)*cos(th1)**2*cos(th4)**2*cos(th5) + 2751*cos(th1)**2*cos(th4)**2*cos(th5)**2 +
295*cos(th1)**2*cos(th4)**2 + 8611*cos(th2)**2*cos(th3)**2*cos(th4)**2) + 4.0e-11*ddth4 + 1.0e-5*ddth5*(-6442.0*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)**2*cos(th3) -
6811.4*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)*cos(th3)*cos(th5) - 2751*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th5)**2 +
2376*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4) + 3405.8*sin(th1)**2*sin(th2)**2*sin(th4)*sin(th5)**2*cos(th3)**2*cos(th4) -
3691.0*sin(th1)**2*sin(th2)**2*sin(th4)*sin(th5)*cos(th3)**2*cos(th4)*cos(th5) - 3405.6*sin(th1)**2*sin(th2)**2*sin(th4)*cos(th3)**2*cos(th4)*cos(th5)**2 +
2376*sin(th1)**2*sin(th2)**2*cos(th3)**2*cos(th4)**2*cos(th5) - 6442.0*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)**2*cos(th2) -
6811.4*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)*cos(th2)*cos(th5) - 2751*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th5)**2 +
2376*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4) + 6811.6*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5)**2*cos(th2)*cos(th3)*cos(th4) -
7382.0*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3)*cos(th4)*cos(th5) - 6811.2*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th2)*cos(th3)*cos(th4)*cos(th5)**2 +
4752*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)**2*cos(th5) + 6442.0*sin(th2)*sin(th3)*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3)**2 - 6442.0*sin(th2)*sin(th3)*sin(th4)*sin(th5)**2*cos(th2)*cos(th3)**2 +
6811.4*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th5) - 6811.4*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3)*cos(th5) +
2751*sin(th2)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th5)**2 - 2751*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th5)**2 - 2376*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4) +
2376*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4) + 3405.8*sin(th3)**2*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th4) -
3691.0*sin(th3)**2*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th4)*cos(th5) - 3405.6*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th4)*cos(th5)**2 +
2376*sin(th3)**2*cos(th1)**2*cos(th2)**2*cos(th4)**2*cos(th5) + 6442.0*sin(th3)*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th3) - 6442.0*sin(th3)*sin(th4)*sin(th5)**2*cos(th2)**2*cos(th3) +
6811.4*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th5) - 6811.4*sin(th3)*sin(th4)*sin(th5)*cos(th2)**2*cos(th3)*cos(th5) +
```

```
    2751*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th5)**2 - 2751*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th5)**2 - 2376*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4) +
    2376*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4) + 2376*sin(th4)**2*cos(th1)**2*cos(th5) - 3405.8*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th4) +
    3405.8*sin(th4)*sin(th5)**2*cos(th2)**2*cos(th3)**2*cos(th4) + 3691.0*sin(th4)*sin(th5)*cos(th1)**2*cos(th4)*cos(th5) - 3691.0*sin(th4)*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5) +
    3405.6*sin(th4)*cos(th1)**2*cos(th4)*cos(th5)**2 - 3405.6*sin(th4)*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5)**2 + 2376*cos(th2)**2*cos(th3)**2*cos(th4)**2*cos(th5)) + 15.90107805*sin(th1)*sin(th4) +
    7.39551375*sin(th1)*sin(th5)*cos(th4) + 4.174155*sin(th1)*cos(th4)*cos(th5) + 0.378666*sin(th1)*cos(th4) + 7.39551375*sin(th2)*sin(th4)*sin(th5)*cos(th1)*cos(th3) +
    4.174155*sin(th2)*sin(th4)*cos(th1)*cos(th3)*cos(th5) + 0.378666*sin(th2)*sin(th4)*cos(th1)*cos(th3) - 15.90107805*sin(th2)*cos(th1)*cos(th3)*cos(th4) +
    7.39551375*sin(th3)*sin(th4)*sin(th5)*cos(th1)*cos(th2) + 4.174155*sin(th3)*sin(th4)*cos(th1)*cos(th2)*cos(th5) + 0.378666*sin(th3)*sin(th4)*cos(th1)*cos(th2) -
    15.90107805*sin(th3)*cos(th1)*cos(th2)*cos(th4)

tau5 = 1.0e-5*ddth1*(-9193.0*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5)**2 + 6385.875*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5) - 9193.0*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th5)**2 -
    11308.125*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th5) + 2376*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th4) + 2376*sin(th1)**2*sin(th2)*sin(th4)**2*cos(th3)*cos(th5) +
    1873.19*sin(th1)**2*sin(th2)*sin(th4)*sin(th5)*cos(th3) + 5108.7*sin(th1)**2*sin(th2)*sin(th4)*sin(th5) - 3317.05*sin(th1)**2*sin(th2)*sin(th4)*cos(th3)*cos(th5) -
    9046.5*sin(th1)**2*sin(th2)*sin(th4)*cos(th5) + 2376*sin(th1)**2*sin(th2)*cos(th3)*cos(th4)**2*cos(th5) + 2376*sin(th1)**2*sin(th3)*sin(th4)**2*cos(th2)*cos(th5) +
    1873.19*sin(th1)**2*sin(th3)*sin(th4)*sin(th5)*cos(th2) - 3317.05*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)*cos(th5) + 2376*sin(th1)**2*sin(th3)*cos(th2)*cos(th4)**2*cos(th5) +
    9193.0*sin(th1)**2*sin(th4)*sin(th5)**2*cos(th2)*cos(th3) - 6385.875*sin(th1)**2*sin(th4)*sin(th5)*cos(th2)*cos(th3) - 2000.9075*sin(th1)**2*sin(th4)*sin(th5) +
    9193.0*sin(th1)**2*sin(th4)*cos(th2)*cos(th3)*cos(th5)**2 + 11308.125*sin(th1)**2*sin(th4)*cos(th2)*cos(th3)*cos(th5) + 3543.2125*sin(th1)**2*sin(th4)*cos(th5) -
    2376*sin(th1)**2*sin(th5)*cos(th2)*cos(th3)*cos(th4) - 9193.0*sin(th2)*sin(th3)*sin(th4)*sin(th5)**2*cos(th1)**2 + 6385.875*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2 -
    9193.0*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th5)**2 - 11308.125*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th5) + 2376*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th4) +
    2376*sin(th2)*sin(th4)**2*cos(th1)**2*cos(th3)*cos(th5) + 1873.19*sin(th2)*sin(th4)*sin(th5)*cos(th1)**2*cos(th3) + 5108.7*sin(th2)*sin(th4)*sin(th5)*cos(th1)**2 -
    3317.05*sin(th2)*sin(th4)*cos(th1)**2*cos(th3)*cos(th5) - 9046.5*sin(th2)*sin(th4)*cos(th1)**2*cos(th5) + 2376*sin(th2)*cos(th1)**2*cos(th3)*cos(th4)**2*cos(th5) +
    2376*sin(th3)*sin(th4)**2*cos(th1)**2*cos(th2)*cos(th5) + 1873.19*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2) - 3317.05*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)*cos(th5) +
    2376*sin(th3)*cos(th1)**2*cos(th2)*cos(th4)**2*cos(th5) + 9193.0*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3) - 6385.875*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3) -
    2000.9075*sin(th4)*sin(th5)*cos(th1)**2 + 9193.0*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5)**2 + 11308.125*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) + 3543.2125*sin(th4)*cos(th1)**2*cos(th5)
    - 2376*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)) + 1.0e-5*ddth2*(-2376*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th4)*sin(th5) - 6442.0*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th5)**2*cos(th4) -
    6811.4*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th4)*cos(th5) + 3317.05*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th5) - 2751*sin(th1)**2*sin(th2)**2*sin(th3)**2*cos(th4)*cos(th5)**2 +
    1873.19*sin(th1)**2*sin(th2)**2*sin(th3)**2*cos(th5) - 2376*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th4)*cos(th5) +
    3405.8*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th3)*cos(th4)**2 + 3405.6*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th3) -
    3691.0*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4)**2*cos(th5) - 1873.19*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4) -
    3691.0*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th5) - 11308.125*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3) -
    3405.6*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th4)**2*cos(th5)**2 + 3317.05*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th4)*cos(th5) -
    3405.8*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th5)**2 - 6385.875*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th5) - 2751*sin(th1)**2*sin(th2)**2*sin(th5)**2*cos(th3)**2*cos(th4) +
    6811.4*sin(th1)**2*sin(th2)**2*sin(th5)*cos(th3)**2*cos(th4)*cos(th5) + 6385.875*sin(th1)**2*sin(th2)**2*sin(th5)*cos(th3)**2*cos(th4) - 6442.0*sin(th1)**2*sin(th2)**2*cos(th3)**2*cos(th4)*cos(th5)**2 -
     11308.125*sin(th1)**2*sin(th2)**2*cos(th3)**2*cos(th4)*cos(th5) - 2376*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th4)*cos(th5) +
    3405.8*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th2)*cos(th4)**2 + 3405.6*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th2) -
    3691.0*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4)**2*cos(th5) - 1873.19*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4) -
    3691.0*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th5) - 11308.125*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2) -
    3405.6*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th4)**2*cos(th5)**2 + 3317.05*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th4)*cos(th5) -
    3405.8*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th5)**2 - 6385.875*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th5) + 4752*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3) +
    7382.0*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)**2*cos(th2)*cos(th3)*cos(th4) + 27245.6*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4)*cos(th5) +
    12771.75*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4) - 6634.1*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3) - 9046.5*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2) -
    7382.0*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)*cos(th5)**2 - 22616.25*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)*cos(th5) -
    3746.38*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th5) - 5108.7*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th5) + 2376*sin(th1)**2*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) -
    3405.8*sin(th1)**2*sin(th2)*sin(th5)**2*cos(th2)*cos(th3)**2*cos(th4)**2 - 3405.6*sin(th1)**2*sin(th2)*sin(th5)**2*cos(th2)*cos(th3)**2 +
    3691.0*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4)**2*cos(th5) + 1873.19*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4) +
    3691.0*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th5) + 11308.125*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2 + 5108.7*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)*cos(th4) +
    3405.6*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)**2*cos(th5)**2 - 3317.05*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) +
```

```
3405.8*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th5)**2 + 6385.875*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th5) - 9046.5*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)*cos(th4)*cos(th5) -
2751*sin(th1)**2*sin(th3)**2*sin(th5)**2*cos(th2)**2*cos(th4) + 6811.4*sin(th1)**2*sin(th3)**2*sin(th5)*cos(th2)**2*cos(th4)*cos(th5) + 6385.875*sin(th1)**2*sin(th3)**2*sin(th5)*cos(th2)**2*cos(th4) -
6442.0*sin(th1)**2*sin(th3)**2*cos(th2)**2*cos(th4)*cos(th5)**2 - 11308.125*sin(th1)**2*sin(th3)**2*cos(th2)**2*cos(th4)*cos(th5) +
2376*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) - 3405.8*sin(th1)**2*sin(th3)*sin(th5)**2*cos(th2)**2*cos(th3)*cos(th4)**2 -
3405.6*sin(th1)**2*sin(th3)*sin(th5)**2*cos(th2)**2*cos(th3) + 3691.0*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4)**2*cos(th5) +
1873.19*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4) + 3691.0*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th5) + 11308.125*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th3) +
5108.7*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th4) + 3405.6*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th4)**2*cos(th5)**2 - 3317.05*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5)
 + 3405.8*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th5)**2 + 6385.875*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th5) - 9046.5*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th4)*cos(th5) -
2376*sin(th1)**2*sin(th4)*sin(th5)*cos(th2)**2*cos(th3)**2 - 6442.0*sin(th1)**2*sin(th5)**2*cos(th2)**2*cos(th3)**2*cos(th4) - 6811.4*sin(th1)**2*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5) +
3317.05*sin(th1)**2*sin(th5)*cos(th2)**2*cos(th3)**2 + 9046.5*sin(th1)**2*sin(th5)*cos(th2)**2*cos(th3) - 2751*sin(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5)**2 +
1873.19*sin(th1)**2*cos(th2)**2*cos(th3)**2*cos(th5) + 5108.7*sin(th1)**2*cos(th2)**2*cos(th3)*cos(th5) - 2376*sin(th2)**2*sin(th3)**2*sin(th4)*sin(th5)*cos(th1)**2 -
6442.0*sin(th2)**2*sin(th3)**2*sin(th5)**2*cos(th1)**2*cos(th4) - 2751*sin(th2)**2*sin(th3)**2*sin(th5)**2*cos(th4) - 6811.4*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th4)*cos(th5) +
3317.05*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th1)**2 + 6811.4*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th4)*cos(th5) + 6385.875*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th4) -
2751*sin(th2)**2*sin(th3)**2*cos(th1)**2*cos(th4)*cos(th5)**2 + 1873.19*sin(th2)**2*sin(th3)**2*cos(th1)**2*cos(th5) - 6442.0*sin(th2)**2*sin(th3)**2*cos(th4)*cos(th5)**2 -
11308.125*sin(th2)**2*sin(th3)**2*cos(th4)*cos(th5) - 2376*sin(th2)**2*sin(th3)*sin(th4)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5) + 2376*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th4)*cos(th5) +
3405.8*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th3)*cos(th4)**2 + 3405.6*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th3) - 3405.8*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th3)*cos(th4)**2 -
3405.6*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th3) - 3691.0*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3)*cos(th4)**2*cos(th5) - 1873.19*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3)*cos(th4) -
3691.0*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3)*cos(th5) - 11308.125*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3) + 3691.0*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4)**2*cos(th5) +
1873.19*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4) + 3691.0*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th5) + 11308.125*sin(th2)**2*sin(th3)*sin(th5)*cos(th3) +
5108.7*sin(th2)**2*sin(th3)*sin(th5)*cos(th4) - 3405.6*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th4)**2*cos(th5)**2 + 3317.05*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5) -
3405.8*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th5)**2 - 6385.875*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th5) + 3405.6*sin(th2)**2*sin(th3)*cos(th3)*cos(th4)**2*cos(th5)**2 -
3317.05*sin(th2)**2*sin(th3)*cos(th3)*cos(th4)*cos(th5) + 3405.8*sin(th2)**2*sin(th3)*cos(th3)*cos(th5)**2 + 6385.875*sin(th2)**2*sin(th3)*cos(th3)*cos(th5) -
9046.5*sin(th2)**2*sin(th3)*cos(th4)*cos(th5) - 2376*sin(th2)**2*sin(th4)*sin(th5)*cos(th3)**2 - 2751*sin(th2)**2*sin(th5)**2*cos(th1)**2*cos(th3)**2*cos(th4) -
6442.0*sin(th2)**2*sin(th5)**2*cos(th3)**2*cos(th4) + 6811.4*sin(th2)**2*sin(th5)*cos(th1)**2*cos(th3)**2*cos(th4)*cos(th5) + 6385.875*sin(th2)**2*sin(th5)*cos(th1)**2*cos(th3)**2*cos(th4) -
6811.4*sin(th2)**2*sin(th5)*cos(th3)**2*cos(th4)*cos(th5) + 3317.05*sin(th2)**2*sin(th5)*cos(th3)**2 + 9046.5*sin(th2)**2*sin(th5)*cos(th3) -
6442.0*sin(th2)**2*cos(th1)**2*cos(th3)**2*cos(th4)*cos(th5)**2 - 11308.125*sin(th2)**2*cos(th1)**2*cos(th3)**2*cos(th4)*cos(th5) - 2751*sin(th2)**2*cos(th3)**2*cos(th4)*cos(th5)**2 +
1873.19*sin(th2)**2*cos(th3)**2*cos(th5) + 5108.7*sin(th2)**2*cos(th3)*cos(th5) - 2376*sin(th2)*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2)*cos(th4)*cos(th5) +
2376*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th4)*cos(th5) + 3405.8*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th4)**2 + 3405.6*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th2)*cos(th4)**2 -
3405.8*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th2)*cos(th4)**2 - 3405.6*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th2) - 3691.0*sin(th2)*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th4)**2*cos(th5) -
1873.19*sin(th2)*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th4) - 3691.0*sin(th2)*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th5) - 11308.125*sin(th2)*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2) +
3691.0*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4)**2*cos(th5) + 1873.19*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4) + 3691.0*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th5) +
11308.125*sin(th2)*sin(th3)**2*sin(th5)*cos(th2) - 3405.6*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th4)**2*cos(th5)**2 + 3317.05*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th4)*cos(th5) -
3405.8*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th5)**2 - 6385.875*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th5) + 3405.6*sin(th2)*sin(th3)**2*cos(th2)*cos(th4)**2*cos(th5)**2 -
3317.05*sin(th2)*sin(th3)**2*cos(th2)*cos(th4)*cos(th5) + 3405.8*sin(th2)*sin(th3)**2*cos(th2)*cos(th5)**2 + 6385.875*sin(th2)*sin(th3)**2*cos(th2)*cos(th5) +
4752*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3) - 4752*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3) +
7382.0*sin(th2)*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) - 7382.0*sin(th2)*sin(th3)*sin(th5)**2*cos(th2)*cos(th3)*cos(th4) +
27245.6*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5) + 12771.75*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) -
6634.1*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3) - 9046.5*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2) - 27245.6*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4)*cos(th5) -
12771.75*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4) + 6634.1*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3) + 9046.5*sin(th2)*sin(th3)*sin(th5)*cos(th2) -
7382.0*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5)**2 - 22616.25*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5) -
3746.38*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) - 5108.7*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th5) + 7382.0*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)*cos(th5)**2 +
22616.25*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)*cos(th5) + 3746.38*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th5) + 5108.7*sin(th2)*sin(th3)*cos(th2)*cos(th5) +
2376*sin(th2)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) - 2376*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) -
3405.8*sin(th2)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)**2 - 3405.6*sin(th2)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3)**2 + 3405.8*sin(th2)*sin(th5)**2*cos(th2)*cos(th3)**2*cos(th4)**2 +
3405.6*sin(th2)*sin(th5)**2*cos(th2)*cos(th3)**2 + 3691.0*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)**2*cos(th5) + 1873.19*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4) +
3691.0*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th5) + 11308.125*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4) + 5108.7*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4) -
3317.05*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) + 3405.8*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th5)**2 + 6385.875*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th5) -
9046.5*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5) - 3405.6*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)**2*cos(th5)**2 + 3317.05*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) -
3405.8*sin(th2)*cos(th2)*cos(th3)**2*cos(th5)**2 - 6385.875*sin(th2)*cos(th2)*cos(th3)**2*cos(th5) + 9046.5*sin(th2)*cos(th2)*cos(th3)*cos(th4)*cos(th5) - 2376*sin(th3)**2*sin(th4)*sin(th5)*cos(th2)**2
- 2751*sin(th3)**2*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th4) - 6442.0*sin(th3)**2*sin(th5)**2*cos(th2)**2*cos(th4) + 6811.4*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th4)*cos(th5) +
6385.875*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th4) - 6811.4*sin(th3)**2*sin(th5)*cos(th2)**2*cos(th4)*cos(th5) + 3317.05*sin(th3)**2*sin(th5)*cos(th2)**2 -
6442.0*sin(th3)**2*cos(th1)**2*cos(th2)**2*cos(th4)*cos(th5)**2 - 11308.125*sin(th3)**2*cos(th1)**2*cos(th2)**2*cos(th4)*cos(th5) - 2751*sin(th3)**2*cos(th2)**2*cos(th4)*cos(th5)**2 +
1873.19*sin(th3)**2*cos(th2)**2*cos(th5) + 2376*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) - 2376*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) -
3405.8*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)**2 - 3405.6*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th3) + 3405.8*sin(th3)*sin(th5)**2*cos(th2)**2*cos(th3)*cos(th4)**2 +
3405.6*sin(th3)*sin(th5)**2*cos(th2)**2*cos(th3) + 3691.0*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)**2*cos(th5) + 1873.19*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4) +
3691.0*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th5) + 11308.125*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3) + 5108.7*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th4) -
3691.0*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4)**2*cos(th5) - 1873.19*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4) - 3691.0*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th5) -
11308.125*sin(th3)*sin(th5)*cos(th2)**2*cos(th3) + 3405.6*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)**2*cos(th5)**2 - 3317.05*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) +
3405.8*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th5)**2 + 6385.875*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th5) - 9046.5*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th4)*cos(th5) -
3405.6*sin(th3)*cos(th2)**2*cos(th3)*cos(th4)**2*cos(th5)**2 + 3317.05*sin(th3)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) - 3405.8*sin(th3)*cos(th2)**2*cos(th3)*cos(th5)**2 -
6385.875*sin(th3)*cos(th2)**2*cos(th3)*cos(th5) - 2376*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)**2 - 6442.0*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4) -
2751*sin(th5)**2*cos(th2)**2*cos(th3)**2*cos(th4) - 6811.4*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5) + 3317.05*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)**2 +
9046.5*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3) + 6811.4*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5) + 6385.875*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th4) -
2751*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5)**2 + 1873.19*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th5) + 5108.7*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th5) -
6442.0*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5)**2 - 11308.125*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5)) + 1.0e-5*ddth3*(-2376*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th4)*sin(th5) -
6442.0*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th5)**2*cos(th4) - 6811.4*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th4)*cos(th5) + 3317.05*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th5) -
```

```
2751*sin(th1)**2*sin(th2)**2*sin(th3)**2*cos(th4)*cos(th5)**2 + 1873.19*sin(th1)**2*sin(th2)**2*sin(th3)**2*cos(th5) - 2376*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th4)*cos(th5) +
3405.8*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th3)*cos(th4)**2 + 3405.6*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th3) -
3691.0*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4)**2*cos(th5) - 1873.19*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4) -
3691.0*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th5) - 11308.125*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3) -
3405.6*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th4)**2*cos(th5)**2 + 3317.05*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th4)*cos(th5) -
3405.8*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th5)**2 - 6385.875*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th5) - 2751*sin(th1)**2*sin(th2)**2*sin(th5)**2*cos(th3)**2*cos(th4) +
6811.4*sin(th1)**2*sin(th2)**2*sin(th5)*cos(th3)**2*cos(th4)*cos(th5) + 6385.875*sin(th1)**2*sin(th2)**2*sin(th5)*cos(th3)**2*cos(th4) - 6442.0*sin(th1)**2*sin(th2)**2*cos(th3)**2*cos(th4)*cos(th5)**2 -
 11308.125*sin(th1)**2*sin(th2)**2*cos(th3)**2*cos(th4)*cos(th5) - 2376*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th4)*cos(th5) +
3405.8*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th2)*cos(th4)**2 + 3405.6*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th2) -
3691.0*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4)**2*cos(th5) - 1873.19*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4) -
3691.0*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th5) - 11308.125*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2) -
3405.6*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th4)**2*cos(th5)**2 + 3317.05*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th4)*cos(th5) -
3405.8*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th5)**2 - 6385.875*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th5) + 4752*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3) +
7382.0*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)**2*cos(th2)*cos(th3)*cos(th4) + 27245.6*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4)*cos(th5) +
12771.75*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4) - 6634.1*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3) -
7382.0*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)*cos(th5)**2 - 22616.25*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)*cos(th5) -
3746.38*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th5) + 2376*sin(th1)**2*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) -
3405.8*sin(th1)**2*sin(th2)*sin(th5)**2*cos(th2)*cos(th3)**2*cos(th4)**2 - 3405.6*sin(th1)**2*sin(th2)*sin(th5)**2*cos(th2)*cos(th3)**2 +
3691.0*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4)**2*cos(th5) + 1873.19*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4) +
3691.0*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th5) + 11308.125*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2 +
3405.6*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)**2*cos(th5)**2 - 3317.05*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) +
3405.8*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th5)**2 + 6385.875*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th5) - 2751*sin(th1)**2*sin(th3)**2*sin(th5)**2*cos(th2)**2*cos(th4) +
6811.4*sin(th1)**2*sin(th3)**2*sin(th5)*cos(th2)**2*cos(th4)*cos(th5) + 6385.875*sin(th1)**2*sin(th3)**2*sin(th5)*cos(th2)**2*cos(th4) - 6442.0*sin(th1)**2*sin(th3)**2*cos(th2)**2*cos(th4)*cos(th5)**2 -
 11308.125*sin(th1)**2*sin(th3)**2*cos(th2)**2*cos(th4)*cos(th5) + 2376*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) -
3405.8*sin(th1)**2*sin(th3)*sin(th5)**2*cos(th2)**2*cos(th3)*cos(th4)**2 - 3405.6*sin(th1)**2*sin(th3)*sin(th5)**2*cos(th2)**2*cos(th3) +
3691.0*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4)**2*cos(th5) + 1873.19*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4) +
3691.0*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th5) + 11308.125*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th3) +
3405.6*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th4)**2*cos(th5)**2 - 3317.05*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) +
3405.8*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th5)**2 + 6385.875*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th5) - 2376*sin(th1)**2*sin(th4)*sin(th5)*cos(th2)**2*cos(th3)**2 -
6442.0*sin(th1)**2*sin(th5)**2*cos(th2)**2*cos(th3)**2*cos(th4) - 6811.4*sin(th1)**2*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5) + 3317.05*sin(th1)**2*sin(th5)*cos(th2)**2*cos(th3)**2 -
2751*sin(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5)**2 + 1873.19*sin(th1)**2*cos(th2)**2*cos(th3)**2*cos(th5) - 2376*sin(th2)**2*sin(th3)**2*sin(th4)*sin(th5)*cos(th1)**2 -
6442.0*sin(th2)**2*sin(th3)**2*sin(th5)**2*cos(th1)**2*cos(th4) - 2751*sin(th2)**2*sin(th3)**2*sin(th5)**2*cos(th4) - 6811.4*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th4)*cos(th5) +
3317.05*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th1)**2 + 6811.4*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th4)*cos(th5) + 6385.875*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th4) -
2751*sin(th2)**2*sin(th3)**2*cos(th1)**2*cos(th4)*cos(th5)**2 + 1873.19*sin(th2)**2*sin(th3)**2*cos(th1)**2*cos(th5) - 6442.0*sin(th2)**2*sin(th3)**2*cos(th4)*cos(th5)**2 -
11308.125*sin(th2)**2*sin(th3)**2*cos(th4)*cos(th5) - 2376*sin(th2)**2*sin(th3)*sin(th4)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5) +
3405.8*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th3)*cos(th4)**2 + 3405.6*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th3) - 3405.8*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th3)*cos(th4)**2 -
3405.6*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th3) - 3691.0*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3)*cos(th4)**2*cos(th5) - 1873.19*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3)*cos(th4) -
3691.0*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3)*cos(th5) - 11308.125*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3) + 3691.0*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4)**2*cos(th5) +
1873.19*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4) + 3691.0*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th5) + 11308.125*sin(th2)**2*sin(th3)*sin(th5)*cos(th3) -
3405.6*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th4)**2*cos(th5)**2 + 3317.05*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5) -
3405.8*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th5)**2 - 6385.875*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th5) + 3405.6*sin(th2)**2*sin(th3)*cos(th3)*cos(th4)**2*cos(th5)**2 -
3317.05*sin(th2)**2*sin(th3)*cos(th3)*cos(th4)*cos(th5) + 3405.8*sin(th2)**2*sin(th3)*cos(th3)*cos(th5)**2 + 6385.875*sin(th2)**2*sin(th3)*cos(th3)*cos(th5) -
2376*sin(th2)**2*sin(th4)*sin(th5)*cos(th3)**2 - 2751*sin(th2)**2*sin(th5)**2*cos(th1)**2*cos(th3)**2*cos(th4) - 6442.0*sin(th2)**2*sin(th5)**2*cos(th3)**2*cos(th4) +
6811.4*sin(th2)**2*sin(th5)*cos(th1)**2*cos(th3)**2*cos(th4)*cos(th5) + 6385.875*sin(th2)**2*sin(th5)*cos(th1)**2*cos(th3)**2*cos(th4) - 6811.4*sin(th2)**2*sin(th5)*cos(th3)**2*cos(th4)*cos(th5) +
3317.05*sin(th2)**2*sin(th5)*cos(th3)**2 - 6442.0*sin(th2)**2*cos(th1)**2*cos(th3)**2*cos(th4)*cos(th5)**2 - 11308.125*sin(th2)**2*cos(th1)**2*cos(th3)*cos(th4)*cos(th5) -
2751*sin(th2)**2*cos(th3)**2*cos(th4)*cos(th5)**2 + 1873.19*sin(th2)**2*cos(th3)**2*cos(th5) - 2376*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th4)*cos(th5) +
2376*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th4)*cos(th5) + 3405.8*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th4)**2 + 3405.6*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th1)**2*cos(th2) -
3405.8*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th2)*cos(th4)**2 - 3405.6*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th2) - 3691.0*sin(th2)*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th4)**2*cos(th5) -
1873.19*sin(th2)*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th4) - 3691.0*sin(th2)*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th5) - 11308.125*sin(th2)*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2) +
3691.0*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4)**2*cos(th5) + 1873.19*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4) + 3691.0*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th5) +
11308.125*sin(th2)*sin(th3)**2*sin(th5)*cos(th2) - 3405.6*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th4)**2*cos(th5)**2 + 3317.05*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th4)*cos(th5) -
3405.8*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th5)**2 - 6385.875*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th5) + 3405.6*sin(th2)*sin(th3)**2*cos(th2)*cos(th4)**2*cos(th5)**2 -
3317.05*sin(th2)*sin(th3)*cos(th2)*cos(th4)*cos(th5) + 3405.8*sin(th2)*sin(th3)*cos(th2)*cos(th5)**2 + 6385.875*sin(th2)*sin(th3)*cos(th2)*cos(th5) +
4752*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3) - 4752*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3) +
7382.0*sin(th2)*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) - 7382.0*sin(th2)*sin(th3)*sin(th5)**2*cos(th2)*cos(th3)*cos(th4) +
27245.6*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5) + 12771.75*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4) -
6634.1*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3) - 27245.6*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4)*cos(th5) - 12771.75*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4)
+ 6634.1*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3) - 7382.0*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5)**2 -
22616.25*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)*cos(th5) - 3746.38*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) +
7382.0*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)*cos(th5)**2 + 22616.25*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)*cos(th5) + 3746.38*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th5) +
2376*sin(th2)*sin(th4)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) - 2376*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) -
3405.8*sin(th2)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)**2 - 3405.6*sin(th2)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3)**2 + 3405.8*sin(th2)*sin(th5)**2*cos(th2)*cos(th3)**2*cos(th4)**2 +
3405.6*sin(th2)*sin(th5)**2*cos(th2)*cos(th3)**2 + 3691.0*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) + 1873.19*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4) +
3691.0*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th5) + 11308.125*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2 - 3691.0*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4)**2*cos(th5) -
1873.19*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4) - 3691.0*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th5) - 11308.125*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2 +
3405.6*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)**2*cos(th5)**2 - 3317.05*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) +
3405.8*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th5)**2 + 6385.875*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th5) - 3405.6*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)**2*cos(th5)**2 +
3317.05*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) - 3405.8*sin(th2)*cos(th2)*cos(th3)**2*cos(th5)**2 - 6385.875*sin(th2)*cos(th2)*cos(th3)**2*cos(th5) -
```

```
2376*sin(th3)**2*sin(th4)*sin(th5)*cos(th2)**2 - 2751*sin(th3)**2*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th4) - 6442.0*sin(th3)**2*sin(th5)**2*cos(th2)**2*cos(th4) +
6811.4*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th4)*cos(th5) + 6385.875*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th4) - 6811.4*sin(th3)**2*sin(th5)*cos(th2)**2*cos(th4)*cos(th5) +
3317.05*sin(th3)**2*sin(th5)*cos(th2)**2 - 6442.0*sin(th3)**2*cos(th1)**2*cos(th2)**2*cos(th4)*cos(th5)**2 - 11308.125*sin(th3)**2*cos(th1)**2*cos(th2)**2*cos(th4)*cos(th5) -
2751*sin(th3)**2*cos(th2)**2*cos(th4)*cos(th5)**2 + 1873.19*sin(th3)**2*cos(th2)**2*cos(th5) + 2376*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) -
2376*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) - 3405.8*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)**2 - 3405.6*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th3) +
3405.8*sin(th3)*sin(th5)**2*cos(th2)**2*cos(th3)*cos(th4)**2 + 3405.6*sin(th3)*sin(th5)**2*cos(th2)**2*cos(th3) + 3691.0*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)**2*cos(th5) +
1873.19*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4) + 3691.0*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th5) + 11308.125*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3) -
3691.0*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4)**2*cos(th5) - 1873.19*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4) - 3691.0*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th5) -
11308.125*sin(th3)*sin(th5)*cos(th2)**2*cos(th3) + 3405.6*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)**2*cos(th5)**2 - 3317.05*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) +
3405.8*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th5)**2 + 6385.875*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th5) - 3405.6*sin(th3)*cos(th2)**2*cos(th3)*cos(th4)**2*cos(th5)**2 +
3317.05*sin(th3)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) - 3405.8*sin(th3)*cos(th2)**2*cos(th3)*cos(th5)**2 - 6385.875*sin(th3)*cos(th2)**2*cos(th3)*cos(th5) -
2376*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)**2 - 6442.0*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4) - 2751*sin(th5)**2*cos(th2)**2*cos(th3)**2*cos(th4) -
6811.4*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5) + 3317.05*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)**2 + 6811.4*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5) +
6385.875*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th4) - 2751*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5)**2 + 1873.19*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th5) -
6442.0*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5)**2 - 11308.125*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5)) + 1.0e-5*ddth4*(-6442.0*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)**2*cos(th3) -
6811.4*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)*cos(th3)*cos(th5) - 2751*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th5)**2 +
2376*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4) + 3405.8*sin(th1)**2*sin(th2)**2*sin(th4)*sin(th5)**2*cos(th3)**2*cos(th4) -
3691.0*sin(th1)**2*sin(th2)**2*sin(th4)*sin(th5)*cos(th3)**2*cos(th4)*cos(th5) - 3405.6*sin(th1)**2*sin(th2)**2*sin(th4)*cos(th3)**2*cos(th4)*cos(th5)**2 +
2376*sin(th1)**2*sin(th2)**2*cos(th3)**2*cos(th4)**2*cos(th5) - 6442.0*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)**2*cos(th2) -
6811.4*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)*cos(th2)*cos(th5) - 2751*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th4)*cos(th2)*cos(th5)**2 +
2376*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4) + 6811.6*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5)**2*cos(th2)*cos(th3)*cos(th4) -
7382.0*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3)*cos(th4)*cos(th5) - 6811.2*sin(th1)**2*sin(th2)*sin(th3)*sin(th4)*cos(th2)*cos(th3)*cos(th4)*cos(th5)**2 +
4752*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)**2*cos(th5) + 6442.0*sin(th1)**2*sin(th2)*sin(th4)*sin(th5)**2*cos(th2)*cos(th3)**2 +
6811.4*sin(th1)**2*sin(th2)*sin(th4)*sin(th5)*cos(th2)*cos(th3)**2*cos(th5) + 2751*sin(th1)**2*sin(th2)*sin(th4)*cos(th2)*cos(th3)**2*cos(th5)**2 -
2376*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4) + 3405.8*sin(th1)**2*sin(th3)**2*sin(th4)*sin(th5)**2*cos(th2)**2*cos(th4) -
3691.0*sin(th1)**2*sin(th3)**2*sin(th4)*sin(th5)*cos(th2)**2*cos(th4)*cos(th5) - 3405.6*sin(th1)**2*sin(th3)**2*sin(th4)*cos(th2)**2*cos(th4)*cos(th5)**2 +
2376*sin(th1)**2*sin(th3)**2*cos(th2)**2*cos(th4)**2*cos(th5) + 6442.0*sin(th1)**2*sin(th3)*sin(th4)*sin(th5)**2*cos(th2)**2*cos(th3) +
6811.4*sin(th1)**2*sin(th3)*sin(th4)*sin(th5)*cos(th2)*cos(th3)**2*cos(th5) + 2751*sin(th1)**2*sin(th3)*sin(th4)*cos(th2)*cos(th3)**2*cos(th5)**2 -
2376*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4) + 2376*sin(th1)**2*sin(th4)**2*cos(th5) - 3405.8*sin(th1)**2*sin(th4)*sin(th5)**2*cos(th4) +
3691.0*sin(th1)**2*sin(th4)*sin(th5)*cos(th4)*cos(th5) + 3405.6*sin(th1)**2*sin(th4)*cos(th4)*cos(th5)**2 + 3405.8*sin(th2)**2*sin(th3)**2*sin(th4)*sin(th5)**2*cos(th4) -
3691.0*sin(th2)**2*sin(th3)**2*sin(th4)*sin(th5)*cos(th4)*cos(th5) - 3405.6*sin(th2)**2*sin(th3)**2*sin(th4)*cos(th4)*cos(th5)**2 + 2376*sin(th2)**2*sin(th3)**2*cos(th4)**2*cos(th5) -
6442.0*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th3) + 6442.0*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)**2*cos(th3) -
6811.4*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th3)*cos(th5) + 6811.4*sin(th2)**2*sin(th3)*sin(th4)*sin(th5)*cos(th3)*cos(th5) -
2751*sin(th2)**2*sin(th3)*sin(th4)*cos(th1)**2*cos(th3)*cos(th5)**2 + 2751*sin(th2)**2*sin(th3)*sin(th4)*cos(th3)*cos(th5)**2 + 2376*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3)*cos(th4) -
2376*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4) + 3405.8*sin(th2)**2*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th3)**2*cos(th4) -
3691.0*sin(th2)**2*sin(th4)*sin(th5)*cos(th1)**2*cos(th3)**2*cos(th4)*cos(th5) - 3405.6*sin(th2)**2*sin(th4)*cos(th1)**2*cos(th3)**2*cos(th4)*cos(th5)**2 +
2376*sin(th2)**2*cos(th1)**2*cos(th3)**2*cos(th4)**2*cos(th5) - 6442.0*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th4) - 6442.0*sin(th2)*sin(th3)**2*sin(th4)*sin(th5)**2*cos(th2)**2*cos(th3) +
6811.4*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th5) - 6811.4*sin(th2)*sin(th3)*sin(th4)*sin(th5)*cos(th2)**2*cos(th3)*cos(th5) +
2751*sin(th2)*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th5)**2 - 2751*sin(th2)*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th5)**2 - 2376*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4) +
2376*sin(th2)*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4) + 3405.8*sin(th3)**2*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th4) -
3691.0*sin(th3)**2*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th4)*cos(th5) - 3405.6*sin(th3)**2*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th4)*cos(th5)**2 +
2376*sin(th3)**2*cos(th1)**2*cos(th2)**2*cos(th4)**2*cos(th5) + 6442.0*sin(th3)*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th3) - 6442.0*sin(th3)*sin(th4)*sin(th5)**2*cos(th2)**2*cos(th3) +
6811.4*sin(th3)*sin(th4)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th5) - 6811.4*sin(th3)*sin(th4)*sin(th5)*cos(th2)**2*cos(th3)*cos(th5) +
2751*sin(th3)*sin(th4)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th5)**2 - 2751*sin(th3)*sin(th4)*cos(th2)**2*cos(th3)*cos(th5)**2 - 2376*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4) +
2376*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4) + 2376*sin(th4)**2*cos(th1)**2*cos(th5) - 3405.8*sin(th4)*sin(th5)**2*cos(th1)**2*cos(th4) +
3405.8*sin(th4)*sin(th5)**2*cos(th2)**2*cos(th3)**2*cos(th4) + 3691.0*sin(th4)*sin(th5)*cos(th1)**2*cos(th4)*cos(th5) - 3691.0*sin(th4)*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5) +
3405.6*sin(th4)*cos(th1)**2*cos(th4)*cos(th5)**2 - 3405.6*sin(th4)*cos(th2)**2*cos(th3)**2*cos(th4)*cos(th5)**2 + 2376*cos(th2)**2*cos(th3)**2*cos(th4)**2*cos(th5)) +
1.0e-5*ddth5*(6442.0*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th5)**2 + 6811.4*sin(th1)**2*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th5) + 2751*sin(th1)**2*sin(th2)**2*sin(th3)**2*cos(th5)**2 -
6811.4*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th3)*cos(th4) + 7382.0*sin(th1)**2*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4)*cos(th5) +
6811.4*sin(th1)**2*sin(th2)**2*sin(th3)*cos(th3)*cos(th4)*cos(th5)**2 + 2751*sin(th1)**2*sin(th2)**2*sin(th5)**2*cos(th3)**2*cos(th4)**2 -
6811.4*sin(th1)**2*sin(th2)**2*sin(th5)*cos(th3)**2*cos(th4)**2*cos(th5) + 6442.0*sin(th1)**2*sin(th2)**2*cos(th3)**2*cos(th4)**2*cos(th5)**2 -
6811.4*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th2)*cos(th4) + 7382.0*sin(th1)**2*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4)*cos(th5) +
6811.4*sin(th1)**2*sin(th2)*sin(th3)**2*cos(th2)*cos(th4)*cos(th5)**2 + 5502*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)**2*cos(th2)*cos(th3)*cos(th4)**2 -
12884.0*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)**2*cos(th2)*cos(th3) - 13622.8*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4)**2*cos(th5) -
13622.8*sin(th1)**2*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th5) + 12884.0*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)**2*cos(th5)**2 -
5502*sin(th1)**2*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th5)**2 + 6811.4*sin(th1)**2*sin(th2)*sin(th5)**2*cos(th2)*cos(th3)**2*cos(th4) -
7382.0*sin(th1)**2*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) - 6811.4*sin(th1)**2*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5)**2 +
2751*sin(th1)**2*sin(th3)**2*sin(th5)**2*cos(th2)**2*cos(th4)**2 - 6811.4*sin(th1)**2*sin(th3)**2*sin(th5)*cos(th2)**2*cos(th4)**2*cos(th5) +
6442.0*sin(th1)**2*sin(th3)**2*cos(th2)**2*cos(th4)**2*cos(th5)**2 + 6811.4*sin(th1)**2*sin(th3)*sin(th5)**2*cos(th2)**2*cos(th3)*cos(th4) -
7382.0*sin(th1)**2*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) - 6811.4*sin(th1)**2*sin(th3)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5)**2 + 2751*sin(th1)**2*sin(th4)**2*sin(th5)**2 -
6811.4*sin(th1)**2*sin(th4)**2*sin(th5)*cos(th5) + 6442.0*sin(th1)**2*sin(th4)**2*cos(th5)**2 + 6442.0*sin(th1)**2*sin(th5)**2*cos(th2)**2*cos(th3)**2 +
```

```
6811.4*sin(th1)**2*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th5) + 2751*sin(th1)**2*cos(th2)**2*cos(th3)**2*cos(th5)**2 + 6442.0*sin(th2)**2*sin(th3)**2*sin(th5)**2*cos(th1)**2 +
2751*sin(th2)**2*sin(th3)**2*sin(th5)**2*cos(th4)**2 + 6811.4*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th5) - 6811.4*sin(th2)**2*sin(th3)**2*sin(th5)*cos(th4)**2*cos(th5) +
2751*sin(th2)**2*sin(th3)**2*cos(th1)**2*cos(th5)**2 + 6442.0*sin(th2)**2*sin(th3)**2*cos(th4)**2*cos(th5)**2 - 6811.4*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th3)*cos(th4) +
6811.4*sin(th2)**2*sin(th3)*sin(th5)**2*cos(th3)*cos(th4) + 7382.0*sin(th2)**2*sin(th3)*sin(th5)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5) - 7382.0*sin(th2)**2*sin(th3)*sin(th5)*cos(th3)*cos(th4)*cos(th5)
+ 6811.4*sin(th2)**2*sin(th3)*cos(th1)**2*cos(th3)*cos(th4)*cos(th5)**2 - 6811.4*sin(th2)**2*sin(th3)*cos(th3)*cos(th4)*cos(th5)**2 + 2751*sin(th2)**2*sin(th5)**2*cos(th1)**2*cos(th3)**2*cos(th4)**2 +
6442.0*sin(th2)**2*sin(th5)**2*cos(th3)**2 - 6811.4*sin(th2)**2*sin(th5)*cos(th1)**2*cos(th3)**2*cos(th4)**2*cos(th5) + 6811.4*sin(th2)**2*sin(th5)*cos(th3)**2*cos(th5) +
6442.0*sin(th2)**2*cos(th1)**2*cos(th3)**2*cos(th4)**2*cos(th5)**2 + 2751*sin(th2)**2*cos(th3)**2*cos(th5)**2 - 6811.4*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th4) +
6811.4*sin(th2)*sin(th3)**2*sin(th5)**2*cos(th2)*cos(th4) + 7382.0*sin(th2)*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)*cos(th4)*cos(th5) - 7382.0*sin(th2)*sin(th3)**2*sin(th5)*cos(th2)*cos(th4)*cos(th5)
+ 6811.4*sin(th2)*sin(th3)**2*cos(th1)**2*cos(th2)*cos(th4)*cos(th5)**2 - 6811.4*sin(th2)*sin(th3)**2*cos(th2)*cos(th4)*cos(th5)**2 +
5502*sin(th2)*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)**2 - 12884.0*sin(th2)*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3) -
5502*sin(th2)*sin(th3)*sin(th5)**2*cos(th2)*cos(th3)*cos(th4)**2 + 12884.0*sin(th2)*sin(th3)*sin(th5)**2*cos(th2)*cos(th3) -
13622.8*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)**2*cos(th5) - 13622.8*sin(th2)*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5) +
13622.8*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th4)**2*cos(th5) + 13622.8*sin(th2)*sin(th3)*sin(th5)*cos(th2)*cos(th3)*cos(th5) +
12884.0*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th4)**2*cos(th5)**2 - 5502*sin(th2)*sin(th3)*cos(th1)**2*cos(th2)*cos(th3)*cos(th5)**2 -
12884.0*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th4)**2*cos(th5)**2 + 5502*sin(th2)*sin(th3)*cos(th2)*cos(th3)*cos(th5)**2 + 6811.4*sin(th2)*sin(th5)**2*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4) -
6811.4*sin(th2)*sin(th5)**2*cos(th2)*cos(th3)**2*cos(th4) - 7382.0*sin(th2)*sin(th5)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)*cos(th5) + 7382.0*sin(th2)*sin(th5)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5)
- 6811.4*sin(th2)*cos(th1)**2*cos(th2)*cos(th3)**2*cos(th4)*cos(th5)**2 + 6811.4*sin(th2)*cos(th2)*cos(th3)**2*cos(th4)*cos(th5)**2 + 2751*sin(th3)**2*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th4)**2 +
6442.0*sin(th3)**2*sin(th5)**2*cos(th2)**2 - 6811.4*sin(th3)**2*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th4)**2*cos(th5) + 6811.4*sin(th3)**2*sin(th5)*cos(th2)**2*cos(th5) +
6442.0*sin(th3)**2*cos(th1)**2*cos(th2)**2*cos(th4)**2*cos(th5)**2 + 2751*sin(th3)**2*cos(th2)**2*cos(th5)**2 + 6811.4*sin(th3)*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4) -
6811.4*sin(th3)*sin(th5)**2*cos(th2)**2*cos(th3)*cos(th4) - 7382.0*sin(th3)*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5) + 7382.0*sin(th3)*sin(th5)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5)
- 6811.4*sin(th3)*cos(th1)**2*cos(th2)**2*cos(th3)*cos(th4)*cos(th5)**2 + 6811.4*sin(th3)*cos(th2)**2*cos(th3)*cos(th4)*cos(th5)**2 + 2751*sin(th4)**2*sin(th5)**2*cos(th1)**2 -
6811.4*sin(th4)**2*sin(th5)*cos(th1)**2*cos(th5) + 6442.0*sin(th4)**2*cos(th1)**2*cos(th5)**2 + 6442.0*sin(th5)**2*cos(th1)**2*cos(th2)**2*cos(th3)**2 +
2751*sin(th5)**2*cos(th2)**2*cos(th3)**2*cos(th4)**2 + 6811.4*sin(th5)*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th5) - 6811.4*sin(th5)*cos(th2)**2*cos(th3)**2*cos(th4)**2*cos(th5) +
2751*cos(th1)**2*cos(th2)**2*cos(th3)**2*cos(th5)**2 + 6442.0*cos(th2)**2*cos(th3)**2*cos(th4)**2*cos(th5)**2) + 1.0e-13*ddth5 - 4.174155*sin(th1)*sin(th4)*sin(th5) +
7.39551375*sin(th1)*sin(th4)*cos(th5) - 7.39551375*sin(th2)*sin(th3)*sin(th5)*cos(th1) - 4.174155*sin(th2)*sin(th3)*cos(th1)*cos(th5) + 4.174155*sin(th2)*sin(th5)*cos(th1)*cos(th3)*cos(th4) -
7.39551375*sin(th2)*cos(th1)*cos(th3)*cos(th4)*cos(th5) + 4.174155*sin(th3)*sin(th5)*cos(th1)*cos(th2)*cos(th4) - 7.39551375*sin(th3)*cos(th1)*cos(th2)*cos(th4)*cos(th5) +
7.39551375*sin(th5)*cos(th1)*cos(th2)*cos(th3) + 4.174155*cos(th1)*cos(th2)*cos(th3)*cos(th5)
```